

G-Est Software
version 2.03

Small-Area Estimation

**Area-Level Model
with EBLUP Estimation**

Methodology Specifications

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Table of Contents

INTRODUCTION	1
TERMS	2
NOTATION	3
OVERVIEW	5
THE MODEL FOR SMALL-AREA ESTIMATION.....	6
DETERMINING THE METHODS AND AREA GROUPS	8
USE OF SAMPLE COUNTS AND POPULATION COUNTS	10
SMOOTHING OF THE DIRECT VARIANCE ESTIMATES	11
SMOOTHING BY THE METHOD OF GENERALIZED VARIANCE FUNCTIONS	11
THE SMALL-AREA ESTIMATION METHODS.....	13
METHOD OF ADJUSTED DENSITY MAXIMIZATION	13
Processing steps for the ADM method	13
METHOD OF RESTRICTED MAXIMUM LIKELIHOOD.....	22
Processing steps for the REML method	22
METHOD OF FAY-HERRIOT	30
Processing steps for the FH method	30
METHOD OF WANG-FULLER	38
Processing steps for the WF method	38
BENCHMARKING	47
PROOF OF BENCHMARKING BASED ON AUGMENTED VECTOR	48
PROOF OF BENCHMARKING BASED ON DIFFERENCE ADJUSTMENT	49
RESULTS FOR MSE ESTIMATION	51
INCLUSION OF THE MODEL ERROR COEFFICIENT	51
ESTIMATION OF THE MSE IN THE WANG-FULLER METHOD.....	52
DIAGNOSTICS.....	53
PLOTS ON THE GVF SMOOTHED VARIANCES.....	53
Transformed Direct Variance and GVF Smoothed Variance By Area Number For Each Area	54
Transformed Direct Variance vs. GVF Smoothed Variance	54
PLOTS ON THE DIRECT ESTIMATES AND THE SMALL-AREA ESTIMATES	55
Direct Estimate and SAE Estimate By Area Number For Each Area	56
Direct Estimate and SAE Estimate By Area Number Sorted By Direct CV.....	56
Direct Estimate and SAE Estimate By Population Size for Each Area	57
Direct Estimate and SAE Estimate By Sample Size for Each Area	57
SAE Estimate vs. Direct Estimate	58
Ratio of SAE Estimate to Direct Estimate By Area Number For Each Area.....	59
Ratio of SAE Estimate to Direct Estimate By Area Number Sorted By Direct CV	59
Ratio of SAE Estimate to Direct Estimate By Population Size for Each Area	60
Ratio of SAE Estimate to Direct Estimate By Sample Size for Each Area	60
PLOTS ON THE DIRECT VARIANCES AND THE SAE MSEs.....	61
Direct Variance and SAE MSE By Area Number For Each Area.....	61
Direct Variance and SAE MSE By Area Number Sorted By Direct CV	62
Direct Variance and SAE MSE By Population Size for Each Area	62
Direct Variance and SAE MSE By Sample Size for Each Area	63
SAE MSE vs. Direct Variance	63
Ratio of SAE MSE to Direct Variance By Area Number For Each Area	64
Ratio of SAE MSE to Direct Variance By Area Number Sorted By Direct CV	65
Ratio of SAE MSE to Direct Variance By Population Size for Each Area	65
Ratio of SAE MSE to Direct Variance By Sample Size for Each Area	66
PLOTS ON THE DIRECT CVs AND THE SAE RRMSEs	66
Direct CV and SAE RRMSE By Area Number For Each Area	67
Direct CV and SAE RRMSE By Area Number Sorted By Direct CV	67
Direct CV and SAE RRMSE By Population Size for Each Area	68
Direct CV and SAE RRMSE By Sample Size for Each Area	68
SAE RRMSE vs. Direct CV	69
Ratio of SAE RRMSE to Direct CV By Area Number For Each Area	70
Ratio of SAE RRMSE to Direct CV By Area Number Sorted By Direct CV	70
Ratio of SAE RRMSE to Direct CV By Population Size for Each Area	71
Ratio of SAE RRMSE to Direct CV By Sample Size for Each Area	71
PLOTS ON THE FIT OF THE MODEL.....	72
Transformed Direct Estimate vs. Model Predicted Value	74

<i>R-squared statistic</i>	74
Difference Between Transformed Direct Estimate and Transformed SAE Estimate vs. Model Predicted Value	75
Model Standardized Residual vs. Model Standardized Predicted Value	76
Square of Model Standardized Residual vs. Model Standardized Predicted Value	77
QQ-plot of standardized residuals vs. standard Normal quantiles	77
<i>Shapiro-Wilk W test of the standardized residuals</i>	78
<i>Sign test on the standardized residuals</i>	79
Influence measures	79
<i>Influence measure of each area</i>	79
REFERENCES	81

Introduction

This document provides a description of the methodology specifications for small-area estimation based on area-level modeling and empirical best linear unbiased prediction (EBLUP). These specifications have provided the framework for the development of software to produce estimates for small areas based on the Fay-Herriot link model. This is a linear mixed model at the area level with fixed and random effects. The basic concepts behind linear mixed model estimation are described in McCulloch and Searle (2001). The applications of linear mixed models to small-area estimation are given in Rao (2003) and Fuller (2009). The Fay-Herriot link model is described in this document.

The programs that have been developed are primarily intended to produce estimates for small-area means and totals but they can be used for other parameters of interest. The methodology that was implemented is described in this document. A description of the programs and their use are given in a separate document titled *G-Est Software - Small-Area Estimation - Area-Level Model with EBLUP Estimation - Description of Function Parameters and User Guide*.

Terms

This document uses terms in the following list:

small area (or area)	a small domain of interest for estimation
parameter	a characteristic of interest (a mean, total or other) of the small area
auxiliary variable	an independent variable in the fixed effects component of the model
direct estimator or design-based estimator	an estimator produced using design-based probability sampling methods
synthetic estimator	a predictive estimator produced from the model
direct estimate	an estimate produced by a direct estimator
sampling variance	the variance of a direct estimator
EBLUP	estimation based on empirical best linear unbiased prediction
EBLUP estimator	an estimator obtained using empirical best linear unbiased prediction
scoring algorithm	an iterative process for solving (an equation involving) the unknown model variance of the areas
error	a random term in the model with specific distributional properties
residual	an estimate of the random error term calculated from the sample
ADM method	the method of adjusted density maximization for model estimation
REML method	the method of restricted maximum likelihood for model estimation
FH method	the method of Fay and Herriot for model estimation
WF method	the method of Wang and Fuller for model estimation

The terms ADM, REML, FH and WF are short forms used to refer to the four methods of estimation of the unknown quantities in the small-area model.

Notation

The following table provides a summary of the notation used in this document. More details are given throughout the document.

notation	definition
A	the set of mutually exclusive groups of areas denoted by $A = A_{EXC} \cup A_{VAL} \cup A_{NUL}$
A_{EXC}	the areas excluded from the small-area model
A_{VAL}	the areas with valid values for the direct estimates and variances
A_{NUL}	the areas with invalid or missing values for the direct estimates and/or variances
m	the number of areas in A_{VAL}
i	the identifier of each area
N_i	the number of units in the population of area i
n_i	the number of sample units in area i
θ_i	the parameter of interest for small area i - a mean ($\theta_i = \bar{Y}_i$), a total ($\theta_i = Y_i$), or other
$\hat{\theta}_i^{DIR}$	a direct (design-based) estimate of θ_i
ψ_i	the sampling variance of the direct (design-based) estimator for area i
$\hat{\psi}_i$	a design-based estimate of the sampling variance for area i
$\tilde{\psi}_i$	a smoothed variance estimate for area i
$\dot{\psi}_i$	the input variance for area i for ADM, REML and FH: $\dot{\psi}_i = \begin{cases} \hat{\psi}_i & \text{if direct} \\ \tilde{\psi}_i & \text{if smoothed} \end{cases}$ for WF: $\dot{\psi}_i = \hat{\psi}_i$
$\hat{\theta}_i^{EBLUP}$	the EBLUP estimate of θ_i ($\hat{\theta}_i^{EBLUP}$ is $\hat{\theta}_i^{ADM}$, $\hat{\theta}_i^{REML}$, $\hat{\theta}_i^{FH}$ or $\hat{\theta}_i^{WF}$ corresponding to the different methods)
$\hat{\theta}_i^{EBLUP,b}$	the small-area benchmarked estimate of θ_i obtained from $\hat{\theta}_i^{EBLUP}$ by applying the method of difference adjustment

notation	definition
$\hat{\theta}_i^{DIR} = \theta_i + e_i$	the sampling model assumed for areas $i \in A_{VAL} \cup A_{NUL}$
$\theta_i = \mathbf{z}_i^T \boldsymbol{\beta} + b_i v_i$	the linear linking model assumed for areas $i \in A_{VAL} \cup A_{NUL}$
$\hat{\theta}_i^{DIR} = \mathbf{z}_i^T \boldsymbol{\beta} + b_i v_i + e_i$	the small-area model assumed for areas $i \in A_{VAL} \cup A_{NUL}$
$\mathbf{z}_i^T \boldsymbol{\beta}$	the fixed term or fixed effect for area i
$b_i v_i$	the random effect for area i
$\mathbf{z}_i$	the vector of known auxiliary values for area i
p	the (current) number of auxiliary variables in $\mathbf{z}_i$
$\boldsymbol{\beta}$	the regression coefficient for the fixed term in the model
$\hat{\boldsymbol{\beta}}$	an estimate of $\boldsymbol{\beta}$
b_i	the model error coefficient (a known positive constant) for area i
v_i	the model error for area i with $v_i \stackrel{ind}{\sim} (0, \sigma_v^2)$
e_i	the sampling error for area i where $e_i \stackrel{ind}{\sim} (0, \psi_i)$
σ_v^2	the model variance
$\hat{\sigma}_v^2$	an estimate of the model variance
$\hat{\theta}^{DIR}$	the known or estimated value of the parameter over the population in all areas A (this value is used for benchmarking a mean or a total and should not be confused with $\hat{\theta}_i^{DIR}$, the direct estimate in area i)

Methodology Note:

- The set of auxiliary variables $\mathbf{z}_i$ should normally include an intercept term with value 1 for all areas.

Overview

The programs for the area-level model allow you to produce small-area estimates for parameters such as means, totals or other. Four methods have been implemented in the current version. These methods vary slightly in the techniques used for the estimation of the unknown quantities in the small-area model. The methods are as follows.

- Adjusted Density Maximization (ADM)
- Restricted Maximum Likelihood (REML)
- Fay-Herriot (FH)
- Wang-Fuller (WF)

The model variance is defined through an equation known as a scoring function. All of these methods use a different scoring function. The solution to the scoring function provides us with the estimate of the model variance. If the model does not fit well then the scoring function may not lead to a valid (positive) estimate of the model variance. In these situations, no small-area estimates are produced unless we are using the Wang-Fuller method and the Fuller max rule is specified to force a positive estimate of the model variance.

The accuracy and efficiency of the small-area estimates are strongly dependent on the specified model. It is always important to check the validity of the model before accepting the results. Diagnostics are provided to do this. When the model is inappropriate, the methods may produce invalid estimates or estimates that are more inefficient than the direct estimates.

The model for small-area estimation

The theory for small-area estimation is based on Fay-Herriot (FH) model. This model is represented by the following two equations:

- Sampling model: $\hat{\theta}_i^{DIR} = \theta_i + e_i$
- Linking model: $\theta_i = \mathbf{z}_i^T \boldsymbol{\beta} + b_i v_i$

The sampling model expresses the direct estimate $\hat{\theta}_i^{DIR}$ as the unknown parameter value θ_i plus a random error e_i due to sampling. The linking model assumes the parameter value θ_i can be expressed as a fixed effect $\mathbf{z}_i^T \boldsymbol{\beta}$ plus an area random effect $b_i v_i$. The fixed effect is a linear combination of the auxiliary variables $\mathbf{z}_i$. The sampling error and the model error are assumed to have the following properties.

$$e_i \stackrel{id}{\sim} (0, \psi_i) \quad e_i \text{ are independently distributed with mean 0 and variance } \psi_i$$

$$v_i \stackrel{iid}{\sim} (0, \sigma_v^2) \quad v_i \text{ are independent and identically distributed with mean 0 and unknown variance } \sigma_v^2$$

The two equations in the Fay-Herriot model can be combined into a single (linear mixed model) equation reflecting the variability due to sampling (through the e_i term) and the variability due to the random effect ($b_i v_i$) for the areas.

$$\hat{\theta}_i^{DIR} = \mathbf{z}_i^T \boldsymbol{\beta} + b_i v_i + e_i$$

The set of small areas A , consists of three mutually exclusive groups, A_{EXC} , A_{VAL} and A_{NUL} , where $A = A_{EXC} \cup A_{VAL} \cup A_{NUL}$. These are determined based on the logic shown in Table 1 in the next section.

The group denoted by A_{EXC} are the exclusion areas. We assume the above small-area model is applicable to the non-exclusion areas in $A_{VAL} \cup A_{NUL}$. The areas in the exclusion group A_{EXC} are not included in the model. No estimation is carried out and no changes are made to the estimates in the areas of this group. The regression coefficient $\boldsymbol{\beta}$ and the model variance σ_v^2 are estimated based on the model applied to the areas in A_{VAL} . These are the valid areas for which we have direct estimates for the parameter and its variance. The group denoted by A_{NUL} consists of areas with invalid or improper estimates for the parameter or its variance. The small-area estimates for these areas are synthetic estimates. They are the predicted values from the model.

The theory for small-area estimation has been developed under the assumption that ψ_i is known and σ_v^2 is unknown. Thus, the various small-area estimation methods focus on the efficient estimation of σ_v^2 and β . The formulas for $\hat{\theta}_i^{EBLUP}$ and $mse(\hat{\theta}_i^{EBLUP})$ are functions of ψ_i so in practice, we need to replace ψ_i by an appropriate estimate. The variance ψ_i is estimated by a sample-based estimate $\hat{\psi}_i$ or its smoothed value $\tilde{\psi}_i$. The logic for providing these inputs for each of the methods is as follows.

- For the ADM, REML and FH methods, you provide either $\tilde{\psi}_i$ or $\hat{\psi}_i$.
- For the WF method, you must provide $\hat{\psi}_i$.

The resulting small-area estimates for each area are as follows:

- For the areas $i \in A_{EXC}$, the small-area estimate is the direct estimate – no change is made.
- For the areas $i \in A_{NUL}$, the small-area estimate is the synthetic estimate.
- For the areas $i \in A_{VAL}$, the small-area estimate is the EBLUP estimate - a linear combination of the direct estimate and the synthetic estimate.

Determining the methods and area groups

This section provides a summary on how your data inputs and specifications are used to determine the methods that can be applied and the area groups for each method. You specify the methods you want to use, the type of input variances you are providing and whether or not you want smoothing. We use Table 1 to establish the methods that can be applied from your list of specified methods and to identify the area groups for each method.

Table 1 Determining the small-area estimation methods and the area groups for each method

input variance	smoothing	possible methods	area group	classification to area groups	additional process
$\tilde{\psi}_i$	Not Applicable	ADM REML FH	A_{EXC}	$\hat{\theta}_i^{DIR} \neq . \ \& \ \tilde{\psi}_i = 0$	
			A_{VAL}	$\hat{\theta}_i^{DIR} \neq . \ \& \ \tilde{\psi}_i > 0$	
			A_{NUL}	otherwise	
$\hat{\psi}_i$	YES	ADM REML FH	A_{EXC}	$\hat{\theta}_i^{DIR} \neq . \ \& \ \hat{\psi}_i = 0$	
			A_{VAL}	$\hat{\theta}_i^{DIR} \neq . \ \& \ \hat{\psi}_i > 0$	$\hat{\psi}_i \rightarrow \tilde{\psi}_i$ (GVF smoothing)
			A_{NUL}	otherwise	
	NO	WF	A_{EXC}	$\hat{\theta}_i^{DIR} \neq . \ \& \ \hat{\psi}_i = 0$	
			A_{VAL}	$\hat{\theta}_i^{DIR} \neq . \ \& \ \hat{\psi}_i > 0 \ \& \ n_i \geq 2$	
			A_{NUL}	otherwise	

Notes:

- If your input variances are smoothed variances $\tilde{\psi}_i$ then you can only use the ADM, REML and FH methods. You cannot apply the WF method. Furthermore, you cannot smooth these variances since they are already smoothed. The option for smoothing is ignored.
- If your input variances are direct variances $\hat{\psi}_i$ then you have the option to smooth them using the method of Generalized Variance Functions.
 - If you choose to smooth them then you can only use the ADM, REML and FH methods. Smoothing is applied only to the direct variances for the areas in group A_{VAL} . After smoothing the direct variances, we continue using only the smoothed variances in this group. We do not modify the direct variances in group A_{EXC} and we do not use the direct variances in group A_{NUL} .
 - If you choose not to smooth them then you can only use the WF method.

- The sample sizes n_i are only required if you want to apply the WF method. The sample sizes must be greater than or equal to 2 for every area in group A_{VAL} .
- The population sizes N_i are not required to determine the area groups.

Use of sample counts and population counts

The table below provides a summary on how the sample counts n_i and population counts N_i are used in small-area estimation.

input...	is required...	to...
n_i	over all areas $A = A_{EXC} \cup A_{VAL} \cup A_{NUL}$	apply the WF method
N_i	over all areas $A = A_{EXC} \cup A_{VAL} \cup A_{NUL}$	benchmark means
	over all areas in $A_{VAL} \cup A_{NUL}$	create a MEAN model for the small-area estimation of totals

Notes:

- The sample counts are required over all areas to classify the areas into groups for the WF method. The sample counts in area group A_{VAL} are also used to produce the MSE estimates for this method. If you do not provide the sample counts for all areas then WF method cannot be applied.
- The population counts are required to benchmark parameters of type MEAN. You must provide these counts for all areas including the areas in the exclusion group A_{EXC} . If you cannot provide all of these counts then we cannot benchmark any parameters of type MEAN.
- You have the option to create a MEAN model for the small-area estimation of totals. If you want to do this then you must provide the population counts for all areas in the union of A_{VAL} and A_{NUL} . We use these values to change the inputs so they correspond to a mean rather than a total. This usually leads to a better fit for the regression model. The approach is as follows.
 - First, we divide the direct estimates ($\hat{\theta}_i^{DIR} = \hat{Y}_i^{DIR}$) for the totals by N_i and their input variances (either $\hat{\psi}_i$ or $\tilde{\psi}_i$) by N_i^2 . We also divide the vector of auxiliary variables $\mathbf{z}_i$ by N_i .
 - Then we treat this 'new' parameter like a mean and apply the specified methods and options to produce the small-area estimates.
 - At the end, we multiply the small-area estimates by N_i and their mean square errors by N_i^2 . This is done for all areas in the union A_{VAL} and A_{NUL} . No changes are made to the areas in the exclusion group A_{EXC} .
- If you choose not to use a MEAN model for the small-area estimation of totals or if you do not provide population counts for all areas in the union A_{VAL} and A_{NUL} then we work with a regression model based on the totals $\hat{\theta}_i^{DIR} = \hat{Y}_i^{DIR}$. This can lead to a poor regression fit. In some cases, it may not be possible to calculate an estimate of the model variance σ_v^2 .

Smoothing of the direct variance estimates

You can provide smoothed variances $\tilde{\psi}_i$ or direct variance estimates $\hat{\psi}_i$. If you provide smoothed variances then we can only apply the ADM, REML and FH methods. If you provide direct variance estimates, you can choose to smooth them by selecting the smoothing option. In this case, we create smoothed variances $\tilde{\psi}_i$ from the $\hat{\psi}_i$ by applying the method of Generalized Variance Functions to the areas in group A_{VAL} and we continue working with $\tilde{\psi}_i$ instead of $\hat{\psi}_i$ in these areas. If you provide direct variance estimates $\hat{\psi}_i$ and choose not to smooth them then we can only apply the WF method. The variability in the use of $\hat{\psi}_i$ introduces an extra term (g_{4i}) in the estimation of $mse(\hat{\theta}_i^{WF})$.

Smoothing is done to dampen the variability of the input variances $\hat{\psi}_i$. The resulting variances $\tilde{\psi}_i$ generally have less variability and fewer outliers. This usually leads to a better fit of the small-area model and tends to avoid problems in the estimation of σ_v^2 .

We apply the method of Generalized Variance Functions (GVF) to the areas in group A_{VAL} to produce smoothed variances $\tilde{\psi}_i$ only for the areas in this group. A description of this method is given below.

Smoothing by the method of Generalized Variance Functions

Smoothing is used to reduce the variability of the direct variances in order to achieve a better fit of the model. Smoothing is only applied to the areas in the area group A_{VAL} . This group has valid estimates $\hat{\theta}_i^{DIR} \neq .$ and variances $\hat{\psi}_i > 0$. To obtain the smoothed variances, we apply the following linear model with $y_i = \log(\hat{\psi}_i)$ using a set of r auxiliary variables x_i for $i \in A_{VAL}$.

$$y_i = \mathbf{x}_i^T \mathbf{B} + \varepsilon_i$$

The auxiliary variables x_i are defined independently of the variables z_i used in the small-area estimation model. Therefore, x_i may or may not include any of the variables in z_i . We assume the ε_i are Normal random errors with the following expectation and variance under this linear model: $E(\varepsilon_i) = 0$ and $Var(\varepsilon_i) = \sigma^2$.

The steps to produce the smoothed variances $\tilde{\psi}_i$ from the direct variances $\hat{\psi}_i$ are as follows:

1. If the number of areas in A_{VAL} is too small ($m \leq r$) then we cannot apply GVF smoothing and produce small-area estimates. Otherwise, we continue to the next step.
2. Regress $y_i = \log(\hat{\psi}_i)$ on x_i for $i \in A_{VAL}$ to obtain an estimate of the regression coefficient $\hat{\mathbf{B}}$ as follows.

$$\hat{\mathbf{B}} = \left(\sum_{i \in A_{VAL}} \mathbf{x}_i \mathbf{x}_i^T \right)^{-1} \sum_{i \in A_{VAL}} \mathbf{x}_i y_i$$

Methodology Note:

- We automatically add an intercept to the set of auxiliary variables $\mathbf{x}_i$ in the smoothing model.

3. Produce the smoothed variances as $\tilde{\psi}_i = \exp(\mathbf{x}_i^T \hat{\mathbf{B}}) \frac{\sum_{i \in A_{VAL}} \hat{\psi}_i}{\sum_{i \in A_{VAL}} \exp(\mathbf{x}_i^T \hat{\mathbf{B}})}$ for $i \in A_{VAL}$.

The smoothed variances $\tilde{\psi}_i$ are then used instead of the input variances $\hat{\psi}_i$ in the estimation of the unknown parameters in the small-area model.

The small-area estimation methods

This section provides a description of the process in each of the small-area methods. These are described in step form. We use process notes and implementation notes to provide further details on the process and possible issues in the implementation. Methodology notes provide references to the methodology.

Method of Adjusted Density Maximization

Processing steps for the ADM method

- Step 0 Use Table 1 (on page 8) to determine if the ADM method can be applied based on the type of input variance and specified smoothing. Continue only if it is possible to apply the method with the given specifications.
- Step 1 Use Table 1 to determine the 3 groups of small areas (A_{EXC} , A_{VAL} and A_{NUL}) for the parameter.
- Step 2 If the parameter is a total then determine if we can apply a MEAN model to produce the small-area estimates for the area totals.

If all of the following conditions are met:

- You choose the option to create a MEAN model for totals.
- The population sizes N_i are known for all areas $i \in A_{VAL} \cup A_{NUL}$.

Then apply a MEAN model for totals as follows:

- Transform the estimates $\hat{\theta}_i^{DIR}$ to $\hat{\theta}_i^{DIR}/N_i$ for $i \in A_{VAL}$.
- Transform the input variances $\dot{\psi}_i$.
 - If the input variances are direct estimates ($\hat{\psi}_i$) then replace them with $\hat{\psi}_i/N_i^2$ for $i \in A_{VAL}$.
 - If the input variances are smoothed estimates ($\tilde{\psi}_i$) then replace them with $\tilde{\psi}_i/N_i^2$ for $i \in A_{VAL}$.
- Replace the vector of auxiliary variables $\mathbf{z}_i$ with $\mathbf{z}_i/N_i$ for $i \in A_{VAL} \cup A_{NUL}$.

If the above conditions are not met then continue by applying a model without any transformations.

Process Notes:

- To process a total under a MEAN model we change $\hat{\theta}_i^{DIR}$ and either $\hat{\psi}_i$ or $\tilde{\psi}_i$ (using the above transformations) so they correspond to a mean. Then we continue to treat the parameter like a mean in steps 3 to 13. In step 14, we transform the SAE estimates back to a total.
- We transform $\mathbf{z}_i \rightarrow \mathbf{z}_i / N_i$ for $i \in A_{VAL} \cup A_{NUL}$, not just for $i \in A_{VAL}$.
- We do not transform the b_i values.
- We need N_i for $i \in A_{VAL} \cup A_{NUL}$.

Step 3 If an intercept is specified for the model then augment the auxiliary variables by including a variable with value 1 for all areas.

Methodology Notes:

- If we have applied a MEAN model in the previous step then the changes to the auxiliary vector can be represented symbolically as follows: $\mathbf{z}_i \rightarrow \mathbf{z}_i / N_i \rightarrow (1, \mathbf{z}_i / N_i)$.
- Otherwise, the inclusion of the intercept term is represented as follows: $\mathbf{z}_i \rightarrow (1, \mathbf{z}_i)$.
- Since the changes to the auxiliary variables are always done in place, it is appropriate to continue to refer to the resulting auxiliary vector as $\mathbf{z}_i$.

Step 4 Determine if we apply smoothing to the variances.

- If we are working with direct variances $\hat{\psi}_i$, we smooth the direct variances $\hat{\psi}_i$ for $i \in A_{VAL}$ using the method of Generalized Variance Functions to obtain $\tilde{\psi}_i$ for $i \in A_{VAL}$.
- If we are working with smoothed variances $\tilde{\psi}_i$ then we do not smooth them again. We leave them unchanged.

Process Note:

- At this point, we have smoothed variances $\tilde{\psi}_i$ for $i \in A_{VAL}$. We may or may not have any areas in A_{NUL} . If there are areas in A_{NUL} , we do not need to worry about the values of the variances for these areas. These variances are not used in the process that follows.

Step 5 If benchmarking is required then determine the method of benchmarking using the logic in Table 2 (on page 47).

Process Note:

- If benchmarking is possible then we apply either the method of augmented vector at step 6 or the method of difference adjustment at step 12. However, we never apply both methods.

- Step 6 If benchmarking by the method of augmented auxiliary vector then transform the auxiliary vector $\mathbf{z}_i$ by including another variable ($\omega_i \tilde{\psi}_i$) as follows:

$$\mathbf{z}_i \rightarrow \begin{pmatrix} \mathbf{z}_i \\ \omega_i \tilde{\psi}_i \end{pmatrix} \text{ for } i \in A_{VAL} \text{ where } \omega_i \text{ is defined in the table below.}$$

parameter type	definition of ω_i for $i \in A$
TOTAL	$\omega_i = 1$
MEAN	$\omega_i = N_i / N$ where $N = \sum_{i \in A} N_i$

Process Note:

- To benchmark a mean we require N_i for $i \in A$ so we can calculate $N = \sum_{i \in A} N_i$.
However, we only augment the auxiliary vector $\mathbf{z}_i$ for $i \in A_{VAL}$.

- Step 7 Use the following (iterative) scoring algorithm as the basis for obtaining $\hat{\sigma}_v^2$, an estimate of the model variance σ_v^2 .

$$\hat{\sigma}_v^{2(a+1)} = \hat{\sigma}_v^{2(a)} + \left[\text{tr}(\mathbf{P}\mathbf{B}\mathbf{P}\mathbf{B}) - \frac{1}{2} \text{tr}(\mathbf{V}^{-2} \mathbf{B}^2) + \frac{1}{(\hat{\sigma}_v^{2(a)})^2} \right]^{-1} \left[\frac{1}{2} \mathbf{y}^T \mathbf{P}\mathbf{B}\mathbf{P}\mathbf{y} - \frac{1}{2} \text{tr}(\mathbf{V}^{-1} \mathbf{B}) + \frac{1}{\hat{\sigma}_v^{2(a)}} \right]$$

where $\mathbf{B} = \text{diag}(b_i^2)$ is a diagonal matrix over $i \in A_{VAL}$

$$\mathbf{P} = \mathbf{V}^{-1} - \mathbf{V}^{-1} \mathbf{Z} (\mathbf{Z}^T \mathbf{V}^{-1} \mathbf{Z})^{-1} \mathbf{Z}^T \mathbf{V}^{-1}$$

$\mathbf{y}$ = is a column vector with entries $\hat{\theta}_i^{DIR}$ over $i \in A_{VAL}$

$\mathbf{Z}$ = is a matrix with row entries $\mathbf{z}_i^T$ over $i \in A_{VAL}$

and $\mathbf{V} = \text{diag}(\tilde{\psi}_i + b_i^2 \hat{\sigma}_v^{2(a)})$ is a diagonal matrix over $i \in A_{VAL}$

Implementation Notes:

- We solve for the required value of $\hat{\sigma}_v^2$ by using the related function $\frac{1}{2} \mathbf{y}^T \mathbf{PBP} \mathbf{y} - \frac{1}{2} \text{tr}(\mathbf{V}^{-1} \mathbf{B}) + \frac{1}{\hat{\sigma}_v^2(a)}$. This function must be 0 at the solution to the scoring algorithm.
- We solve for this value using a two-step process. First, we evaluate this function at equidistant points on the interval $[\hat{\sigma}_{v,min}^2, \hat{\sigma}_{v,max}^2]$, where $\hat{\sigma}_{v,min}^2 = \varepsilon$ is a small positive value close to 0 and $\hat{\sigma}_{v,max}^2$ is a large positive value. We look for a change in sign over two consecutive points.
 - If we do not observe a change in sign then we stop and set the estimated model variance to missing.
 - If we observe a change in sign then the two points define an interval containing the solution. We then continue to the second step and apply repeated bisection to solve for it. The solution is the estimate of the model variance.
- We start with $\hat{\sigma}_{v,max}^2 = \{ \sum_{i \in A_{VAL}} (\hat{\theta}_i^{DIR})^2 \} / \text{Min}(b_i^2) \big|_{i \in A_{VAL}}$ and continue evaluating right to left towards $\hat{\sigma}_{v,min}^2 = \varepsilon$, where ε is the specified epsilon value.
 - The interior points are determined based on the distance $d = (\hat{\sigma}_{v,max}^2 - \hat{\sigma}_{v,min}^2) / a_{max}$ where a_{max} is the specified maximum number of iterations.
 - We evaluate the function at the points $\hat{\sigma}_{v,max}^2, \hat{\sigma}_{v,max}^2 - d, \hat{\sigma}_{v,max}^2 - 2d$, and so on, until we observe a change in sign or we reach $\hat{\sigma}_{v,min}^2$.
- We terminate the algorithm when it meets one of the following conditions:
 - We reach $\hat{\sigma}_{v,min}^2$ in the initial search and we do not observe a change in sign. We conclude there is no solution and assign a missing value to $\hat{\sigma}_v^2$.
 - We observe a change in sign over two consecutive points. We then apply repeated bisection to find a smaller interval $(\hat{\sigma}_{v,left}^2, \hat{\sigma}_{v,right}^2)$ for which we accept $\hat{\sigma}_v^2 = (\hat{\sigma}_{v,left}^2 + \hat{\sigma}_{v,right}^2) / 2$ as the solution when it satisfies the following relative error:

$$(\hat{\sigma}_{v,right}^2 - \hat{\sigma}_{v,left}^2) \leq \varepsilon \left| \hat{\sigma}_v^2 \right|$$

Methodology Notes:

- The inclusion of the matrix $\mathbf{B}$ in the ADM scoring algorithm is based on communications with P. Lahiri in 2010.
- The ADM method always admits a positive solution for the estimated model variance. Therefore, we use $\hat{\sigma}_{v,min}^2 = \varepsilon$, where ε is a small positive value.
- The value $\hat{\sigma}_{v,max}^2$ is based on the following results where the expectation is with respect to the model.

$$\frac{\sum_{i \in A_{VAL}} (\hat{\theta}_i^{DIR})^2}{\text{Min}(b_i^2) \Big|_{i \in A_{VAL}}} \geq \sum_{i \in A_{VAL}} \frac{(\hat{\theta}_i^{DIR})^2}{b_i^2} \geq \sum_{i \in A_{VAL}} \frac{(\hat{\theta}_i^{DIR} - \mathbf{z}_i^T \boldsymbol{\beta})^2}{b_i^2} = \sum_{i \in A_{VAL}} (v_i + \frac{e_i}{b_i})^2$$

$$\text{and } E \left(\sum_{i \in A_{VAL}} (v_i + \frac{e_i}{b_i})^2 \right) > \sigma_v^2$$

Step 8 If there is no solution for $\hat{\sigma}_v^2$ then we stop and no small-area estimates are produced with the ADM method. Else, we continue to next step.

Methodology Note:

- The estimate of the model variance is always positive so the above condition should never occur. However, we may not find the solution if it is smaller than $\hat{\sigma}_{v,min}^2 = \varepsilon$. In this case, the model variance is effectively 0.

Step 9 Produce the estimated asymptotic variance of $\hat{\sigma}_v^2$ as follows.

$$\text{var}(\hat{\sigma}_v^2) = 2 \left[\sum_{i \in A_{VAL}} \frac{b_i^4}{(\tilde{\psi}_i + b_i^2 \hat{\sigma}_v^2)^2} \right]^{-1} = \frac{2}{\text{tr}(\mathbf{V}^{-2} \mathbf{B}^2)}$$

Process Note:

- The estimated variance $\text{var}(\hat{\sigma}_v^2)$ is used in step 13.

Methodology Note:

- The reference to this formula is the SAE Specifications by Yong You (December 2008).

Step 10 Calculate the estimated regression coefficient for the fixed effects and its estimated variance-covariance.

$$\hat{\beta} = \left[\sum_{i \in A_{VAL}} \frac{\mathbf{z}_i \mathbf{z}_i^T}{\tilde{\psi}_i + b_i^2 \hat{\sigma}_v^2} \right]^{-1} \left[\sum_{i \in A_{VAL}} \frac{\mathbf{z}_i \hat{\theta}_i^{DIR}}{\tilde{\psi}_i + b_i^2 \hat{\sigma}_v^2} \right]$$

$$\text{var}(\hat{\beta}) = \left[\sum_{i \in A_{VAL}} \frac{\mathbf{z}_i \mathbf{z}_i^T}{\tilde{\psi}_i + b_i^2 \hat{\sigma}_v^2} \right]^{-1}$$

Implementation Note:

➤ We use a generalized inverse (GINV function in SAS IML) to compute

$\left[\sum_{i \in A_{VAL}} \frac{\mathbf{z}_i \mathbf{z}_i^T}{\tilde{\psi}_i + b_i^2 \hat{\sigma}_v^2} \right]^{-1}$ in case the matrix is singular and cannot be inverted using the INV function in SAS IML.

Step 11 Calculate the ADM estimate $\hat{\theta}_i^{ADM}$ for each small area.

$$\hat{\theta}_i^{ADM} = \begin{cases} \hat{\theta}_i^{DIR} & \text{for } i \in A_{EXC} \\ \hat{\gamma}_i \hat{\theta}_i^{DIR} + (1 - \hat{\gamma}_i) \mathbf{z}_i^T \hat{\beta} & \text{for } i \in A_{VAL} \\ \mathbf{z}_i^T \hat{\beta} & \text{for } i \in A_{NUL} \end{cases}$$

$$\text{with } \hat{\gamma}_i = \frac{b_i^2 \hat{\sigma}_v^2}{\tilde{\psi}_i + b_i^2 \hat{\sigma}_v^2} \quad \text{for } i \in A_{VAL}$$

Methodology Note:

➤ The references are formulas 7.1.2, 7.1.3 and 7.1.4 in JNK Rao (2003).

Step 12 If benchmarking by using the method of difference adjustment then adjust the estimates $\hat{\theta}_i^{ADM}$ as follows:

$$\hat{\theta}_i^{ADM,b} = \begin{cases} \hat{\theta}_i^{ADM} & \text{for } i \in A_{EXC} \\ \hat{\theta}_i^{ADM} + \alpha_i (\hat{\theta}^* - \sum_{d \in A} \omega_d \hat{\theta}_d^{ADM}) & \text{for } i \in A_{VAL} \\ \hat{\theta}_i^{ADM} & \text{for } i \in A_{NUL} \end{cases}$$

$$\text{with } \alpha_i = \frac{\omega_i (\tilde{\psi}_i + b_i^2 \hat{\sigma}_v^2)}{\sum_{i \in A_{VAL}} \omega_i^2 (\tilde{\psi}_i + b_i^2 \hat{\sigma}_v^2)} \quad \text{for } i \in A_{VAL}$$

where ω_i is defined in the table below.

parameter type	definition of ω_i for $i \in A$
TOTAL	$\omega_i = 1$
MEAN	$\omega_i = N_i / N$ where $N = \sum_{i \in A} N_i$

Process Note:

- The term $\hat{\theta}^*$ is either $\hat{\theta}^{DIR}$ (a population value provided by the user) or a value calculated from the parameter estimates as $\sum_{i \in A} \omega_i \hat{\theta}_i^{DIR}$ with $\omega_i = 1$ for a total and $\omega_i = N_i/N$ for a mean. Either way $\hat{\theta}^*$ is treated as a constant in the above calculation.

Implementation Note:

- If we are processing a total under a MEAN model and the user has provided the population total $\hat{\theta}^{DIR}$ for benchmarking then we have to adjust this value. We need to keep in mind that if the user provides a population value $\hat{\theta}^{DIR}$ for benchmarking then that value corresponds to the type of parameter. In this case, the parameter is a total (although it is processed under a MEAN model), so the user provides the overall population total for benchmarking. Because we are working with a MEAN model, we need to divide this value by $N = \sum_{i \in A} N_i$ before using it in the above formula.

Methodology Note:

- The reference to the method of benchmarking by difference adjustment is the November 2007 notes from Fuller.

Step 13a Calculate $mse(\hat{\theta}_i^{ADM})$, the estimated MSE of $\hat{\theta}_i^{ADM}$.

$$mse(\hat{\theta}_i^{ADM}) = \begin{cases} \dot{\psi}_i & \text{for } i \in A_{EXC} \\ g_{0i} + g_{1i} + g_{2i} + 2g_{3i} & \text{for } i \in A_{VAL} \\ \mathbf{z}_i^T \text{var}(\hat{\boldsymbol{\beta}}) \mathbf{z}_i + b_i^2 \hat{\sigma}_v^2 & \text{for } i \in A_{NUL} \end{cases}$$

$$\text{with } g_{0i} = -b_i^2(1 - \hat{\gamma}_i)^2 \cdot \left\{ \frac{[tr(\mathbf{PB} - \mathbf{V}^{-1}\mathbf{B}) + (2/\hat{\sigma}_v^2)]}{tr(\mathbf{V}^{-2}\mathbf{B}^2)} \right\}$$

where $\mathbf{B} = \text{diag}(b_i^2)$ is a diagonal matrix over $i \in A_{VAL}$

$$\mathbf{P} = \mathbf{V}^{-1} - \mathbf{V}^{-1}\mathbf{Z}(\mathbf{Z}^T\mathbf{V}^{-1}\mathbf{Z})^{-1}\mathbf{Z}^T\mathbf{V}^{-1}$$

$\mathbf{Z}$ = is a matrix with row entries $\mathbf{z}_i^T$ over $i \in A_{VAL}$

and $\mathbf{V} = \text{diag}(\tilde{\psi}_i + b_i^2 \hat{\sigma}_v^{2(a)})$ is a diagonal matrix over $i \in A_{VAL}$

$$g_{1i} = \hat{\gamma}_i \tilde{\psi}_i$$

$$g_{2i} = (1 - \hat{\gamma}_i)^2 \mathbf{z}_i^T \left[\sum_{i \in A_{VAL}} \mathbf{z}_i \mathbf{z}_i^T / (\tilde{\psi}_i + b_i^2 \hat{\sigma}_v^2) \right]^{-1} \mathbf{z}_i$$

$$g_{3i} = \frac{(b_i^2 \tilde{\psi}_i)^2}{(\tilde{\psi}_i + b_i^2 \hat{\sigma}_v^2)^3} \text{var}(\hat{\sigma}_v^2)$$

Methodology Notes:

- The references are the SAE Specifications by Yong You (December 2008), Li and Lahiri (2010) and formulas 7.1.7, 7.1.8 and 7.1.21 in JNK Rao (2003).
- The g_{0i} term is known as the bias correction term.
- If we have benchmarked in step 12 then we take $mse(\hat{\theta}_i^{ADM})$ as an approximation to $mse(\hat{\theta}_i^{ADM,b})$.
- The form of the estimated variance $var(\hat{\beta})$ is given in step 10.

Step 13b If $mse(\hat{\theta}_i^{ADM}) < 0$ for one or more $i \in A_{VAL}$ then recalculate $mse(\hat{\theta}_i^{ADM})$ for all $i \in A$ using the following alternative formula that drops the bias correction term g_{0i} .

$$mse(\hat{\theta}_i^{ADM}) = \begin{cases} \dot{\psi}_i & \text{for } i \in A_{EXC} \\ g_{1i} + g_{2i} + g_{3i} & \text{for } i \in A_{VAL} \\ \mathbf{z}_i^T var(\hat{\beta}) \mathbf{z}_i + b_i^2 \hat{\sigma}_v^2 & \text{for } i \in A_{NUL} \end{cases}$$

Methodology Notes:

- This alternative formula is described in Li and Lahiri (2010).
- The g_{0i} term can be negative and it may dominate $mse(\hat{\theta}_i^{ADM})$.
- The terms g_{1i} , g_{2i} and g_{3i} are the same as in step 13a.
- The form of the estimated variance $var(\hat{\beta})$ is given in step 10.

Step 14 If the parameter is a total and we have used a MEAN model then we need to adjust the estimates $\hat{\theta}_i^{ADM}$ in step 11 or $\hat{\theta}_i^{ADM,b}$ in step 12 if the parameter has been benchmarked using the method of difference adjustment) as well as the $mse(\hat{\theta}_i^{ADM})$ in step 13. This is done for all $i \in A_{VAL} \cup A_{NUL}$ as follows.

$$\hat{\theta}_i^{ADM} = \begin{cases} \hat{\theta}_i^{DIR} & \text{for } i \in A_{EXC} \\ N_i \hat{\theta}_i^{ADM} \text{ (or } N_i \hat{\theta}_i^{ADM,b} \text{)} & \text{for } i \in A_{VAL} \cup A_{NUL} \end{cases}$$

$$mse(\hat{\theta}_i^{ADM}) = \begin{cases} \dot{\psi}_i & \text{for } i \in A_{EXC} \\ \begin{cases} g_{0i}^* + g_{1i}^* + g_{2i}^* + 2g_{3i}^* & \text{(if Step 13a is used)} \\ g_{1i}^* + g_{2i}^* + g_{3i}^* & \text{(if Step 13b is used)} \end{cases} & \text{for } i \in A_{VAL} \\ N_i^2 \left\{ \mathbf{z}_i^T \text{var}(\hat{\boldsymbol{\beta}}) \mathbf{z}_i + b_i^2 \hat{\sigma}_v^2 \right\} & \text{for } i \in A_{NUL} \end{cases}$$

where $g_{0i}^* = N_i^2 g_{0i}$
 $g_{1i}^* = N_i^2 g_{1i}$
 $g_{2i}^* = N_i^2 g_{2i}$
 $g_{3i}^* = N_i^2 g_{3i}$

Implementation Note:

- To obtain the required estimates for the total we simply do the following:
 - The SAE estimate for the total is simply obtained by multiplying $\hat{\theta}_i^{ADM}$ from step 11 or $\hat{\theta}_i^{ADM,b}$ from step 12 by N_i for $i \in A_{VAL} \cup A_{NUL}$.
 - The estimated MSE for the total is obtained by multiplying the MSE components in step 13 by N_i^2 for $i \in A_{VAL} \cup A_{NUL}$.

Method of Restricted Maximum Likelihood

Processing steps for the REML method

- Step 0 Use Table 1 (on page 8) to determine if the REML method can be applied based on the type of input variance and specified smoothing. Continue only if it is possible to apply the method with the given specifications.
- Step 1 Use Table 1 to determine the 3 groups of small areas (A_{EXC} , A_{VAL} and A_{NUL}) for the parameter.
- Step 2 If the parameter is a total then determine if we can apply a MEAN model to produce the small-area estimates for the area totals.

If all of the following conditions are met:

- You choose the option to create a MEAN model for totals.
- The population sizes N_i are known for all areas $i \in A_{VAL} \cup A_{NUL}$.

Then apply a MEAN model for totals as follows:

- Transform the estimates $\hat{\theta}_i^{DIR}$ to $\hat{\theta}_i^{DIR}/N_i$ for $i \in A_{VAL}$.
- Transform the input variances $\hat{\psi}_i$.
 - If the input variances are direct estimates ($\hat{\psi}_i$) then replace them with $\hat{\psi}_i/N_i^2$ for $i \in A_{VAL}$.
 - If the input variances are smoothed estimates ($\tilde{\psi}_i$) then replace them with $\tilde{\psi}_i/N_i^2$ for $i \in A_{VAL}$.
- Replace the vector of auxiliary variables $\mathbf{z}_i$ with $\mathbf{z}_i/N_i$ for $i \in A_{VAL} \cup A_{NUL}$.

If the above conditions are not met then continue by applying a model without any transformations.

Process Notes:

- To process a total under a MEAN model we change $\hat{\theta}_i^{DIR}$ and either $\hat{\psi}_i$ or $\tilde{\psi}_i$ (using the above transformations) so they correspond to a mean. Then we continue to treat the parameter like a mean in steps 3 to 13. In step 14, we transform the SAE estimates back to a total.
- We transform $\mathbf{z}_i \rightarrow \mathbf{z}_i/N_i$ for $i \in A_{VAL} \cup A_{NUL}$, not just for $i \in A_{VAL}$.
- We do not transform the b_i values.
- We need N_i for $i \in A_{VAL} \cup A_{NUL}$.

- Step 3 If an intercept is specified for the model then augment the auxiliary variables by including a variable with value 1 for all areas.

Methodology Notes:

- If we have applied a MEAN model in the previous step then the changes to the auxiliary vector can be represented symbolically as follows: $\mathbf{z}_i \rightarrow \mathbf{z}_i / N_i \rightarrow (1, \mathbf{z}_i / N_i)$.
- Otherwise, the inclusion of the intercept term is represented as follows: $\mathbf{z}_i \rightarrow (1, \mathbf{z}_i)$.
- Since the changes to the auxiliary variables are always done in place, it is appropriate to continue to refer to the resulting auxiliary vector as $\mathbf{z}_i$.

Step 4 Determine if we apply smoothing to the input variances.

- If we are working with direct variances $\hat{\psi}_i$, we smooth the direct variances $\hat{\psi}_i$ for $i \in A_{VAL}$ using the method of Generalized Variance Functions to obtain $\tilde{\psi}_i$ for $i \in A_{VAL}$.
- If we are working with smoothed variances $\tilde{\psi}_i$ then we do not smooth them again. We leave them unchanged.

Process Note:

- At this point, we have smoothed variances $\tilde{\psi}_i$ for $i \in A_{VAL}$. We may or may not have any areas in A_{NUL} . If there are areas in A_{NUL} , we do not need to worry about the values of the variances for these areas. These variances are not used in the process that follows.

Step 5 If benchmarking is required then determine the method of benchmarking using the logic in Table 2 (on page 47).

Process Note:

- If benchmarking is possible then we apply either the method of augmented vector at step 6 or the method of difference adjustment at step 12. However, we never apply both methods.

Step 6 If benchmarking by the method of augmented auxiliary vector then transform the auxiliary vector $\mathbf{z}_i$ by including another variable ($\omega_i \tilde{\psi}_i$) as follows:

$$\mathbf{z}_i \rightarrow \begin{pmatrix} \mathbf{z}_i \\ \omega_i \tilde{\psi}_i \end{pmatrix} \text{ for } i \in A_{VAL} \text{ where } \omega_i \text{ is defined in the table below.}$$

parameter type	definition of ω_i for $i \in A$
TOTAL	$\omega_i = 1$
MEAN	$\omega_i = N_i / N$ where $N = \sum_{i \in A} N_i$

Process Note:

- To benchmark a mean we require N_i for $i \in A$ so we can calculate $N = \sum_{i \in A} N_i$. However, we only augment the auxiliary vector $\mathbf{z}_i$ for $i \in A_{VAL}$.

Step 7 Use the following (iterative) scoring algorithm as the basis for obtaining $\hat{\sigma}_v^2$, an estimate of the model variance σ_v^2 .

$$\hat{\sigma}_v^{2(a+1)} = \hat{\sigma}_v^{2(a)} + \left[\frac{1}{2} \text{tr}(\mathbf{PBPB}) \right]^{-1} \left[\frac{1}{2} \mathbf{y}^T \mathbf{PBP} \mathbf{y} - \frac{1}{2} \text{tr}(\mathbf{PB}) \right]$$

where $\mathbf{B} = \text{diag}(b_i^2)$ is a diagonal matrix over $i \in A_{VAL}$

$$\mathbf{P} = \mathbf{V}^{-1} - \mathbf{V}^{-1} \mathbf{Z} (\mathbf{Z}^T \mathbf{V}^{-1} \mathbf{Z})^{-1} \mathbf{Z}^T \mathbf{V}^{-1}$$

$\mathbf{y} =$ is a column vector with entries $\hat{\theta}_i^{DIR}$ over $i \in A_{VAL}$

$\mathbf{Z} =$ is a matrix with row entries $\mathbf{z}_i^T$ over $i \in A_{VAL}$

and $\mathbf{V} = \text{diag}(\tilde{\psi}_i + b_i^2 \hat{\sigma}_v^{2(a)})$ is a diagonal matrix over $i \in A_{VAL}$

Implementation Notes:

- We solve for the required value of $\hat{\sigma}_v^2$ by using the related function $\frac{1}{2} \mathbf{y}^T \mathbf{PBP} \mathbf{y} - \frac{1}{2} \text{tr}(\mathbf{PB})$. This function must be 0 at the solution to the scoring algorithm.
- We solve for this value using a two-step process. First, we evaluate this function at equidistant points on the interval $[\hat{\sigma}_{v,min}^2, \hat{\sigma}_{v,max}^2]$, where $\hat{\sigma}_{v,min}^2$ is typically a negative value and $\hat{\sigma}_{v,max}^2$ is a large positive value. We look for a change in sign over two consecutive points.
 - If we do not observe a change in sign then we stop and set the estimated model variance to missing.
 - If we observe a change in sign then the two points define an interval containing the solution. We then continue to the second step and apply repeated bisection to solve for it. The solution is the estimate of the model variance.
- We start with $\hat{\sigma}_{v,max}^2 = \{ \sum_{i \in A_{VAL}} (\hat{\theta}_i^{DIR})^2 \} / \text{Min}(b_i^2) |_{i \in A_{VAL}}$ and continue evaluating right to left towards $\hat{\sigma}_{v,min}^2 = \text{Max}(-\tilde{\psi}_i / b_i^2) + \varepsilon |_{i \in A_{VAL}}$, where ε is the specified epsilon value.
 - The interior points are determined based on the distance $d = (\hat{\sigma}_{v,max}^2 - \hat{\sigma}_{v,min}^2) / a_{max}$ where a_{max} is the specified maximum number of iterations.
 - We evaluate the function at the points $\hat{\sigma}_{v,max}^2, \hat{\sigma}_{v,max}^2 - d, \hat{\sigma}_{v,max}^2 - 2d$, and so on, until we observe a change in sign or we reach $\hat{\sigma}_{v,min}^2$.
- We terminate the algorithm when it meets one of the following conditions:
 - We reach $\hat{\sigma}_{v,min}^2$ in the initial search and we do not observe a change in sign. We conclude there is no solution and assign a missing value to $\hat{\sigma}_v^2$.
 - We observe a change in sign over two consecutive points. We then apply repeated bisection to find a smaller interval $(\hat{\sigma}_{v,left}^2, \hat{\sigma}_{v,right}^2)$ for which we accept $\hat{\sigma}_v^2 = (\hat{\sigma}_{v,left}^2 + \hat{\sigma}_{v,right}^2) / 2$ as the solution when it satisfies the following relative error:

$$(\hat{\sigma}_{v,right}^2 - \hat{\sigma}_{v,left}^2) \leq \varepsilon | \hat{\sigma}_v^2 |$$

Process Note:

- If the solution is negative then we set $\hat{\sigma}_v^2$ to missing and stop. We cannot use a negative value in the calculations that follow.

Methodology Notes:

- The references are formulas 7.1.14 and 7.1.15 in JNK Rao (2003).
- The minimum value $\hat{\sigma}_{v,min}^2$ is determined from the requirement that we have a positive number in the denominator of each term of sums such as $\sum_{i \in A_{VAL}} \frac{\mathbf{z}_i \mathbf{z}_i^T}{\tilde{\psi}_i + b_i^2 \hat{\sigma}_v^2}$, etc.
- The value $\hat{\sigma}_{v,max}^2$ is based on the following results where the expectation is with respect to the model.

$$\frac{\sum_{i \in A_{VAL}} (\hat{\theta}_i^{DIR})^2}{\text{Min}(b_i^2) \Big|_{i \in A_{VAL}}} \geq \sum_{i \in A_{VAL}} \frac{(\hat{\theta}_i^{DIR})^2}{b_i^2} \geq \sum_{i \in A_{VAL}} \frac{(\hat{\theta}_i^{DIR} - \mathbf{z}_i^T \boldsymbol{\beta})^2}{b_i^2} = \sum_{i \in A_{VAL}} \left(v_i + \frac{e_i}{b_i}\right)^2$$

$$\text{and } E\left(\sum_{i \in A_{VAL}} \left(v_i + \frac{e_i}{b_i}\right)^2\right) > \sigma_v^2$$

Step 8 If there is no positive solution for $\hat{\sigma}_v^2$ then we stop and no small-area estimates are produced with the REML method. Else, we continue to next step.

Step 9 Produce the estimated asymptotic variance of $\hat{\sigma}_v^2$ as follows.

$$\text{var}(\hat{\sigma}_v^2) = 2 \left[\sum_{i \in A_{VAL}} \frac{b_i^4}{(\tilde{\psi}_i + b_i^2 \hat{\sigma}_v^2)^2} \right]^{-1} = \frac{2}{\text{tr}(\mathbf{V}^{-2} \mathbf{B}^2)}$$

Process Note:

- The estimated variance $\text{var}(\hat{\sigma}_v^2)$ is used in step 13.

Methodology Note:

- The reference is formula 7.1.16 in JNK Rao (2003).

Step 10 Calculate the estimated regression coefficient for the fixed effects and its estimated variance-covariance.

$$\hat{\boldsymbol{\beta}} = \left[\sum_{i \in A_{VAL}} \frac{\mathbf{z}_i \mathbf{z}_i^T}{\tilde{\psi}_i + b_i^2 \hat{\sigma}_v^2} \right]^{-1} \left[\sum_{i \in A_{VAL}} \frac{\mathbf{z}_i \hat{\theta}_i^{DIR}}{\tilde{\psi}_i + b_i^2 \hat{\sigma}_v^2} \right]$$

$$\text{var}(\hat{\boldsymbol{\beta}}) = \left[\sum_{i \in A_{VAL}} \frac{\mathbf{z}_i \mathbf{z}_i^T}{\tilde{\psi}_i + b_i^2 \hat{\sigma}_v^2} \right]^{-1}$$

Implementation Note:

- We use a generalized inverse (GINV function in SAS IML) to compute

$\left[\sum_{i \in A_{VAL}} \frac{\mathbf{z}_i \mathbf{z}_i^T}{\tilde{\psi}_i + b_i^2 \hat{\sigma}_v^2} \right]^{-1}$ in case the matrix is singular and cannot be inverted using the INV function in SAS IML.

Step 11 Calculate the REML estimate $\hat{\theta}_i^{REML}$ for each small area.

$$\hat{\theta}_i^{REML} = \begin{cases} \hat{\theta}_i^{DIR} & \text{for } i \in A_{EXC} \\ \hat{\gamma}_i \hat{\theta}_i^{DIR} + (1 - \hat{\gamma}_i) \mathbf{z}_i^T \hat{\boldsymbol{\beta}} & \text{for } i \in A_{VAL} \\ \mathbf{z}_i^T \hat{\boldsymbol{\beta}} & \text{for } i \in A_{NUL} \end{cases}$$

with $\hat{\gamma}_i = \frac{b_i^2 \hat{\sigma}_v^2}{\tilde{\psi}_i + b_i^2 \hat{\sigma}_v^2}$ for $i \in A_{VAL}$

Methodology Note:

- The references are formulas 7.1.2, 7.1.3 and 7.1.4 in JNK Rao (2003).

Step 12 If benchmarking by using the method of difference adjustment then adjust the estimates $\hat{\theta}_i^{REML}$ as follows:

$$\hat{\theta}_i^{REML,b} = \begin{cases} \hat{\theta}_i^{REML} & \text{for } i \in A_{EXC} \\ \hat{\theta}_i^{REML} + \alpha_i (\hat{\theta}^* - \sum_{d \in A} \omega_d \hat{\theta}_d^{REML}) & \text{for } i \in A_{VAL} \\ \hat{\theta}_i^{REML} & \text{for } i \in A_{NUL} \end{cases}$$

with $\alpha_i = \frac{\omega_i (\tilde{\psi}_i + b_i^2 \hat{\sigma}_v^2)}{\sum_{i \in A_{VAL}} \omega_i^2 (\tilde{\psi}_i + b_i^2 \hat{\sigma}_v^2)}$ for $i \in A_{VAL}$

where ω_i is defined in the table below.

parameter type	definition of ω_i for $i \in A$
TOTAL	$\omega_i = 1$
MEAN	$\omega_i = N_i / N$ where $N = \sum_{i \in A} N_i$

Process Note:

- The term $\hat{\theta}^*$ is either $\hat{\theta}^{DIR}$ (a population value provided by the user) or a value calculated from the parameter estimates as $\sum_{i \in A} \omega_i \hat{\theta}_i^{DIR}$ with $\omega_i = 1$ for a total and $\omega_i = N_i / N$ for a mean. Either way $\hat{\theta}^*$ is treated as a constant in the above calculation.

Implementation Note:

- If we are processing a total under a MEAN model and the user has provided the population total $\hat{\theta}^{DIR}$ for benchmarking then we have to adjust this value. We need to keep in mind that if the user provides a population value $\hat{\theta}^{DIR}$ for benchmarking then that value corresponds to the type of parameter. In this case, the parameter is a total (although it is processed under a MEAN model), so the user provides the overall population total for benchmarking. Because we are working with a MEAN model, we need to divide this value by $N = \sum_{i \in A} N_i$ before using it in the above formula.

Methodology Note:

- The reference is the November 2007 notes from Fuller.

Step 13 Calculate $mse(\hat{\theta}_i^{REML})$, the estimated MSE of $\hat{\theta}_i^{REML}$.

$$mse(\hat{\theta}_i^{REML}) = \begin{cases} \hat{\psi}_i & \text{for } i \in A_{EXC} \\ g_{1i} + g_{2i} + 2g_{3i} & \text{for } i \in A_{VAL} \\ \mathbf{z}_i^T var(\hat{\boldsymbol{\beta}}) \mathbf{z}_i + b_i^2 \hat{\sigma}_v^2 & \text{for } i \in A_{NUL} \end{cases}$$

where $g_{1i} = \hat{\gamma}_i \hat{\psi}_i$

$$g_{2i} = (1 - \hat{\gamma}_i)^2 \mathbf{z}_i^T \left[\sum_{i \in A_{VAL}} \mathbf{z}_i \mathbf{z}_i^T / (\hat{\psi}_i + b_i^2 \hat{\sigma}_v^2) \right]^{-1} \mathbf{z}_i$$

$$g_{3i} = \frac{(b_i^2 \hat{\psi}_i)^2}{(\hat{\psi}_i + b_i^2 \hat{\sigma}_v^2)^3} var(\hat{\sigma}_v^2)$$

Methodology Notes:

- The references are formulas 7.1.7, 7.1.8 and 7.1.21 in JNK Rao (2003) or formula 11 in Datta, Rao and Smith (2005).
- If we have benchmarked in step 12 then we take $mse(\hat{\theta}_i^{REML})$ as an approximation to $mse(\hat{\theta}_i^{REML,b})$.
- The form of the estimated variance $var(\hat{\boldsymbol{\beta}})$ is given in step 10.

Step 14 If the parameter is a total and we have used a MEAN model then we need to adjust the estimates $\hat{\theta}_i^{REML}$ in step 11 or $\hat{\theta}_i^{REML,b}$ in step 12 if the parameter has been benchmarked using the method of difference adjustment) as well as the $mse(\hat{\theta}_i^{REML})$ in step 13. This is done for all $i \in A_{VAL} \cup A_{NUL}$ as follows.

$$\hat{\theta}_i^{REML} = \begin{cases} \hat{\theta}_i^{DIR} & \text{for } i \in A_{EXC} \\ N_i \hat{\theta}_i^{REML} \text{ (or } N_i \hat{\theta}_i^{REML,b} \text{)} & \text{for } i \in A_{VAL} \cup A_{NUL} \end{cases}$$

$$mse(\hat{\theta}_i^{REML}) = \begin{cases} \dot{\psi}_i & \text{for } i \in A_{EXC} \\ g_{1i}^* + g_{2i}^* + 2g_{3i}^* & \text{for } i \in A_{VAL} \\ N_i^2 \left\{ \mathbf{z}_i^T \text{var}(\hat{\boldsymbol{\beta}}) \mathbf{z}_i + b_i^2 \hat{\sigma}_v^2 \right\} & \text{for } i \in A_{NUL} \end{cases}$$

where $g_{1i}^* = N_i^2 g_{1i}$
 $g_{2i}^* = N_i^2 g_{2i}$
 $g_{3i}^* = N_i^2 g_{3i}$

Implementation Note:

- To obtain the required estimates for the total we simply do the following:
 - The SAE estimate for the total is simply obtained by multiplying $\hat{\theta}_i^{REML}$ from step 11 or $\hat{\theta}_i^{REML,b}$ from step 12 by N_i for $i \in A_{VAL} \cup A_{NUL}$.
 - The estimated MSE for the total is obtained by multiplying the MSE components in step 13 by N_i^2 for $i \in A_{VAL} \cup A_{NUL}$.

Method of Fay-Herriot

Processing steps for the FH method

- Step 0 Use Table 1 (on page 8) to determine if the FH method can be applied based on the type of input variance and specified smoothing. Continue only if it is possible to apply the method with the given specifications.
- Step 1 Use Table 1 to determine the 3 groups of small areas (A_{EXC} , A_{VAL} and A_{NUL}) for the parameter.
- Step 2 If the parameter is a total then determine if we can apply a MEAN model to produce the small-area estimates for the area totals.

If all of the following conditions are met:

- You choose the option to create a MEAN model for totals.
- The population sizes N_i are known for all areas $i \in A_{VAL} \cup A_{NUL}$.

Then apply a MEAN model for totals as follows:

- Transform the estimates $\hat{\theta}_i^{DIR}$ to $\hat{\theta}_i^{DIR}/N_i$ for $i \in A_{VAL}$.
- Transform the input variances $\hat{\psi}_i$.
 - If the input variances are direct estimates ($\hat{\psi}_i$) then replace them with $\hat{\psi}_i/N_i^2$ for $i \in A_{VAL}$.
 - If the input variances are smoothed estimates ($\tilde{\psi}_i$) then replace them with $\tilde{\psi}_i/N_i^2$ for $i \in A_{VAL}$.
- Replace the vector of auxiliary variables $\mathbf{z}_i$ with $\mathbf{z}_i/N_i$ for $i \in A_{VAL} \cup A_{NUL}$.

If the above conditions are not met then continue by applying a model without any transformations.

Process Notes:

- To process a total under a MEAN model we change $\hat{\theta}_i^{DIR}$ and either $\hat{\psi}_i$ or $\tilde{\psi}_i$ (using the above transformations) so they correspond to a mean. Then we continue to treat the parameter like a mean in steps 3 to 13. In step 14, we transform the SAE estimates back to a total.
- We transform $\mathbf{z}_i \rightarrow \mathbf{z}_i/N_i$ for $i \in A_{VAL} \cup A_{NUL}$, not just for $i \in A_{VAL}$.
- We do not transform the b_i values.
- We need N_i for $i \in A_{VAL} \cup A_{NUL}$.

- Step 3 If an intercept is specified for the model then augment the auxiliary vector by including a variable with value 1 for all areas.

Methodology Notes:

- If we have applied a MEAN model in the previous step then the changes to the auxiliary vector can be represented symbolically as follows: $\mathbf{z}_i \rightarrow \mathbf{z}_i / N_i \rightarrow (1, \mathbf{z}_i / N_i)$.
- Otherwise, the inclusion of the intercept term is represented as follows: $\mathbf{z}_i \rightarrow (1, \mathbf{z}_i)$.
- Since the changes to the auxiliary variables are always done in place, it is appropriate to continue to refer to the resulting auxiliary vector as $\mathbf{z}_i$.

Step 4 Determine if we apply smoothing to the input variances.

- If we are working with direct variances $\hat{\psi}_i$, we smooth the direct variances $\hat{\psi}_i$ for $i \in A_{VAL}$ using the method of Generalized Variance Functions to obtain $\tilde{\psi}_i$ for $i \in A_{VAL}$.
- If we are working with smoothed variances $\tilde{\psi}_i$ then we do not smooth them again. We leave them unchanged.

Process Note:

- At this point, we have smoothed variances $\tilde{\psi}_i$ for $i \in A_{VAL}$. We may or may not have any areas in A_{NUL} . If there are areas in A_{NUL} , we do not need to worry about the values of the variances for these areas. These variances are not used in the process that follows.

Step 5 If benchmarking is required then determine the method of benchmarking using the logic in Table 2 (on page 47).

Process Note:

- If benchmarking is possible then we apply either the method of augmented vector at step 6 or the method of difference adjustment at step 12. However, we never apply both methods.

Step 6 If benchmarking by the method of augmented auxiliary vector then transform the auxiliary vector $\mathbf{z}_i$ by including another variable ($\omega_i \tilde{\psi}_i$) as follows:

$$\mathbf{z}_i \rightarrow \begin{pmatrix} \mathbf{z}_i \\ \omega_i \tilde{\psi}_i \end{pmatrix} \text{ for } i \in A_{VAL} \text{ where } \omega_i \text{ is defined in the table below.}$$

parameter type	definition of ω_i for $i \in A$
TOTAL	$\omega_i = 1$
MEAN	$\omega_i = N_i / N$ where $N = \sum_{i \in A} N_i$

Process Note:

- To benchmark a mean we require N_i for $i \in A$ so we can calculate $N = \sum_{i \in A} N_i$. However, we only augment the auxiliary vector $\mathbf{z}_i$ for $i \in A_{VAL}$.

Step 7 Use the following (iterative) scoring algorithm as the basis for obtaining $\hat{\sigma}_v^2$, an estimate of the model variance σ_v^2 .

$$\hat{\sigma}_v^{2(a+1)} = \hat{\sigma}_v^{2(a)} + \frac{1}{h'_*(\hat{\sigma}_v^{2(a)})} \left[m - p - h(\hat{\sigma}_v^{2(a)}) \right]$$

where
$$h(\hat{\sigma}_v^{2(a)}) = \sum_{i \in A_{VAL}} \frac{(\hat{\theta}_i^{DIR} - \mathbf{z}_i^T \hat{\boldsymbol{\beta}}^{(a)})^2}{(\tilde{\psi}_i + b_i^2 \hat{\sigma}_v^{2(a)})}$$

$$h'_*(\hat{\sigma}_v^{2(a)}) = - \sum_{i \in A_{VAL}} \frac{b_i^2 (\hat{\theta}_i^{DIR} - \mathbf{z}_i^T \hat{\boldsymbol{\beta}}^{(a)})^2}{(\tilde{\psi}_i + b_i^2 \hat{\sigma}_v^{2(a)})^2}$$

$$\hat{\boldsymbol{\beta}}^{(a)} = \left[\sum_{i \in A_{VAL}} \frac{\mathbf{z}_i \mathbf{z}_i^T}{(\tilde{\psi}_i + b_i^2 \hat{\sigma}_v^{2(a)})} \right]^{-1} \left[\sum_{i \in A_{VAL}} \frac{\mathbf{z}_i \hat{\theta}_i^{DIR}}{(\tilde{\psi}_i + b_i^2 \hat{\sigma}_v^{2(a)})} \right]$$

and p = number of auxiliary variables in $\mathbf{z}_i$

Implementation Notes:

- We solve for the required value of $\hat{\sigma}_v^2$ by using the related function $m - p - h(\hat{\sigma}_v^{2(a)})$. This function must be 0 at the solution to the scoring algorithm.
- We solve for this value using a two-step process. First, we evaluate this function at equidistant points on the interval $[\hat{\sigma}_{v,min}^2, \hat{\sigma}_{v,max}^2]$, where $\hat{\sigma}_{v,min}^2$ is typically a negative value and $\hat{\sigma}_{v,max}^2$ is a large positive value. We look for a change in sign over two consecutive points.
 - If we do not observe a change in sign then we stop and set the estimated model variance to missing.
 - If we observe a change in sign then the two points define an interval containing the solution. We then continue to the second step and apply repeated bisection to solve for it. The solution is the estimate of the model variance.
- We start with $\hat{\sigma}_{v,max}^2 = \{ \sum_{i \in A_{VAL}} (\hat{\theta}_i^{DIR})^2 \} / \text{Min}(b_i^2) \big|_{i \in A_{VAL}}$ and continue evaluating right to left towards $\hat{\sigma}_{v,min}^2 = \text{Max}(-\tilde{y}_i / b_i^2) + \varepsilon \big|_{i \in A_{VAL}}$, where ε is the specified epsilon value.
 - The interior points are determined based on the distance $d = (\hat{\sigma}_{v,max}^2 - \hat{\sigma}_{v,min}^2) / a_{max}$ where a_{max} is the specified maximum number of iterations.
 - We evaluate the function at the points $\hat{\sigma}_{v,max}^2$, $\hat{\sigma}_{v,max}^2 - d$, $\hat{\sigma}_{v,max}^2 - 2d$, and so on, until we observe a change in sign or we reach $\hat{\sigma}_{v,min}^2$.
- We terminate the algorithm when it meets one of the following conditions:
 - We reach $\hat{\sigma}_{v,min}^2$ in the initial search and we do not observe a change in sign. We conclude there is no solution and assign a missing value to $\hat{\sigma}_v^2$.
 - We observe a change in sign over two consecutive points. We then apply repeated bisection to find a smaller interval $(\hat{\sigma}_{v,left}^2, \hat{\sigma}_{v,right}^2)$ for which we accept $\hat{\sigma}_v^2 = (\hat{\sigma}_{v,left}^2 + \hat{\sigma}_{v,right}^2) / 2$ as the solution when it satisfies the following relative error:

$$(\hat{\sigma}_{v,right}^2 - \hat{\sigma}_{v,left}^2) \leq \varepsilon \left| \hat{\sigma}_v^2 \right|$$

Process Note:

- If the solution is negative then we set it to missing and stop. We cannot use a negative value in the calculations that follow.

Methodology Notes:

- The reference is formula 7.1.12 in JNK Rao (2003).
- The minimum value $\hat{\sigma}_{v,min}^2$ is determined from the requirement that we have a positive number in the denominator of each term of sums such as $\sum_{i \in A_{VAL}} \frac{\mathbf{z}_i \mathbf{z}_i^T}{\tilde{\psi}_i + b_i^2 \hat{\sigma}_v^2}$, etc.
- The value $\hat{\sigma}_{v,max}^2$ is based on the following results where the expectation is with respect to the model.

$$\frac{\sum_{i \in A_{VAL}} (\hat{\theta}_i^{DIR})^2}{\text{Min}(b_i^2) \Big|_{i \in A_{VAL}}} \geq \sum_{i \in A_{VAL}} \frac{(\hat{\theta}_i^{DIR})^2}{b_i^2} \geq \sum_{i \in A_{VAL}} \frac{(\hat{\theta}_i^{DIR} - \mathbf{z}_i^T \boldsymbol{\beta})^2}{b_i^2} = \sum_{i \in A_{VAL}} \left(v_i + \frac{e_i}{b_i}\right)^2$$

$$\text{and } E \left(\sum_{i \in A_{VAL}} \left(v_i + \frac{e_i}{b_i}\right)^2 \right) > \sigma_v^2$$

Step 8 If there is no positive solution for $\hat{\sigma}_v^2$ then we stop and no small-area estimates are produced with the FH method. Else, we continue to next step.

Step 9 Produce the estimated asymptotic variance of $\hat{\sigma}_v^2$ as follows.

$$\text{var}(\hat{\sigma}_v^2) = 2m \left[\sum_{i \in A_{VAL}} \frac{b_i^2}{(\tilde{\psi}_i + b_i^2 \hat{\sigma}_v^2)} \right]^{-2}$$

Process Note:

- The estimated variance $\text{var}(\hat{\sigma}_v^2)$ is used in step 13.

Methodology Note:

- The reference is formula 7.1.18 in JNK Rao (2003).

Step 10 Calculate the estimated regression coefficient for the fixed effects and its estimated variance-covariance.

$$\hat{\boldsymbol{\beta}} = \left[\sum_{i \in A_{VAL}} \frac{\mathbf{z}_i \mathbf{z}_i^T}{\tilde{\psi}_i + b_i^2 \hat{\sigma}_v^2} \right]^{-1} \left[\sum_{i \in A_{VAL}} \frac{\mathbf{z}_i \hat{\theta}_i^{DIR}}{\tilde{\psi}_i + b_i^2 \hat{\sigma}_v^2} \right]$$

$$\text{var}(\hat{\boldsymbol{\beta}}) = \left[\sum_{i \in A_{VAL}} \frac{\mathbf{z}_i \mathbf{z}_i^T}{\tilde{\psi}_i + b_i^2 \hat{\sigma}_v^2} \right]^{-1}$$

Implementation Note:

- We use a generalized inverse (GINV function in SAS IML) to compute

$\left[\sum_{i \in A_{VAL}} \frac{\mathbf{z}_i \mathbf{z}_i^T}{\tilde{\psi}_i + b_i^2 \hat{\sigma}_v^2} \right]^{-1}$ in case the matrix is singular and cannot be inverted using the INV function in SAS IML.

Step 11 Calculate the FH estimate $\hat{\theta}_i^{FH}$ for each small area.

$$\hat{\theta}_i^{FH} = \begin{cases} \hat{\theta}_i^{DIR} & \text{for } i \in A_{EXC} \\ \hat{\gamma}_i \hat{\theta}_i^{DIR} + (1 - \hat{\gamma}_i) \mathbf{z}_i^T \hat{\boldsymbol{\beta}} & \text{for } i \in A_{VAL} \\ \mathbf{z}_i^T \hat{\boldsymbol{\beta}} & \text{for } i \in A_{NUL} \end{cases}$$

with $\hat{\gamma}_i = \frac{b_i^2 \hat{\sigma}_v^2}{\tilde{\psi}_i + b_i^2 \hat{\sigma}_v^2}$ for $i \in A_{VAL}$

Methodology Note:

- The references are formulas 7.1.2, 7.1.3 and 7.1.4 in JNK Rao (2003).

Step 12 If benchmarking by using the method of difference adjustment then adjust the estimates $\hat{\theta}_i^{FH}$ as follows:

$$\hat{\theta}_i^{FH,b} = \begin{cases} \hat{\theta}_i^{FH} & \text{for } i \in A_{EXC} \\ \hat{\theta}_i^{FH} + \alpha_i (\hat{\theta}^* - \sum_{d \in A} \omega_d \hat{\theta}_d^{FH}) & \text{for } i \in A_{VAL} \\ \hat{\theta}_i^{FH} & \text{for } i \in A_{NUL} \end{cases}$$

with $\alpha_i = \frac{\omega_i (\tilde{\psi}_i + b_i^2 \hat{\sigma}_v^2)}{\sum_{i \in A_{VAL}} \omega_i^2 (\tilde{\psi}_i + b_i^2 \hat{\sigma}_v^2)}$ for $i \in A_{VAL}$

where ω_i is defined in the table below.

parameter type	definition of ω_i for $i \in A$
TOTAL	$\omega_i = 1$
MEAN	$\omega_i = N_i / N$ where $N = \sum_{i \in A} N_i$

Process Note:

- The term $\hat{\theta}^*$ is either $\hat{\theta}^{DIR}$ (a population value provided by the user) or a value calculated from the parameter estimates as $\sum_{i \in A} \omega_i \hat{\theta}_i^{DIR}$ with $\omega_i = 1$ for a total and $\omega_i = N_i / N$ for a mean. Either way $\hat{\theta}^*$ is treated as a constant in the above calculation.

Implementation Note:

- If we are processing a total under a MEAN model and the user has provided the population total $\hat{\theta}^{DIR}$ for benchmarking then we have to adjust this value. We need to keep in mind that if the user provides a population value $\hat{\theta}^{DIR}$ for benchmarking then that value corresponds to the type of parameter. In this case, the parameter is a total (although it is processed under a MEAN model), so the user provides the overall population total for benchmarking. Because we are working with a MEAN model, we need to divide this value by $N = \sum_{i \in A} N_i$ before using it in the above formula.

Methodology Note:

- The reference is the November 2007 notes from Fuller.

Step 13 Calculate $mse(\hat{\theta}_i^{FH})$, the estimated MSE of $\hat{\theta}_i^{FH}$.

$$mse(\hat{\theta}_i^{FH}) = \begin{cases} \dot{\psi}_i & \text{for } i \in A_{EXC} \\ g_{0i} + g_{1i} + g_{2i} + 2g_{3i} & \text{for } i \in A_{VAL} \\ \mathbf{z}_i^T \text{var}(\hat{\boldsymbol{\beta}}) \mathbf{z}_i + b_i^2 \hat{\sigma}_v^2 & \text{for } i \in A_{NUL} \end{cases}$$

$$\text{where } g_{0i} = \frac{-2b_i^2(1-\hat{\gamma}_i)^2 \left\{ m \sum_{i \in A_{VAL}} (\tilde{\psi}_i + b_i^2 \hat{\sigma}_v^2)^{-2} - \left(\sum_{i \in A_{VAL}} (\tilde{\psi}_i + b_i^2 \hat{\sigma}_v^2)^{-1} \right)^2 \right\}}{\left\{ \sum_{i \in A_{VAL}} (\tilde{\psi}_i + b_i^2 \hat{\sigma}_v^2)^{-1} \right\}^3}$$

$$g_{1i} = \hat{\gamma}_i \tilde{\psi}_i$$

$$g_{2i} = (1 - \hat{\gamma}_i)^2 \mathbf{z}_i^T \left[\sum_{i \in A_{VAL}} \mathbf{z}_i \mathbf{z}_i^T / (\tilde{\psi}_i + b_i^2 \hat{\sigma}_v^2) \right]^{-1} \mathbf{z}_i$$

$$g_{3i} = \frac{(b_i^2 \tilde{\psi}_i)^2}{(\tilde{\psi}_i + b_i^2 \hat{\sigma}_v^2)^3} \text{var}(\hat{\sigma}_v^2)$$

Methodology Notes:

- The references are formula (7.1.26) to (7.1.29) in JNK Rao (2003) or formula 16 in Datta, Rao and Smith (2005) or formula 19 in Pfeiffermann (2004).
- The g_{0i} value is known as the bias correction term.
- If we have benchmarked in step 12 then we take $mse(\hat{\theta}_i^{FH})$ as an approximation to $mse(\hat{\theta}_i^{FH,b})$.
- The form of the estimated variance $\text{var}(\hat{\boldsymbol{\beta}})$ is given in step 10.

Step 14 If the parameter is a total and we have used a MEAN model then we need to adjust the estimates $\hat{\theta}_i^{FH}$ in step 11 or $\hat{\theta}_i^{FH,b}$ in step 12 if the parameter has been benchmarked using the method of difference adjustment) as well as the $mse(\hat{\theta}_i^{FH})$ in step 13. This is done for all $i \in A_{VAL} \cup A_{NUL}$ as follows.

$$\hat{\theta}_i^{FH} = \begin{cases} \hat{\theta}_i^{DIR} & \text{for } i \in A_{EXC} \\ N_i \hat{\theta}_i^{FH} \text{ (or } N_i \hat{\theta}_i^{FH,b} \text{)} & \text{for } i \in A_{VAL} \cup A_{NUL} \end{cases}$$

$$mse(\hat{\theta}_i^{FH}) = \begin{cases} \hat{\psi}_i & \text{for } i \in A_{EXC} \\ g_{0i}^* + g_{1i}^* + g_{2i}^* + 2g_{3i}^* & \text{for } i \in A_{VAL} \\ N_i^2 \left\{ \mathbf{z}_i^T \text{var}(\hat{\boldsymbol{\beta}}) \mathbf{z}_i + b_i^2 \hat{\sigma}_v^2 \right\} & \text{for } i \in A_{NUL} \end{cases}$$

$$\begin{aligned} \text{where } g_{0i}^* &= N_i^2 g_{0i} \\ g_{1i}^* &= N_i^2 g_{1i} \\ g_{2i}^* &= N_i^2 g_{2i} \\ g_{3i}^* &= N_i^2 g_{3i} \end{aligned}$$

Implementation Note:

- To obtain the required estimates for the total we simply do the following:
 - The SAE estimate for the total is simply obtained by multiplying $\hat{\theta}_i^{FH}$ from step 11 or $\hat{\theta}_i^{FH,b}$ from step 12 by N_i for $i \in A_{VAL} \cup A_{NUL}$.
 - The estimated MSE for the total is obtained by multiplying the MSE components in step 13 by N_i^2 for $i \in A_{VAL} \cup A_{NUL}$.

Method of Wang-Fuller

Processing steps for the WF method

- Step 0 Use Table 1 (on page 8) to determine if the WF method can be applied based on the type of input variance and specified smoothing. Continue only if it is possible to apply the method with the given specifications.
- Step 1 Use Table 1 to determine the 3 groups of small areas (A_{EXC} , A_{VAL} and A_{NUL}) for the parameter.
- Step 2 If the parameter is a total then determine if we can apply a MEAN model to produce the small-area estimates for the area totals.

If all of the following conditions are met:

- You choose the option to create a MEAN model for totals.
- The population sizes N_i are known for all areas $i \in A_{VAL} \cup A_{NUL}$.

Then apply a MEAN model for totals as follows:

- Transform the estimates $\hat{\theta}_i^{DIR}$ to $\hat{\theta}_i^{DIR}/N_i$ for $i \in A_{VAL}$.
- Transform the input variances $\hat{\psi}_i$ to $\hat{\psi}_i/N_i^2$ for $i \in A_{VAL}$.
- Replace the vector of auxiliary variables $\mathbf{z}_i$ with $\mathbf{z}_i/N_i$ for $i \in A_{VAL} \cup A_{NUL}$.

If the above conditions are not met then continue by applying a model based on totals without any transformations.

Process Notes:

- To process a total under a MEAN model we change $\hat{\theta}_i^{DIR}$ and $\hat{\psi}_i$ (using the above transformations) so they correspond to a mean. Then we continue to treat the parameter like a mean in steps 3 to 14. In step 15, we convert the SAE estimates so they correspond to a total.
- We transform $\mathbf{z}_i \rightarrow \mathbf{z}_i/N_i$ for $i \in A_{VAL} \cup A_{NUL}$, not just for $i \in A_{VAL}$.
- We do not transform the b_i values.
- We need N_i for $i \in A_{VAL} \cup A_{NUL}$.

- Step 3 If an intercept is specified for the model then augment the auxiliary variables by including a variable with value 1 for all areas.

Methodology Notes:

- If we have applied a MEAN model in the previous step then the changes to the auxiliary vector can be represented symbolically as follows: $\mathbf{z}_i \rightarrow \mathbf{z}_i / N_i \rightarrow (1, \mathbf{z}_i / N_i)$.
- Otherwise, the inclusion of the intercept term is represented as follows: $\mathbf{z}_i \rightarrow (1, \mathbf{z}_i)$.
- Since the changes to the auxiliary variables are always done in place, it is appropriate to continue to refer to the resulting auxiliary vector as $\mathbf{z}_i$.

- Step 4 Ignore the smoothing option. No smoothing is applied in this method. We work with $\hat{\psi}_i$ for $i \in A$.

- Step 5 If benchmarking is required then determine the method of benchmarking using the logic in Table 2 (on page 47).

Process Note:

- If benchmarking is possible then we apply either the method of augmented vector at step 6 or the method of difference adjustment at step 13. However, we never apply both methods.

- Step 6 If benchmarking by using the method of augmented auxiliary vector then transform the auxiliary vector $\mathbf{z}_i$ by including another variable ($\omega_i \hat{\psi}_i$) as follows:

$$\mathbf{z}_i \rightarrow \begin{pmatrix} \mathbf{z}_i \\ \omega_i \hat{\psi}_i \end{pmatrix} \text{ for } i \in A_{VAL} \text{ where } \omega_i \text{ is defined in the table below.}$$

parameter type	definition of ω_i for $i \in A$
TOTAL	$\omega_i = 1$
MEAN	$\omega_i = N_i / N$ where $N = \sum_{i \in A} N_i$

Process Note:

- To benchmark a mean we require N_i for $i \in A$ so we can calculate $N = \sum_{i \in A} N_i$. However, we only augment the auxiliary vector $\mathbf{z}_i$ for $i \in A_{VAL}$.

Step 7 Use the following (iterative) scoring algorithm as the basis for obtaining $\hat{\sigma}_v^2$, an estimate of the model variance σ_v^2 .

$$\hat{\sigma}_v^{2(a+1)} = \sum_{i \in A_{VAL}} \kappa_i^{(a)} \frac{1}{b_i^2} \left[\frac{m}{(m-p)} (\hat{\theta}_i^{DIR} - \mathbf{z}_i^T \hat{\boldsymbol{\beta}}^{(a)})^2 - \hat{\psi}_i \right]$$

$$\text{where } \hat{\boldsymbol{\beta}}^{(a)} = \left[\sum_{i \in A_{VAL}} \frac{\mathbf{z}_i \mathbf{z}_i^T}{(\hat{\psi}_i + b_i^2 \hat{\sigma}_v^{2(a)})} \right]^{-1} \left[\sum_{i \in A_{VAL}} \frac{\mathbf{z}_i \hat{\theta}_i^{DIR}}{(\hat{\psi}_i + b_i^2 \hat{\sigma}_v^{2(a)})} \right]$$

$$\kappa_i^{(a)} = \frac{\left[\hat{\sigma}_v^{2(a)} + \frac{(n_i+1)}{(n_i-1)} \frac{\hat{\psi}_i}{b_i^2} \right]^{-1}}{\sum_{i \in A_{VAL}} \left[\hat{\sigma}_v^{2(a)} + \frac{(n_i+1)}{(n_i-1)} \frac{\hat{\psi}_i}{b_i^2} \right]^{-1}}$$

and p = number of auxiliary variables in $\mathbf{z}_i$

Implementation Notes:

- We take the equation from the scoring algorithm and express it as a function of $\hat{\sigma}_v^2$ by writing it as $\hat{\sigma}_v^2 - \sum_{i \in A_{VAL}} \kappa_i \frac{1}{b_i^2} \left[\frac{m}{(m-p)} (\hat{\theta}_i^{DIR} - \mathbf{z}_i^T \hat{\boldsymbol{\beta}})^2 - \hat{\psi}_i \right]$. Clearly, this function must be 0 at the required solution.
- We solve for this value using a two-step process. First, we evaluate this function at equidistant points on the interval $[\hat{\sigma}_{v,min}^2, \hat{\sigma}_{v,max}^2]$, where $\hat{\sigma}_{v,min}^2$ is typically a negative value and $\hat{\sigma}_{v,max}^2$ is a large positive value. We look for a change in sign over two consecutive points.
 - If we do not observe a change in sign then we stop and set the estimated model variance to missing.
 - If we observe a change in sign then the two points define an interval containing the solution. We then continue to the second step and apply repeated bisection to solve for it. The solution is the estimate of the model variance.
- We start with $\hat{\sigma}_{v,max}^2 = \{ \sum_{i \in A_{VAL}} (\hat{\theta}_i^{DIR})^2 \} / \text{Min}(b_i^2) |_{i \in A_{VAL}}$ and continue evaluating right to left towards $\hat{\sigma}_{v,min}^2 = \text{Max}(-\hat{\psi}_i / b_i^2) + \varepsilon |_{i \in A_{VAL}}$, where ε is the specified epsilon value.
 - The interior points are determined based on the distance $d = (\hat{\sigma}_{v,max}^2 - \hat{\sigma}_{v,min}^2) / a_{max}$ where a_{max} is the specified maximum number of iterations.
 - We evaluate the function at the points $\hat{\sigma}_{v,max}^2$, $\hat{\sigma}_{v,max}^2 - d$, $\hat{\sigma}_{v,max}^2 - 2d$, and so on, until we observe a change in sign or we reach $\hat{\sigma}_{v,min}^2$.
- We terminate the algorithm when it meets one of the following conditions:
 - We reach $\hat{\sigma}_{v,min}^2$ in the initial search and we do not observe a change in sign. We conclude there is no solution and assign a missing value to $\hat{\sigma}_v^2$.
 - We observe a change in sign over two consecutive points. We then apply repeated bisection to find a smaller interval $(\hat{\sigma}_{v,left}^2, \hat{\sigma}_{v,right}^2)$ for which we accept $\hat{\sigma}_v^2 = (\hat{\sigma}_{v,left}^2 + \hat{\sigma}_{v,right}^2) / 2$ as the solution when it satisfies the following relative error:

$$(\hat{\sigma}_{v,right}^2 - \hat{\sigma}_{v,left}^2) \leq \varepsilon | \hat{\sigma}_v^2 |$$

Methodology Notes:

- The references to these formulas are found on page 314 of Fuller (2009).
- The minimum value $\hat{\sigma}_{v,min}^2$ is determined from the requirement that we have a positive number in the denominator of each term of sums such as $\sum_{i \in A_{VAL}} \frac{\mathbf{z}_i \mathbf{z}_i^T}{\hat{\psi}_i + b_i^2 \hat{\sigma}_v^2}$, etc.
- The value $\hat{\sigma}_{v,max}^2$ is based on the following results where the expectation is with respect to the model.

$$\frac{\sum_{i \in A_{VAL}} (\hat{\theta}_i^{DIR})^2}{\text{Min}(b_i^2) \Big|_{i \in A_{VAL}}} \geq \sum_{i \in A_{VAL}} \frac{(\hat{\theta}_i^{DIR})^2}{b_i^2} \geq \sum_{i \in A_{VAL}} \frac{(\hat{\theta}_i^{DIR} - \mathbf{z}_i^T \boldsymbol{\beta})^2}{b_i^2} = \sum_{i \in A_{VAL}} \left(v_i + \frac{e_i}{b_i}\right)^2$$

$$\text{and } E\left(\sum_{i \in A_{VAL}} \left(v_i + \frac{e_i}{b_i}\right)^2\right) > \sigma_v^2$$

- The κ_i for $i \in A_{VAL}$ are positive at each iteration. This can be seen from the following set of inequalities and noting that $n_i \geq 2$ for $i \in A_{VAL}$.

$$\frac{(n_i + 1)}{(n_i - 1)} \left(\frac{-\hat{\psi}_i}{b_i^2} \right) < \frac{-\hat{\psi}_i}{b_i^2} \leq \text{Max} \left(\frac{-\hat{\psi}_i}{b_i^2} \right) < \hat{\sigma}_{v,min}^2 \leq \hat{\sigma}_v^{2(a)}$$

Step 8 If there is no positive solution for $\hat{\sigma}_v^2$ and the Fuller max rule is not specified then we stop and no small-area estimates are produced with the Wang-Fuller method. Else, we continue to step 9.

Step 9 If the Fuller max rule is specified then produce a new estimate $\hat{\sigma}_v^2$ as follows.

$$\hat{\sigma}_v^2 = \text{Max}\{ 0.5\sqrt{\text{var}(\hat{\sigma}_v^2)}, \hat{\sigma}_v^2 \}$$

$$\text{where } \text{var}(\hat{\sigma}_v^2) = \sum_{i \in A_{VAL}} 2 \kappa_i^2 \frac{1}{(b_i^2)^2} \left[(\hat{\psi}_i + b_i^2 \hat{\sigma}_v^2)^2 + \frac{(\hat{\psi}_i)^2}{(n_i - 1)} \right]$$

$$\text{with } \kappa_i = \frac{\left[\hat{\sigma}_v^2 + \frac{(n_i + 1)}{(n_i - 1)} \frac{\hat{\psi}_i}{b_i^2} \right]^{-1}}{\sum_{i \in A_{VAL}} \left[\hat{\sigma}_v^2 + \frac{(n_i + 1)}{(n_i - 1)} \frac{\hat{\psi}_i}{b_i^2} \right]^{-1}}$$

Methodology Note:

➤ The formula for $\text{var}(\hat{\sigma}_v^2)$ is obtained from the corresponding one given by $\hat{V}\{\hat{\sigma}_u^2\}$ on page 314 of Fuller (2009). Fuller uses a different notation and he does not have b_i in the model. Using our notation, we have the following results.

- Without the b_i , the formula in the book becomes

$$\sum_{i \in A_{VAL}} \kappa_i^2 \left[2 (\hat{\sigma}_v^2 + \hat{\psi}_i)^2 + \text{var}(\hat{\psi}_i) \right].$$

- We then assume that $(n_i - 1) \frac{\hat{\psi}_i}{\psi_i} \sim \chi_{(n_i - 1)}^2$, a Chi-square distribution on $(n_i - 1)$ degrees of freedom.

- Under this assumption $\text{var}(\hat{\psi}_i) = \frac{2 (\hat{\psi}_i)^2}{(n_i - 1)}$.

- Therefore, without the b_i , we obtain $\text{var}(\hat{\sigma}_v^2) = \sum_{i \in A_{VAL}} 2 \kappa_i^2 \left[(\hat{\sigma}_v^2 + \hat{\psi}_i)^2 + \frac{(\hat{\psi}_i)^2}{(n_i - 1)} \right]$.

- With the b_i in the model, we simply replace $\hat{\psi}_i$ by $\frac{\hat{\psi}_i}{b_i^2}$. This leads to the following formula.

$$\text{var}(\hat{\sigma}_v^2) = \sum_{i \in A_{VAL}} 2 \kappa_i^2 \frac{1}{(b_i^2)^2} \left[(\hat{\psi}_i + b_i^2 \hat{\sigma}_v^2)^2 + \frac{(\hat{\psi}_i)^2}{(n_i - 1)} \right]$$

Process Note:

➤ If there is no solution for $\hat{\sigma}_v^2$ from step 7 ($\hat{\sigma}_v^2 = .$) then we set $\hat{\sigma}_v^2 = 0.5 \sqrt{\text{var}(\hat{\sigma}_v^2)}$ where $\text{var}(\hat{\sigma}_v^2)$ is as follows:

$$\text{var}(\hat{\sigma}_v^2) = \sum_{i \in A_{VAL}} 2 \kappa_i^2 \left[(\hat{\psi}_i)^2 + \frac{(\hat{\psi}_i)^2}{(n_i - 1)} \right]$$

$$\text{where } \kappa_i = \frac{\left[\frac{(n_i + 1)}{(n_i - 1)} \hat{\psi}_i \right]^{-1}}{\sum_{i \in A_{VAL}} \left[\frac{(n_i + 1)}{(n_i - 1)} \hat{\psi}_i \right]^{-1}} \quad \text{for } i \in A_{VAL}$$

Step 10 Calculate the estimated variance $\text{var}(\hat{\sigma}_v^2)$ based on the value of $\hat{\sigma}_v^2$:

$$\text{var}(\hat{\sigma}_v^2) = \sum_{i \in A_{VAL}} 2 \kappa_i^2 \frac{1}{(b_i^2)^2} \left[(\hat{\psi}_i + b_i^2 \hat{\sigma}_v^2)^2 + \frac{(\hat{\psi}_i)^2}{(n_i - 1)} \right]$$

$$\text{with } \kappa_i = \frac{\left[\hat{\sigma}_v^2 + \frac{(n_i + 1)}{(n_i - 1)} \frac{\hat{\psi}_i}{b_i^2} \right]^{-1}}{\sum_{i \in A_{VAL}} \left[\hat{\sigma}_v^2 + \frac{(n_i + 1)}{(n_i - 1)} \frac{\hat{\psi}_i}{b_i^2} \right]^{-1}}$$

Step 11 Calculate the estimated regression coefficient for the fixed effects and its estimated variance-covariance.

$$\hat{\beta} = \left[\sum_{i \in A_{VAL}} \frac{\mathbf{z}_i \mathbf{z}_i^T}{\hat{\psi}_i + b_i^2 \hat{\sigma}_v^2} \right]^{-1} \left[\sum_{i \in A_{VAL}} \frac{\mathbf{z}_i \hat{\theta}_i^{DIR}}{\hat{\psi}_i + b_i^2 \hat{\sigma}_v^2} \right]$$

$$var(\hat{\beta}) = \left[\sum_{i \in A_{VAL}} \frac{\mathbf{z}_i \mathbf{z}_i^T}{\hat{\psi}_i + b_i^2 \hat{\sigma}_v^2} \right]^{-1}$$

Implementation Note:

➤ We use a generalized inverse (GINV function in SAS IML) to compute

$\left[\sum_{i \in A_{VAL}} \frac{\mathbf{z}_i \mathbf{z}_i^T}{\hat{\psi}_i + b_i^2 \hat{\sigma}_v^2} \right]^{-1}$ in case the matrix is singular and cannot be inverted using the INV function in SAS IML.

Step 12 Calculate the Wang-Fuller estimate $\hat{\theta}_i^{WF}$ for each small area.

$$\hat{\theta}_i^{WF} = \begin{cases} \hat{\theta}_i^{DIR} & \text{for } i \in A_{EXC} \\ \hat{\gamma}_i \hat{\theta}_i^{DIR} + (1 - \hat{\gamma}_i) \mathbf{z}_i^T \hat{\beta} & \text{for } i \in A_{VAL} \\ \mathbf{z}_i^T \hat{\beta} & \text{for } i \in A_{NUL} \end{cases}$$

$$\text{with } \hat{\gamma}_i = \frac{b_i^2 \hat{\sigma}_v^2}{\hat{\psi}_i + b_i^2 \hat{\sigma}_v^2} \quad \text{for } i \in A_{VAL}$$

Methodology Note:

➤ The references are formulas 7.1.2, 7.1.3 and 7.1.4 in JNK Rao (2003).

Step 13 If benchmarking by using the method of difference adjustment then adjust the estimates $\hat{\theta}_i^{WF}$ as follows:

$$\hat{\theta}_i^{WF,b} = \begin{cases} \hat{\theta}_i^{WF} & \text{for } i \in A_{EXC} \\ \hat{\theta}_i^{WF} + \alpha_i (\hat{\theta}^* - \sum_{d \in A} \omega_d \hat{\theta}_d^{WF}) & \text{for } i \in A_{VAL} \\ \hat{\theta}_i^{WF} & \text{for } i \in A_{NUL} \end{cases}$$

$$\text{with } \alpha_i = \frac{\omega_i (\hat{\psi}_i + b_i^2 \hat{\sigma}_v^2)}{\sum_{i \in A_{VAL}} \omega_i^2 (\hat{\psi}_i + b_i^2 \hat{\sigma}_v^2)} \quad \text{for } i \in A_{VAL}$$

where ω_i is defined in the table below.

parameter type	definition of ω_i for $i \in A$
TOTAL	$\omega_i = 1$
MEAN	$\omega_i = N_i / N$ where $N = \sum_{i \in A} N_i$

Process Note:

- The term $\hat{\theta}^*$ is either $\hat{\theta}^{DIR}$ (a population value provided by the user) or a value calculated from the parameter estimates as $\sum_{i \in A} \omega_i \hat{\theta}_i^{DIR}$ with $\omega_i = 1$ for a total and $\omega_i = N_i/N$ for a mean. Either way $\hat{\theta}^*$ is treated as a constant in the above calculation.

Implementation Note:

- If we are processing a total under a MEAN model and the user has provided the population total $\hat{\theta}^{DIR}$ for benchmarking then we have to adjust this value. We need to keep in mind that if the user provides a population value $\hat{\theta}^{DIR}$ for benchmarking then that value corresponds to the type of parameter. In this case, the parameter is a total (although it is processed under a MEAN model), so the user provides the overall population total for benchmarking. Because we are working with a MEAN model, we need to divide this value by $N = \sum_{i \in A} N_i$ before using it in the above formula.

Methodology Note:

- The reference to the method of benchmarking by difference adjustment is the November 2007 notes from Fuller.

Step 14 Calculate $mse(\hat{\theta}_i^{WF})$, the estimated MSE of $\hat{\theta}_i^{WF}$.

$$mse(\hat{\theta}_i^{WF}) = \begin{cases} \hat{\psi}_i & \text{for } i \in A_{EXC} \\ g_{1i} + g_{2i} + 2g_{3i} + 4g_{4i} & \text{for } i \in A_{VAL} \\ \mathbf{z}_i^T \text{var}(\hat{\boldsymbol{\beta}}) \mathbf{z}_i + b_i^2 \hat{\sigma}_v^2 & \text{for } i \in A_{NUL} \end{cases}$$

where $g_{1i} = \hat{\gamma}_i \hat{\psi}$

$$g_{2i} = (1 - \hat{\gamma}_i)^2 \mathbf{z}_i^T \left[\sum_{i \in A_{VAL}} \mathbf{z}_i \mathbf{z}_i^T / (\hat{\psi}_i + b_i^2 \hat{\sigma}_v^2) \right]^{-1} \mathbf{z}_i$$

$$g_{3i} = \frac{(b_i^2 \hat{\psi}_i)^2}{(\hat{\psi}_i + b_i^2 \hat{\sigma}_v^2)^3} \text{var}(\hat{\sigma}_v^2)$$

$$g_{4i} = \frac{1}{(n_i - 1)} \frac{(b_i^2 \hat{\psi}_i \hat{\sigma}_v^2)^2}{(\hat{\psi}_i + b_i^2 \hat{\sigma}_v^2)^3}$$

Methodology Notes:

- The reference is formula (5.5.9) on page 314 of the book by Fuller (2009).
- If we have benchmarked in step 13 then we take $mse(\hat{\theta}_i^{WF})$ as an approximation to $mse(\hat{\theta}_i^{WF,b})$.
- The form of the estimated variance $\text{var}(\hat{\boldsymbol{\beta}})$ is given in step 11.

Step 15 If the parameter is a total and we have used a MEAN model then we need to adjust the estimates $\hat{\theta}_i^{WF}$ in step 12 or $\hat{\theta}_i^{WF,b}$ in step 13 if the parameter has been benchmarked using the method of difference adjustment) as well as the $mse(\hat{\theta}_i^{WF})$ in step 14. This is done for all $i \in A_{VAL} \cup A_{NUL}$ as follows.

$$\hat{\theta}_i^{WF} = \begin{cases} \hat{\theta}_i^{DIR} & \text{for } i \in A_{EXC} \\ N_i \hat{\theta}_i^{WF} \text{ (or } N_i \hat{\theta}_i^{WF,b} \text{)} & \text{for } i \in A_{VAL} \cup A_{NUL} \end{cases}$$

$$mse(\hat{\theta}_i^{WF}) = \begin{cases} \hat{\psi}_i & \text{for } i \in A_{EXC} \\ g_{1i}^* + g_{2i}^* + 2g_{3i}^* + 4g_{3i}^* & \text{for } i \in A_{VAL} \\ N_i^2 \left\{ \mathbf{z}_i^T \text{var}(\hat{\boldsymbol{\beta}}) \mathbf{z}_i + b_i^2 \hat{\sigma}_v^2 \right\} & \text{for } i \in A_{NUL} \end{cases}$$

where $g_{0i}^* = N_i^2 g_{0i}$
 $g_{1i}^* = N_i^2 g_{1i}$
 $g_{2i}^* = N_i^2 g_{2i}$
 $g_{3i}^* = N_i^2 g_{3i}$

Implementation Note:

- To obtain the required estimates for the total we simply do the following:
 - The SAE estimate for the total is simply obtained by multiplying $\hat{\theta}_i^{WF}$ from step 12 or $\hat{\theta}_i^{WF,b}$ from step 13 by N_i for $i \in A_{VAL} \cup A_{NUL}$.
 - The estimated MSE for the total is obtained by multiplying the MSE components in step 14 by N_i^2 for $i \in A_{VAL} \cup A_{NUL}$.

Benchmarking

Benchmarking is used to ensure that the resulting small-area estimates add up to a specified population value $\hat{\theta}^{DIR}$ or the weighted sum of the direct estimates $\sum_{i \in A} \omega_i \hat{\theta}_i^{DIR}$ where ω_i is either $\omega_i = 1$ for a total or $\omega_i = N_i/N$ for a mean. Benchmarking can only be applied when the parameter of interest is a total or a mean. If you provide a value $\hat{\theta}^{DIR}$ then this quantity must be the population total or the population mean across all areas including the exclusion areas. If the value $\hat{\theta}^{DIR}$ is provided then we first attempt to benchmark the estimates to this value using the method of difference adjustment. If this is not possible then we attempt to use the method of augmented vector to benchmark the estimates to the weighted sum of the direct estimates $\sum_{i \in A} \omega_i \hat{\theta}_i^{DIR}$. If this is not possible then no benchmarking is done. Table 2 summarizes the logic used in determining the method for benchmarking when you specify this option.

Table 2 Logic for determining the benchmark method

parameter type	conditions			benchmark method	benchmark result
TOTAL	$\hat{\theta}^{DIR}$ is provided			difference adjustment	$\sum_{i \in A} \hat{\theta}_i^{EBLUP,b} = \hat{\theta}^{DIR}$
	$\hat{\theta}^{DIR}$ is not provided	A_{NUL} is empty		augmented auxiliary vector	$\sum_{i \in A} \hat{\theta}_i^{EBLUP} = \sum_{i \in A} \hat{\theta}_i^{DIR}$
		A_{NUL} is not empty	$\hat{\theta}_i^{DIR}$ known for all $i \in A$	difference adjustment	$\sum_{i \in A} \hat{\theta}_i^{EBLUP,b} = \sum_{i \in A} \hat{\theta}_i^{DIR}$
			$\hat{\theta}_i^{DIR}$ not known for all $i \in A$	none	none
MEAN	$\hat{\theta}^{DIR}$ is provided	N_i known for all $i \in A$		difference adjustment	$\sum_{i \in A} \omega_i \hat{\theta}_i^{EBLUP,b} = \hat{\theta}^{DIR}$ where $\omega_i = N_i / \sum_{i \in A} N_i$
		N_i not known for all $i \in A$		none	none
	$\hat{\theta}^{DIR}$ is not provided	A_{NUL} is empty	N_i known for all $i \in A$	augmented auxiliary vector	$\sum_{i \in A} \omega_i \hat{\theta}_i^{EBLUP} = \sum_{i \in A} \omega_i \hat{\theta}_i^{DIR}$ where $\omega_i = N_i / \sum_{i \in A} N_i$
			N_i not known for all $i \in A$	none	none
		A_{NUL} is not empty	N_i and $\hat{\theta}_i^{DIR}$ known for all $i \in A$	difference adjustment	$\sum_{i \in A} \omega_i \hat{\theta}_i^{EBLUP,b} = \sum_{i \in A} \omega_i \hat{\theta}_i^{DIR}$ where $\omega_i = N_i / \sum_{i \in A} N_i$
			N_i and $\hat{\theta}_i^{DIR}$ not known for all $i \in A$	none	none
Other	none			none	none

If benchmarking is required for a total or a mean, then the benchmark method is determined by checking the conditions in Table 2. The benchmark result is shown in the last column of the table.

Note that to benchmark a mean under either method, we must know N_i for all $i \in A$ so we can produce $\omega_i = N_i / \sum_{i \in A} N_i$ for $i \in A$. We do not need the N_i values to benchmark totals. To calculate $\sum_{i \in A} \hat{\theta}_i^{DIR}$ we must know $\hat{\theta}_i^{DIR}$ for all $i \in A$. Similarly, to calculate $\sum_{i \in A} \omega_i \hat{\theta}_i^{DIR}$ with $\omega_i = N_i / \sum_{i \in A} N_i$ we must know both N_i and $\hat{\theta}_i^{DIR}$ for all $i \in A$.

Proof of benchmarking based on augmented vector

This method of benchmarking involves augmenting the auxiliary vector $\mathbf{z}_i$ by adding the extra term $\omega_i \tilde{\psi}_i$ or $\omega_i \hat{\psi}_i$ so $\mathbf{z}_i$ becomes either $\mathbf{z}_i = \begin{pmatrix} \mathbf{z}_i \\ \omega_i \tilde{\psi}_i \end{pmatrix}$ or $\mathbf{z}_i = \begin{pmatrix} \mathbf{z}_i \\ \omega_i \hat{\psi}_i \end{pmatrix}$. The proof that follows uses smoothed variances but the derivation is the same whether we are working with smoothed variances ($\tilde{\psi}_i$) or direct variances ($\hat{\psi}_i$). The requirements for application of this method are as follows.

A_{NUL} is empty so $A = A_{EXC} \cup A_{VAL}$

and ω_i is known for all $i \in A = A_{EXC} \cup A_{VAL}$

For a total, ω_i is simply defined as $\omega_i = 1$ so it is obviously known for all $i \in A = A_{EXC} \cup A_{VAL}$. For a mean, $\omega_i = N_i / \sum_{i \in A} N_i$ so we must know N_i for all $i \in A = A_{EXC} \cup A_{VAL}$. Note that the sum is over all $i \in A = A_{EXC} \cup A_{VAL}$ and not simply $i \in A_{VAL}$.

Prove: We need to show that under the above conditions, the estimates from the augmented model satisfy the corresponding benchmark equation.

Proof: By augmenting the auxiliary vector $\mathbf{z}_i$ with $\omega_i \tilde{\psi}_i$, we now have the following model for $i \in A_{VAL}$.

$$\hat{\theta}_i^{DIR} = \mathbf{z}_i^T \boldsymbol{\beta} + \omega_i \tilde{\psi}_i \delta + b_i v_i + e_i$$

The Generalized Least Squares estimates of $(\boldsymbol{\beta}, \delta)$ are obtained by solving the following equation.

$$\left\{ \sum_{i \in A_{VAL}} \frac{\gamma_i}{b_i^2 \sigma_v^2} \begin{pmatrix} \mathbf{z}_i \\ \omega_i \tilde{\psi}_i \end{pmatrix} \begin{pmatrix} \mathbf{z}_i^T & \omega_i \tilde{\psi}_i \end{pmatrix} \right\} \begin{pmatrix} \hat{\boldsymbol{\beta}} \\ \hat{\delta} \end{pmatrix} = \sum_{i \in A_{VAL}} \frac{\gamma_i}{b_i^2 \sigma_v^2} \begin{pmatrix} \mathbf{z}_i \\ \omega_i \tilde{\psi}_i \end{pmatrix} \hat{\theta}_i^{DIR}$$

The above equation can be split into two parts corresponding to the entries $\mathbf{z}_i$ and $\omega_i \tilde{\psi}_i$. They are as follows.

$$\sum_{i \in A_{VAL}} \frac{\gamma_i}{b_i^2 \sigma_v^2} \mathbf{z}_i^T \hat{\boldsymbol{\beta}} + \omega_i \tilde{\psi}_i \hat{\delta} = \sum_{i \in A_{VAL}} \frac{\gamma_i}{b_i^2 \sigma_v^2} \mathbf{z}_i^T \hat{\theta}_i^{DIR}$$

and $\sum_{i \in A_{VAL}} \frac{\gamma_i}{b_i^2 \sigma_v^2} \omega_i \tilde{\psi}_i (\mathbf{z}_i^T \hat{\boldsymbol{\beta}} + \omega_i \tilde{\psi}_i \hat{\delta}) = \sum_{i \in A_{VAL}} \frac{\gamma_i}{b_i^2 \sigma_v^2} \omega_i \tilde{\psi}_i \hat{\theta}_i^{DIR}$

We can express the equation corresponding to the bottom part as follows.

$$\sum_{i \in A_{VAL}} \frac{\gamma_i}{b_i^2 \sigma_v^2} \omega_i \tilde{\psi}_i (\hat{\theta}_i^{DIR} - \mathbf{z}_i^T \hat{\boldsymbol{\beta}}) = \sum_{i \in A_{VAL}} \frac{\gamma_i}{b_i^2 \sigma_v^2} \omega_i^2 \tilde{\psi}_i^2 \hat{\delta}$$

or equivalently $\sum_{i \in A_{VAL}} \omega_i (1 - \gamma_i) (\hat{\theta}_i^{DIR} - \mathbf{z}_i^T \hat{\boldsymbol{\beta}}) = \sum_{i \in A_{VAL}} (1 - \gamma_i) \omega_i^2 \tilde{\psi}_i \hat{\delta}$

We now show that we obtain this same equation as a result of the benchmark constraint

$\sum_{i \in A} \omega_i \hat{\theta}_i^{EBLUP} = \sum_{i \in A} \omega_i \hat{\theta}_i^{DIR}$. This constraint can be written in terms of the two groups A_{EXC} and A_{VAL} defining $A = A_{EXC} \cup A_{VAL}$ as follows.

$$\sum_{i \in A_{EXC}} \omega_i \hat{\theta}_i^{EBLUP} + \sum_{i \in A_{VAL}} \omega_i \hat{\theta}_i^{EBLUP} = \sum_{i \in A_{EXC}} \omega_i \hat{\theta}_i^{DIR} + \sum_{i \in A_{VAL}} \omega_i \hat{\theta}_i^{DIR}$$

Since $\hat{\theta}_i^{EBLUP} = \hat{\theta}_i^{DIR}$ for $i \in A_{EXC}$, the above benchmark constraint reduces to the following equation.

$$\sum_{i \in A_{VAL}} \omega_i \hat{\theta}_i^{EBLUP} = \sum_{i \in A_{VAL}} \omega_i \hat{\theta}_i^{DIR}$$

We now replace $\hat{\theta}_i^{EBLUP}$ by its form $\gamma_i \hat{\theta}_i^{DIR} + (1 - \gamma_i)(\mathbf{z}_i^T \hat{\boldsymbol{\beta}} + \omega_i \tilde{\psi}_i \hat{\delta})$ under the augmented model. This leads to the following result.

$$\begin{aligned} \sum_{i \in A_{VAL}} \omega_i \hat{\theta}_i^{DIR} &= \sum_{i \in A_{VAL}} \omega_i \hat{\theta}_i^{EBLUP} \\ \sum_{i \in A_{VAL}} \omega_i \hat{\theta}_i^{DIR} &= \sum_{i \in A_{VAL}} \omega_i \{ \gamma_i \hat{\theta}_i^{DIR} + (1 - \gamma_i)(\mathbf{z}_i^T \hat{\boldsymbol{\beta}} + \omega_i \tilde{\psi}_i \hat{\delta}) \} \\ \text{or } \sum_{i \in A_{VAL}} \omega_i (1 - \gamma_i) (\hat{\theta}_i^{DIR} - \mathbf{z}_i^T \hat{\boldsymbol{\beta}}) &= \sum_{i \in A_{VAL}} (1 - \gamma_i) \omega_i^2 \tilde{\psi}_i \hat{\delta} \end{aligned}$$

This equation is the same as that obtained earlier from the estimator of $(\boldsymbol{\beta}, \delta)$. Thus, augmenting the auxiliary vector as $(\mathbf{z}_i^T, \omega_i \tilde{\psi}_i)^T$ leads to a set of small-area estimates satisfying the benchmark constraint.

Proof of benchmarking based on difference adjustment

Under the method of difference adjustment, the benchmarked estimates $\hat{\theta}_i^{EBLUP,b}$ are defined as follows.

$$\hat{\theta}_i^{EBLUP,b} = \begin{cases} \hat{\theta}_i^{EBLUP} & \text{for } i \in A_{EXC} \\ \hat{\theta}_i^{EBLUP} + \alpha_i (\hat{\theta}^* - \sum_{d \in A} \omega_d \hat{\theta}_d^{EBLUP}) & \text{for } i \in A_{VAL} \\ \hat{\theta}_i^{EBLUP} & \text{for } i \in A_{NUL} \end{cases}$$

where $\alpha_i = \frac{\omega_i (\tilde{\psi}_i + b_i^2 \hat{\sigma}_v^2)}{\sum_{d \in A_{VAL}} \omega_d^2 (\tilde{\psi}_d + b_d^2 \hat{\sigma}_v^2)}$ for $i \in A_{VAL}$

The term $\hat{\theta}^*$ is either $\hat{\theta}^{DIR}$ (a population value provided by the user) or a value calculated from the parameter estimates as $\sum_{i \in A} \omega_i \hat{\theta}_i^{DIR}$ with $\omega_i = 1$ for a total and $\omega_i = N_i / \sum_{i \in A} N_i$ for a mean. Either way this value is treated as a constant in the above calculation. It is understood that $\hat{\theta}_i^{EBLUP} = \hat{\theta}_i^{DIR}$ for $i \in A_{EXC}$. This definition simplifies the notation and the proof shown below.

Prove: We need to show that under the above definition, $\sum_{i \in A} \omega_i \hat{\theta}_i^{EBLUP,b} = \hat{\theta}^*$.

Proof: Substituting the value of $\hat{\theta}_i^{EBLUP,b}$ into the left-hand side of the equation leads to the following derivation.

$$\begin{aligned}
& \sum_{i \in A} \omega_i \hat{\theta}_i^{EBLUP,b} \\
&= \sum_{i \in A_{EXC}} \omega_i \hat{\theta}_i^{EBLUP,b} + \sum_{i \in A_{VAL}} \omega_i \hat{\theta}_i^{EBLUP,b} + \sum_{i \in A_{NUL}} \omega_i \hat{\theta}_i^{EBLUP,b} \\
&= \sum_{i \in A_{EXC}} \omega_i \hat{\theta}_i^{EBLUP,b} + \sum_{i \in A_{VAL}} \omega_i \left(\hat{\theta}_i^{EBLUP,b} + \alpha_i (\hat{\theta}^* - \sum_{d \in A} \omega_d \hat{\theta}_d^{EBLUP}) \right) + \sum_{i \in A_{NUL}} \omega_i \hat{\theta}_i^{EBLUP} \\
&= \sum_{i \in A_{EXC}} \omega_i \hat{\theta}_i^{EBLUP} + \sum_{i \in A_{VAL}} \omega_i \hat{\theta}_i^{EBLUP} + \sum_{i \in A_{VAL}} \omega_i \left(\alpha_i (\hat{\theta}^* - \sum_{d \in A} \omega_d \hat{\theta}_d^{EBLUP}) \right) + \sum_{i \in A_{NUL}} \omega_i \hat{\theta}_i^{EBLUP} \\
&= \sum_{i \in A_{EXC}} \omega_i \hat{\theta}_i^{EBLUP} + \sum_{i \in A_{VAL}} \omega_i \hat{\theta}_i^{EBLUP} + \sum_{i \in A_{NUL}} \omega_i \hat{\theta}_i^{EBLUP} + \sum_{i \in A_{VAL}} \omega_i \left(\alpha_i (\hat{\theta}^* - \sum_{d \in A} \omega_d \hat{\theta}_d^{EBLUP}) \right) \\
&= \sum_{i \in A} \omega_i \hat{\theta}_i^{EBLUP} + \sum_{i \in A_{VAL}} \omega_i \left(\alpha_i (\hat{\theta}^* - \sum_{d \in A} \omega_d \hat{\theta}_d^{EBLUP}) \right) \\
&= \sum_{i \in A} \omega_i \hat{\theta}_i^{EBLUP} + \sum_{i \in A_{VAL}} \omega_i \left(\frac{\omega_i (\tilde{\psi}_i + b_i^2 \hat{\sigma}_v^2)}{\sum_{d \in A_{VAL}} \omega_d^2 (\tilde{\psi}_d + b_d^2 \hat{\sigma}_v^2)} (\hat{\theta}^* - \sum_{d \in A} \omega_d \hat{\theta}_d^{EBLUP}) \right) \\
&= \sum_{i \in A} \omega_i \hat{\theta}_i^{EBLUP} + \left(\frac{\sum_{i \in A_{VAL}} \omega_i^2 (\tilde{\psi}_i + b_i^2 \hat{\sigma}_v^2)}{\sum_{d \in A_{VAL}} \omega_d^2 (\tilde{\psi}_d + b_d^2 \hat{\sigma}_v^2)} (\hat{\theta}^* - \sum_{d \in A} \omega_d \hat{\theta}_d^{EBLUP}) \right) \\
&= \sum_{i \in A} \omega_i \hat{\theta}_i^{EBLUP} + (\hat{\theta}^* - \sum_{d \in A} \omega_d \hat{\theta}_d^{EBLUP}) \\
&= \hat{\theta}^*
\end{aligned}$$

This completes the proof and shows that the adjusted values satisfy the benchmark constraint.

Results for MSE estimation

This section contains two important results on the estimation of the MSE for the four small-area methods considered in this paper. The first result shows how to include the model error coefficient b_i in the small-area estimation theory. The second result shows how $mse(\hat{\theta}_i^{EBLUP})$ of the Wang-Fuller small-area estimator can be expressed in the g_{1i} , g_{2i} , g_{3i} and g_{4i} terms.

Inclusion of the model error coefficient

To include the model error coefficient, we start with the derivation of the $MSE(\hat{\theta}_i^{EBLUP})$ as in the Appendix in Wang and Fuller (2003). The expression is shown for the MSE of an estimated mean, but it is also applicable to other parameters that we can linearize in the derivation of the approximate MSE, such as a ratio. In this case, the sampling variance is interpreted as the sampling variance of the corresponding linearized form of the parameter. We adapt the expansion shown in (A.5) of their paper to our notation so $\gamma_i = b_i^2 \sigma_v^2 / (\psi_i + b_i^2 \sigma_v^2)$ and $\hat{\gamma}_i = b_i^2 \hat{\sigma}_v^2 / (\hat{\psi}_i + b_i^2 \hat{\sigma}_v^2)$. Using the same assumptions as given by Wang and Fuller (2003), we obtain the following form for $MSE(\hat{\theta}_i^{EBLUP})$.

$$MSE(\hat{\theta}_i^{EBLUP}) = \gamma_i \psi_i + (1 - \gamma_i)^2 \mathbf{z}_i^T \text{Var}(\hat{\boldsymbol{\beta}}) \mathbf{z}_i + (\psi_i + b_i^2 \sigma_v^2) \text{Var}(\hat{\gamma}_i)$$

The definition of the regression coefficient $\hat{\boldsymbol{\beta}}$ is the same as shown in earlier sections. The term $\text{Var}(\hat{\gamma}_i)$ is approximated as follows. First, we do a Taylor expansion of $(\hat{\gamma}_i - \gamma_i)$ as a function of $\hat{\sigma}_v^2$ and $\hat{\psi}_i$. Then we take the variance of these terms. We keep only the linear terms in the expansion. The other terms are of lower order when the sample size is sufficiently large. Thus, the procedure leads to the following.

$$\begin{aligned} (\hat{\gamma}_i - \gamma) &\equiv \left(\frac{1}{(\psi_i + b_i^2 \sigma_v^2)} \right) \left[(1 - \gamma_i) b_i^2 (\hat{\sigma}_v^2 - \sigma_v^2) + \gamma_i (\hat{\psi}_i - \psi_i) \right] \\ &= \left(\frac{1}{(\psi_i + b_i^2 \sigma_v^2)} \right) \left[\frac{\psi_i}{(\psi_i + b_i^2 \sigma_v^2)} b_i^2 (\hat{\sigma}_v^2 - \sigma_v^2) + \frac{b_i^2 \sigma_v^2}{(\psi_i + b_i^2 \sigma_v^2)} (\hat{\psi}_i - \psi_i) \right] \\ &= \left(\frac{b_i^2}{(\psi_i + b_i^2 \sigma_v^2)^2} \right) \left[\psi_i (\hat{\sigma}_v^2 - \sigma_v^2) + \sigma_v^2 (\hat{\psi}_i - \psi_i) \right] \\ \text{Var}(\hat{\gamma}_i) &\equiv \frac{b_i^4}{(\psi_i + b_i^2 \sigma_v^2)^4} \left[(\psi_i)^2 \text{Var}(\hat{\sigma}_v^2) + (\sigma_v^2)^2 \text{Var}(\hat{\psi}_i) \right] \end{aligned}$$

We now substitute this result into the above expression for the $MSE(\hat{\theta}_i^{EBLUP})$. This produces the required approximation.

$$MSE(\hat{\theta}_i^{EBLUP}) = \gamma_i \psi_i + (1 - \gamma_i)^2 \mathbf{z}_i^T \text{Var}(\hat{\boldsymbol{\beta}}) \mathbf{z}_i + \frac{b_i^4}{(\psi_i + b_i^2 \sigma_v^2)^3} \left[(\psi_i)^2 \text{Var}(\hat{\sigma}_v^2) + (\sigma_v^2)^2 \text{Var}(\hat{\psi}_i) \right]$$

This expression shows the inclusion of the model error coefficient in the various terms either explicitly or implicitly through γ_i and $\hat{\boldsymbol{\beta}}$.

Estimation of the MSE in the Wang-Fuller method

Following Prasad and Rao (1990), Wang and Fuller obtained the following estimator of $MSE(\hat{\theta}_i^{EBLUP})$.

$$mse(\hat{\theta}_i^{EBLUP}) = \hat{\gamma}_i \hat{\psi}_i + (1 - \hat{\gamma}_i)^2 \mathbf{z}_i^T \text{var}(\hat{\boldsymbol{\beta}}) \mathbf{z}_i + 2(\hat{\psi}_i + b_i^2 \hat{\sigma}_v^2) \text{var}(\hat{\gamma}_i)$$

where $\text{var}(\hat{\gamma}_i) = \frac{b_i^4}{(\hat{\psi}_i + b_i^2 \hat{\sigma}_v^2)^4} \left[(\hat{\psi}_i)^2 \text{var}(\hat{\sigma}_v^2) + (\hat{\sigma}_v^2)^2 \text{var}(\hat{\psi}_i) \right]$ is the estimated variance of $\hat{\gamma}_i$. We can rewrite the above expression as follows.

$$\begin{aligned} mse(\hat{\theta}_i^{EBLUP}) &= \hat{\gamma}_i \hat{\psi}_i + (1 - \hat{\gamma}_i)^2 \mathbf{z}_i^T \text{var}(\hat{\boldsymbol{\beta}}) \mathbf{z}_i + 2 \frac{b_i^4}{(\hat{\psi}_i + b_i^2 \hat{\sigma}_v^2)^3} \left[(\hat{\psi}_i)^2 \text{var}(\hat{\sigma}_v^2) + (\hat{\sigma}_v^2)^2 \text{var}(\hat{\psi}_i) \right] \\ &= \hat{\gamma}_i \hat{\psi}_i + (1 - \hat{\gamma}_i)^2 \mathbf{z}_i^T \text{var}(\hat{\boldsymbol{\beta}}) \mathbf{z}_i + \frac{2(b_i^2 \hat{\psi}_i)^2 \text{var}(\hat{\sigma}_v^2)}{(\hat{\psi}_i + b_i^2 \hat{\sigma}_v^2)^3} + \frac{2(b_i^2 \hat{\sigma}_v^2)^2 \text{var}(\hat{\psi}_i)}{(\hat{\psi}_i + b_i^2 \hat{\sigma}_v^2)^3} \end{aligned}$$

We then have the following results:

- The first term $\hat{\gamma}_i \hat{\psi}_i$ is g_{1i} .
- The second term $(1 - \hat{\gamma}_i)^2 \mathbf{z}_i^T \text{var}(\hat{\boldsymbol{\beta}}) \mathbf{z}_i$ is equal to $(1 - \hat{\gamma}_i)^2 \mathbf{z}_i^T \left[\sum_{i \in A_{VAL}} \mathbf{z}_i \mathbf{z}_i^T / (\hat{\psi}_i + b_i^2 \hat{\sigma}_v^2) \right]^{-1} \mathbf{z}_i$, which is the g_{2i} term.
- The third term $\frac{2(b_i^2 \hat{\psi}_i)^2 \text{var}(\hat{\sigma}_v^2)}{(\hat{\psi}_i + b_i^2 \hat{\sigma}_v^2)^3}$ is simply $2g_{3i}$.
- The last term is obtained by assuming $(n_i - 1)\hat{\psi}_i \sim \chi_{n_i-1}^2 \psi_i$. This means $\text{var}(\hat{\psi}_i) = \frac{2\hat{\psi}_i^2}{(n_i - 1)}$.

Substituting this result into the fourth term of $mse(\hat{\theta}_i^{EBLUP})$, we obtain $\frac{4}{(n_i - 1)} \frac{(b_i^2 \hat{\psi}_i \hat{\sigma}_v^2)^2}{(\hat{\psi}_i + b_i^2 \hat{\sigma}_v^2)^3}$. This is our $4g_{4i}$ term.

Diagnostics

This section describes the diagnostic plots and tests used to test the fitness of the small-area model. All plots use only the areas in $A_{VAL} \cup A_{NUL}$. They do not include areas in the exclusion group A_{EXC} because these areas are not used to build the model.

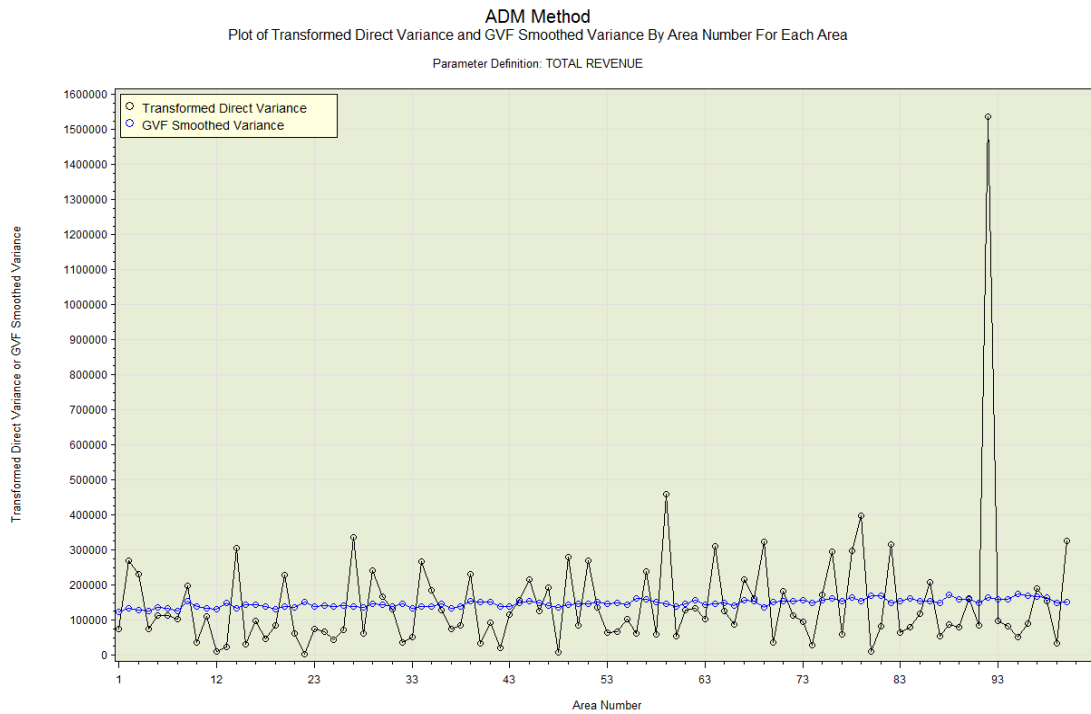
Plots on the GVF smoothed variances

These plots are produced for the ADM, REML and FH methods and only when you provide direct variance estimates to be smoothed using GVF smoothing. These plots compare the direct variance estimates to the smoothed variance estimates. Therefore, if you provide smoothed variance estimates then these plots are not generated since there is nothing to compare.

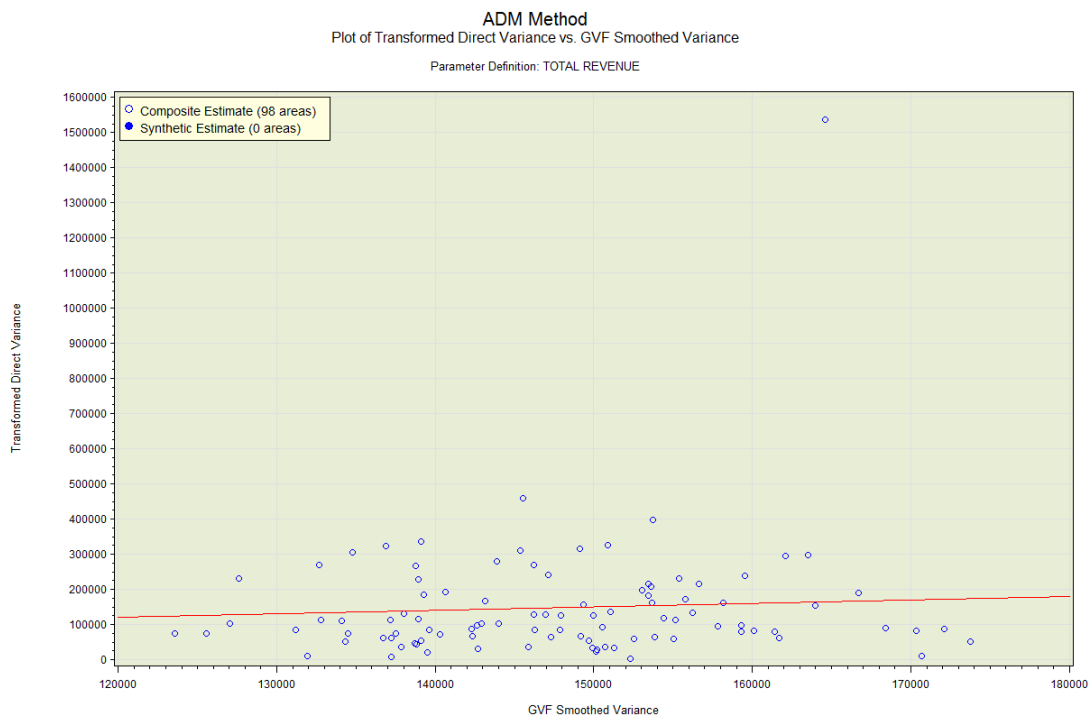
- If the parameter is a total and a MEAN model is applied then we create plots based on the transformed direct variances and the smoothed variances. You provide direct variances for the totals ($\hat{\psi}_i$) and we transform these so they are direct variances for the corresponding means ($\hat{\psi}_i/N_i^2$). The smoothed variances ($\tilde{\psi}_i$) are then produced from the transformed direct variances ($\hat{\psi}_i/N_i^2 \rightarrow \tilde{\psi}_i$). The two plots that are produced have the following titles.
 - Plot of Transformed Direct Variance and GVF Smoothed Variance By Area Number For Each Area
 - Plot of Transformed Direct Variance vs. GVF Smoothed Variance
- Otherwise, we create plots based on the direct variances ($\hat{\psi}_i$) and the smoothed variances ($\tilde{\psi}_i$). The smoothed variances are produced from the direct variances ($\hat{\psi}_i \rightarrow \tilde{\psi}_i$). The two plots that are produced have the following titles.
 - Plot of Direct Variance and GVF Smoothed Variance By Area Number For Each Area
 - Plot of Direct Variance vs. GVF Smoothed Variance

The two plots shown below are from an example involving the estimation of a total and the application of a MEAN model.

Transformed Direct Variance and GVF Smoothed Variance By Area Number For Each Area



Transformed Direct Variance vs. GVF Smoothed Variance



The red line on the plot has an intercept of 0 and a slope of 1 but this may not be evident from the scale shown on the two axes. The scale on the vertical axis may be different from that on the horizontal

axis for two reasons. First, the GVF smoothed variances are generally smaller and tend to have a narrower range of values. The scale may also be different if there are extreme points on the plot, as shown in the above example. Areas with synthetic estimates are only displayed if the direct estimate is missing (for classification to area group A_{NUL}) but the direct variance is not missing (so it can be displayed on the plot).

Plots on the direct estimates and the small-area estimates

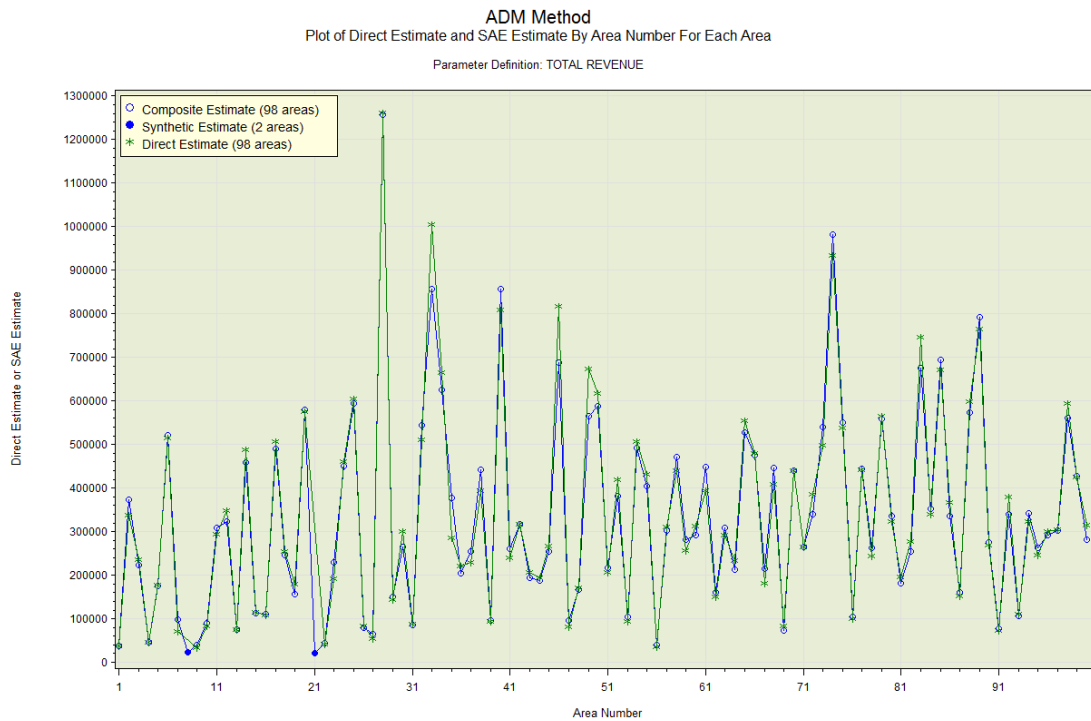
These plots provide a comparison between the direct estimates ($\hat{\theta}_i^{DIR}$) and the small-area estimates ($\hat{\theta}_i^{EBLUP}$). The direct estimates are shown as asterisks (*) connected by a green line. The small-area estimates are shown as blue circles (for areas in area group A_{VAL}) or blue dots (for areas in area group A_{NUL}). They are all connected by a blue line. The areas in area group A_{VAL} have composite estimates while the areas in area group A_{NUL} have synthetic estimates.

- For an area in area group A_{VAL} , we expect the small area estimate to be close to the direct estimate if the following conditions apply:
 - The SAE model provides a good fit.
 - The direct estimate has been produced through unbiased or approximately unbiased design-based estimation methods such as calibration estimation with sufficient number of sample units in the calibration groups and the small area.

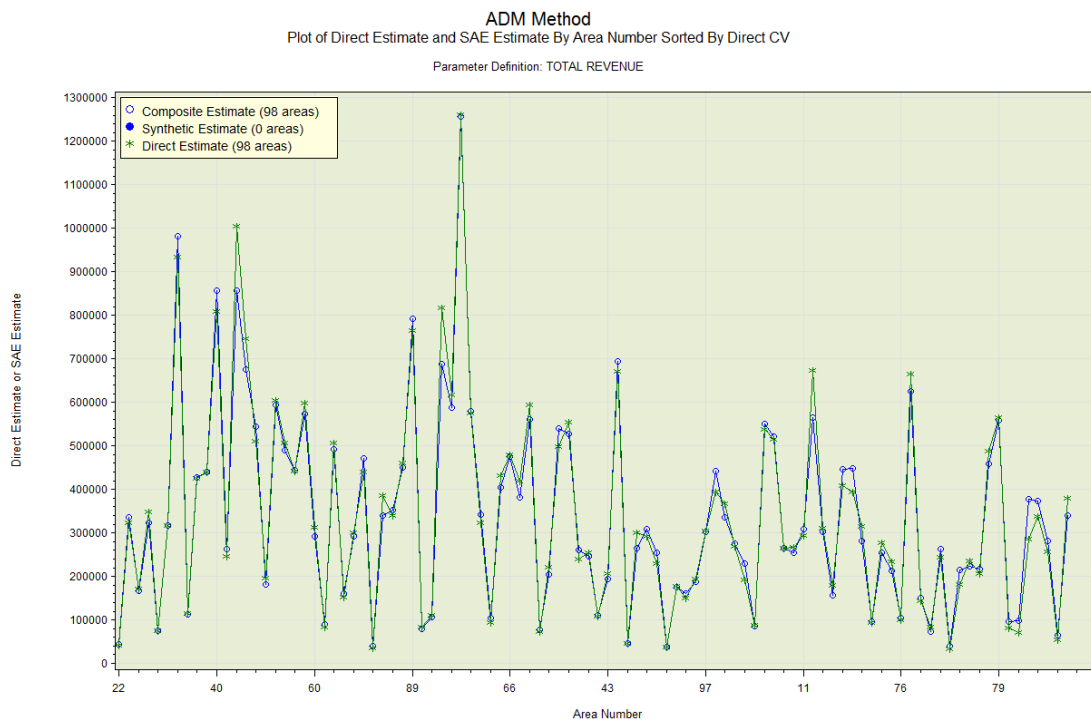
If the direct estimate for an area has poor precision then the corresponding small-area estimate may be quite different. We prefer the small-area estimate if the SAE model provides a good fit.

- For an area in area group A_{NUL} to be displayed on these plots, the direct estimate must be known but its direct variance is missing (for classification to area group A_{NUL}). The small area estimate may not be close to the direct estimate since the missing direct variance may indicate a lack of precision and high variability in the direct estimate. If the SAE model fits well then small-area estimate is preferred over the direct estimate.

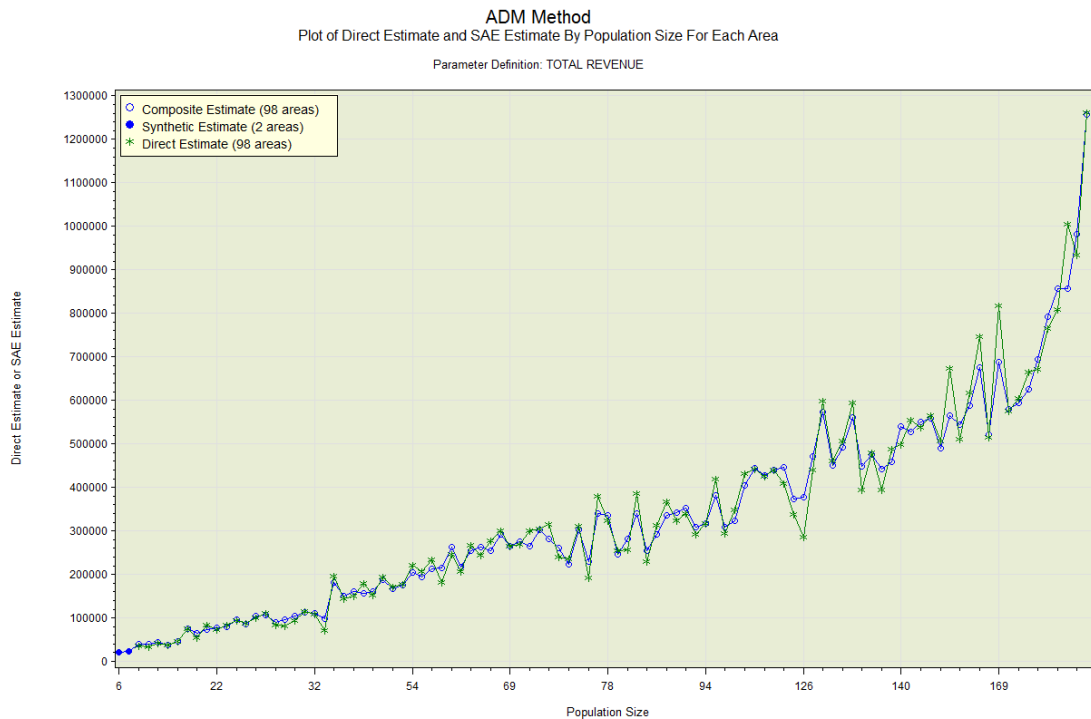
Direct Estimate and SAE Estimate By Area Number For Each Area



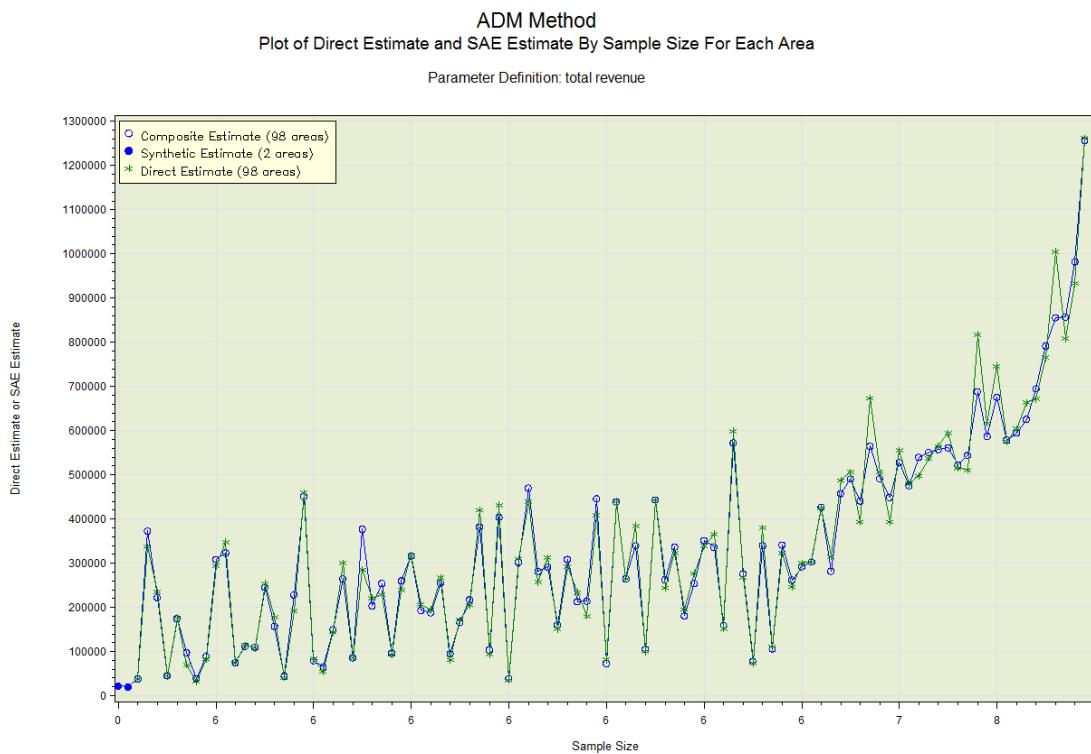
Direct Estimate and SAE Estimate By Area Number Sorted By Direct CV



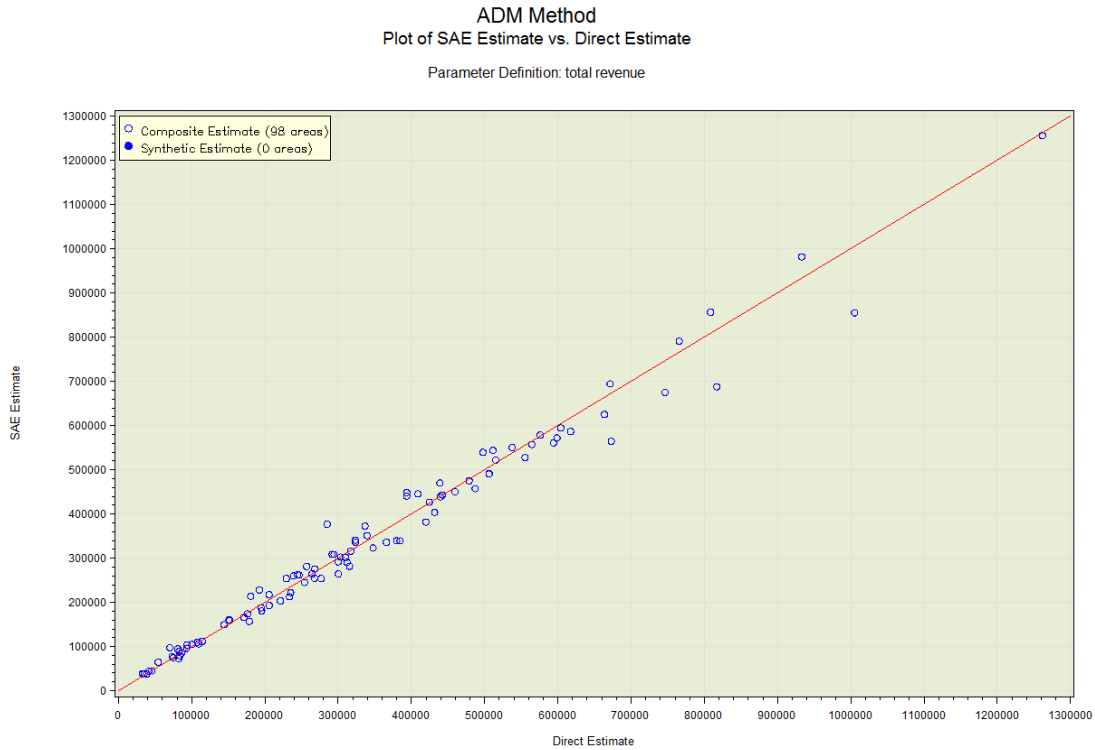
Direct Estimate and SAE Estimate By Population Size for Each Area



Direct Estimate and SAE Estimate By Sample Size for Each Area

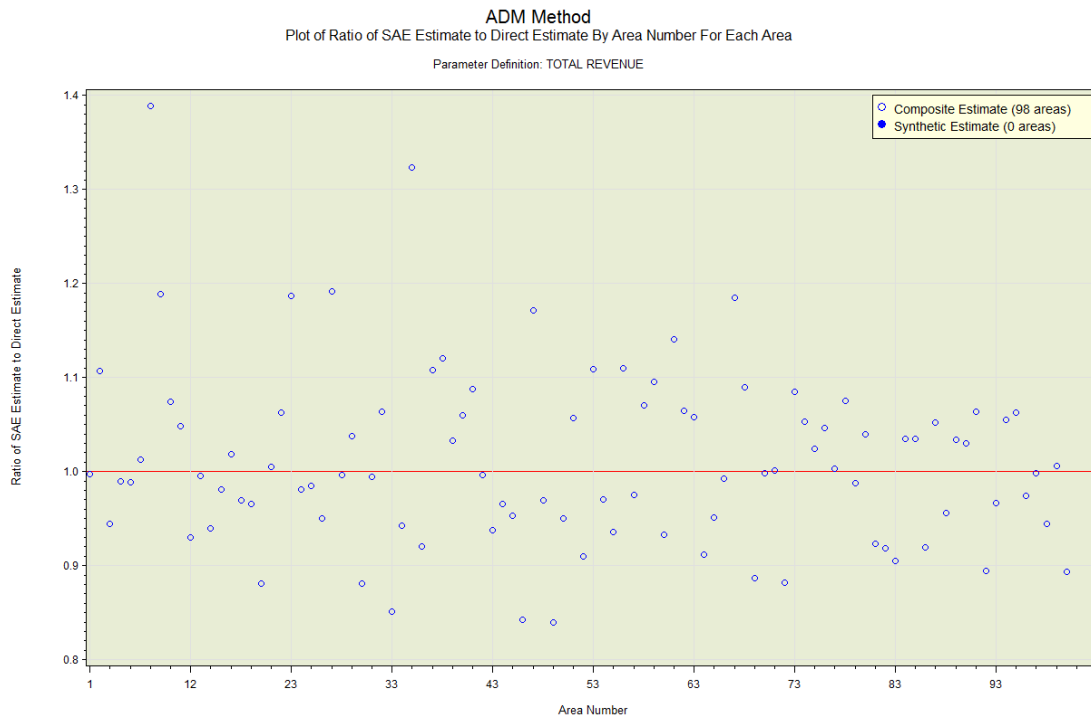


SAE Estimate vs. Direct Estimate

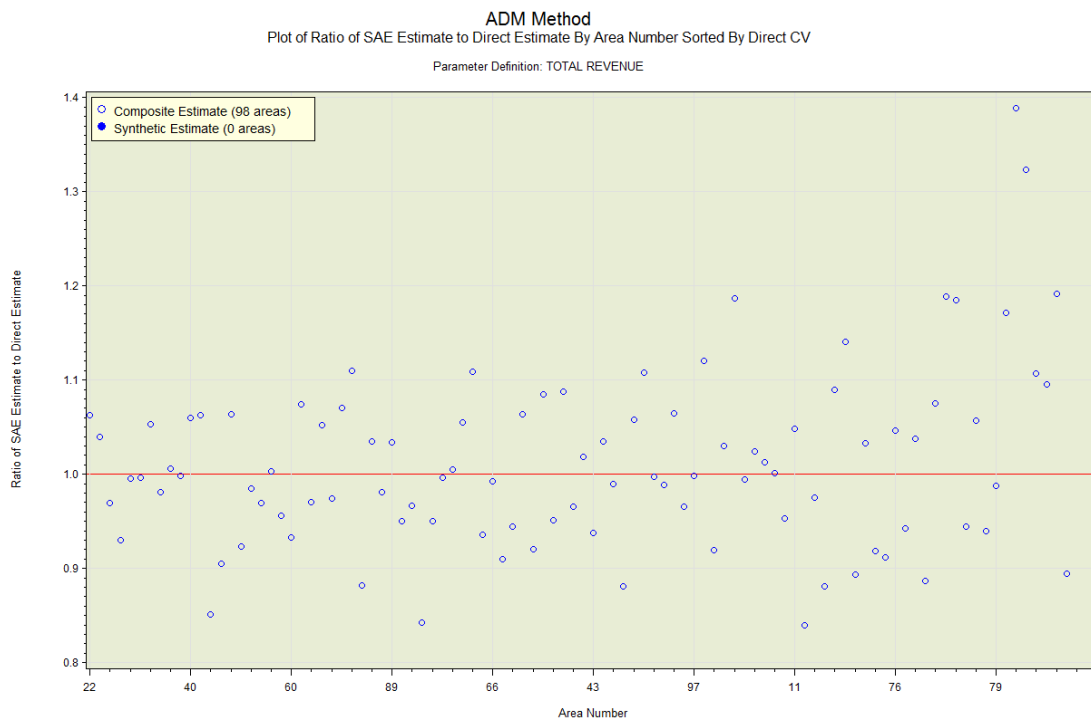


The red reference line on the plot has an intercept of 0 and slope of 1. If the model fits well then the two axis should have similar scale and the points should be close to the line. However, areas in area group A_{NUL} may have synthetic estimates which are considerably different from the direct estimates.

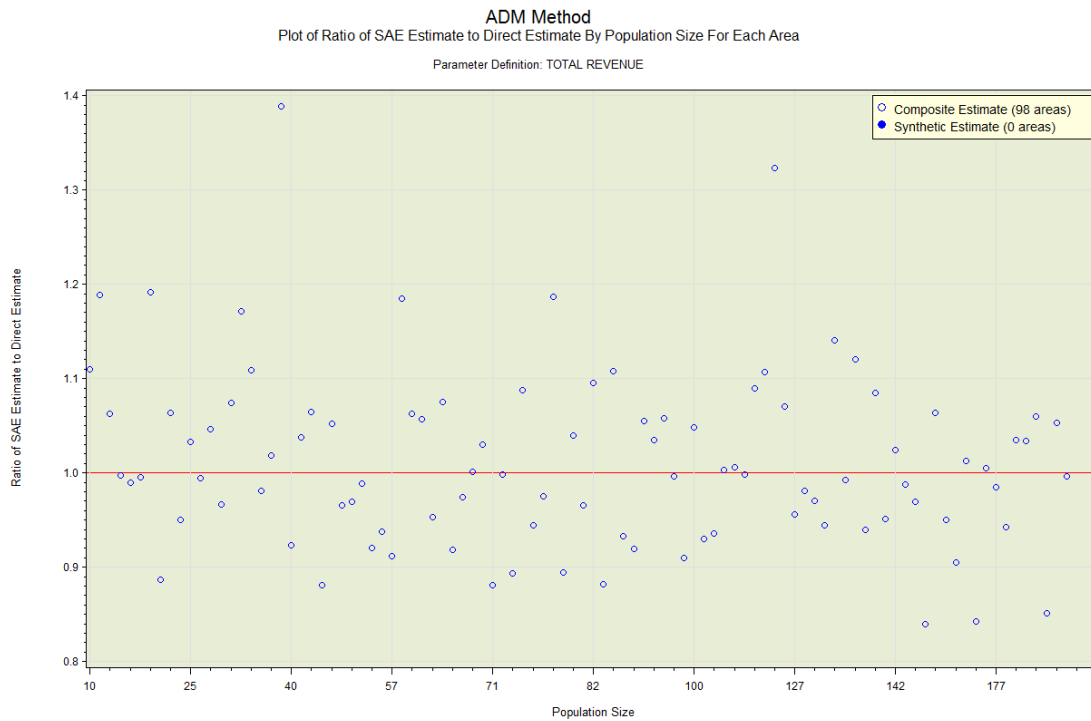
Ratio of SAE Estimate to Direct Estimate By Area Number For Each Area



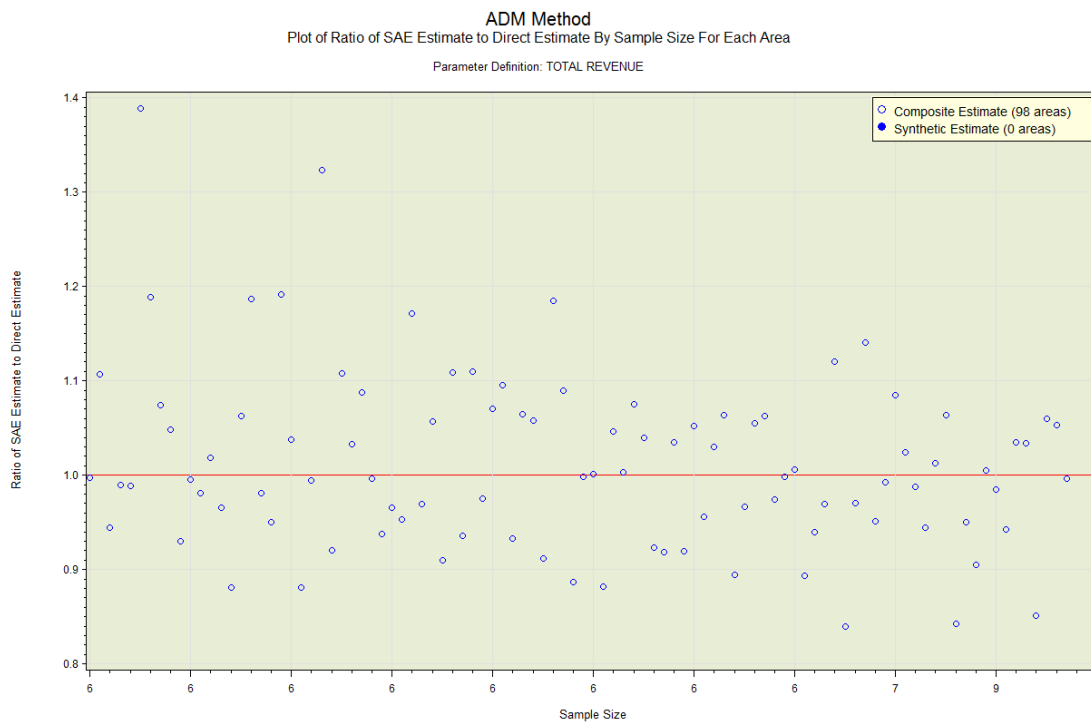
Ratio of SAE Estimate to Direct Estimate By Area Number Sorted By Direct CV



Ratio of SAE Estimate to Direct Estimate By Population Size for Each Area



Ratio of SAE Estimate to Direct Estimate By Sample Size for Each Area



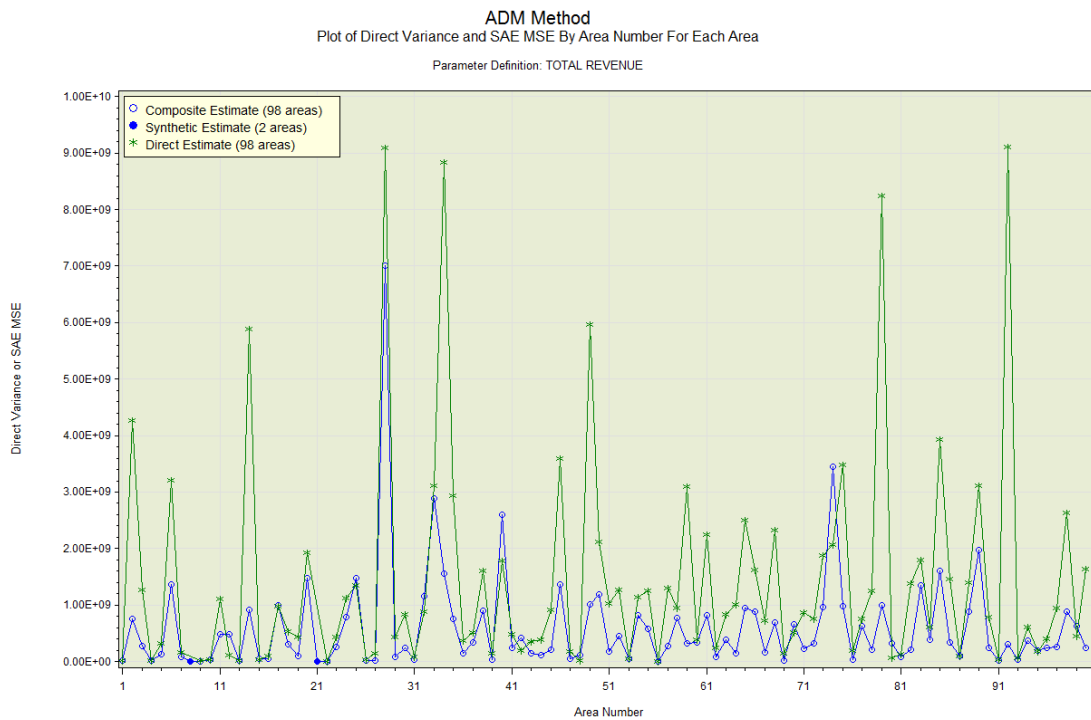
Plots on the direct variances and the SAE MSEs

These plots provide a comparison between the direct variances and the estimated SAE mean square errors ($mse(\hat{\theta}_i^{EBLUP})$). For this set of plots, we use the input variances $\hat{\psi}_i$ that you provide for each of the four methods.

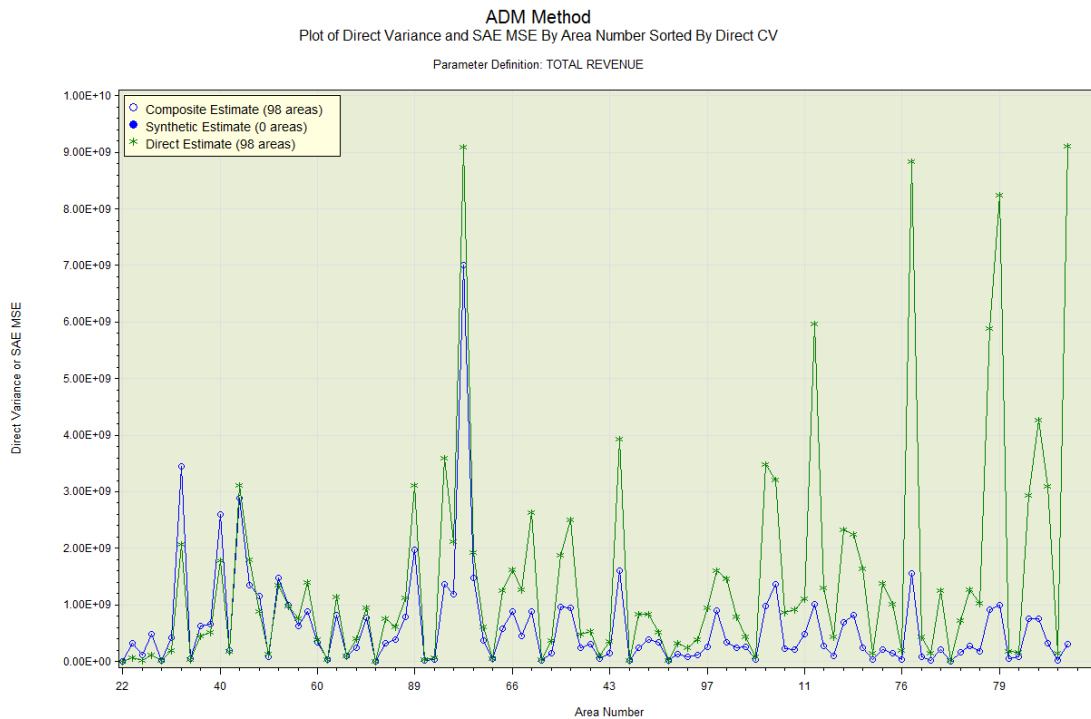
The direct variances are shown as asterisks (*) connected by a green line. The SAE MSEs are shown as blue circles for areas in area group A_{VAL} or blue dots for areas in area group A_{NUL} . They are all connected by a blue line. The areas in area group A_{VAL} have composite estimates while the areas in area group A_{NUL} have synthetic estimates.

- For an area in area group A_{VAL} , we expect the small area MSE to be generally smaller than the direct variance if the SAE model provides a good fit. However, if the direct estimate is based on calibration estimation with a sufficiently large number of sample units in the area then the direct variance should be comparable to the SAE MSE.
- For an area in area group A_{NUL} to be displayed on these plots, the direct variance must be known but its direct estimate is missing (for classification to area group A_{NUL}). This is an unlikely scenario. If the model fits well then the SAE estimate and its MSE are valid estimates for the area.

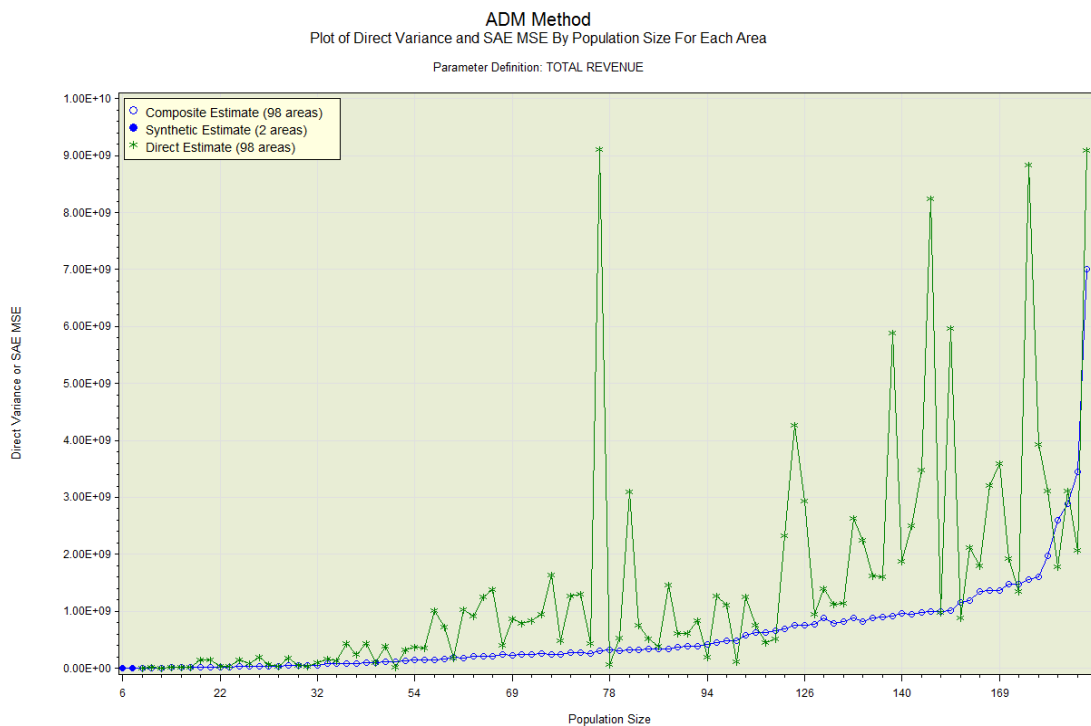
Direct Variance and SAE MSE By Area Number For Each Area



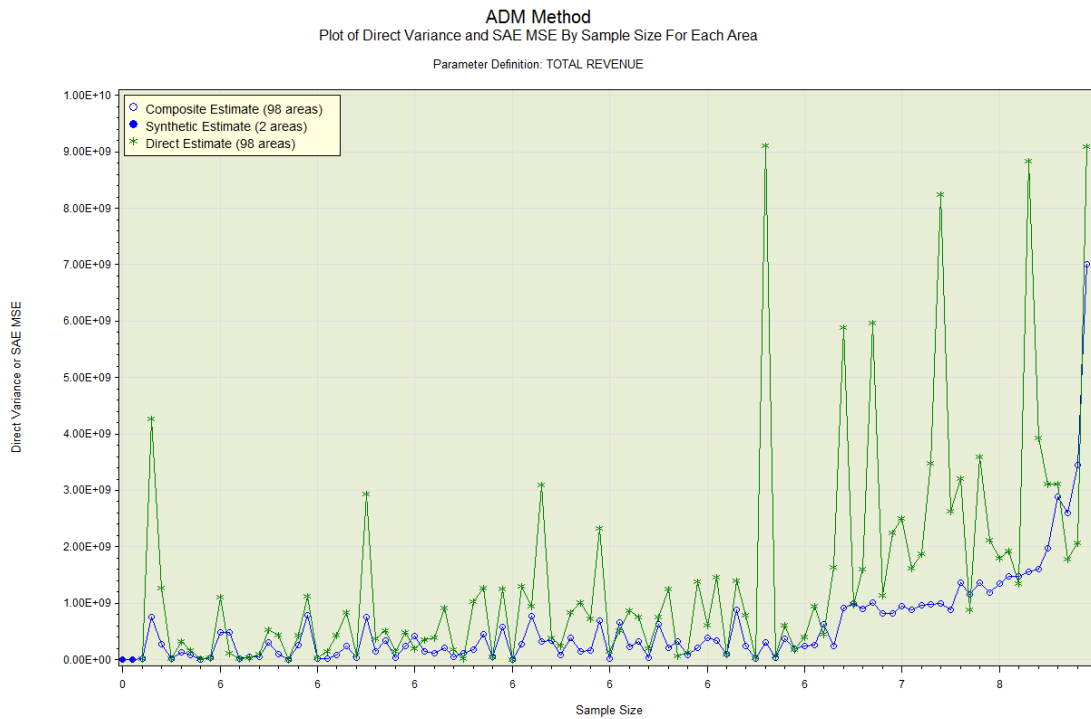
Direct Variance and SAE MSE By Area Number Sorted By Direct CV



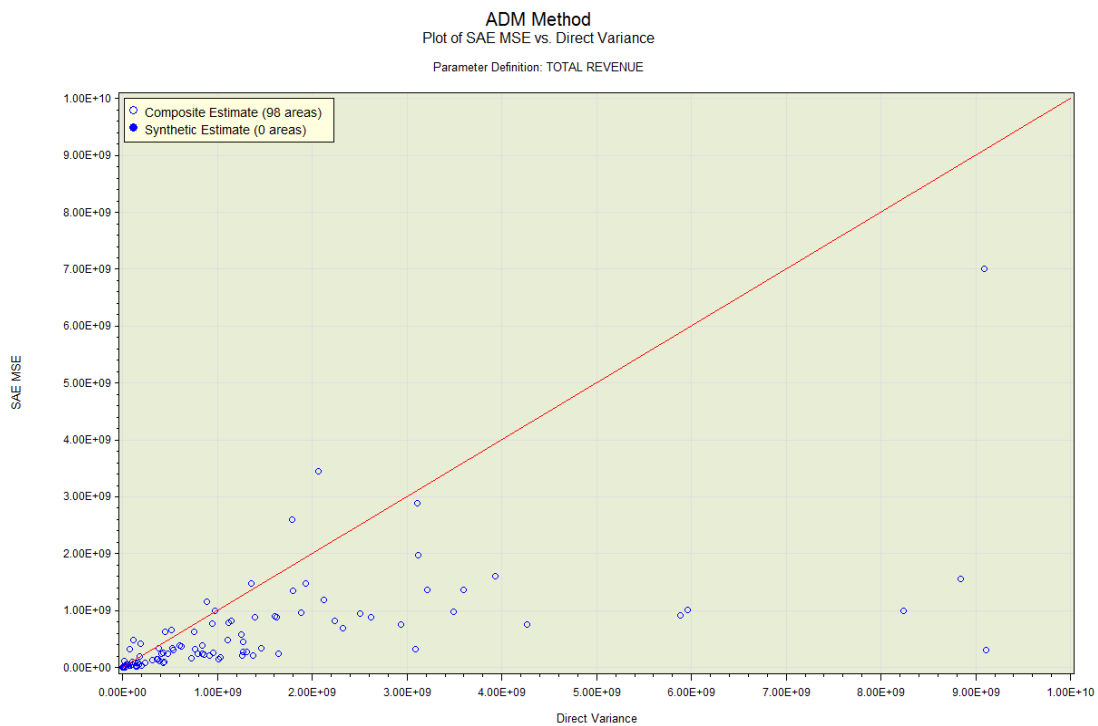
Direct Variance and SAE MSE By Population Size for Each Area



Direct Variance and SAE MSE By Sample Size for Each Area

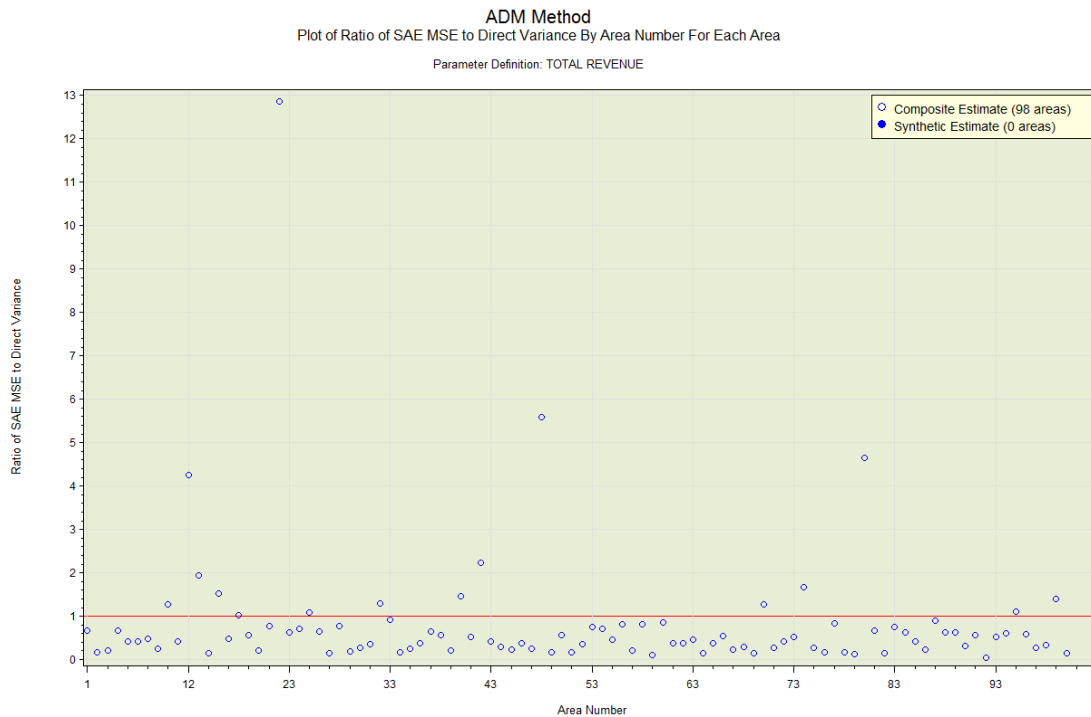


SAE MSE vs. Direct Variance

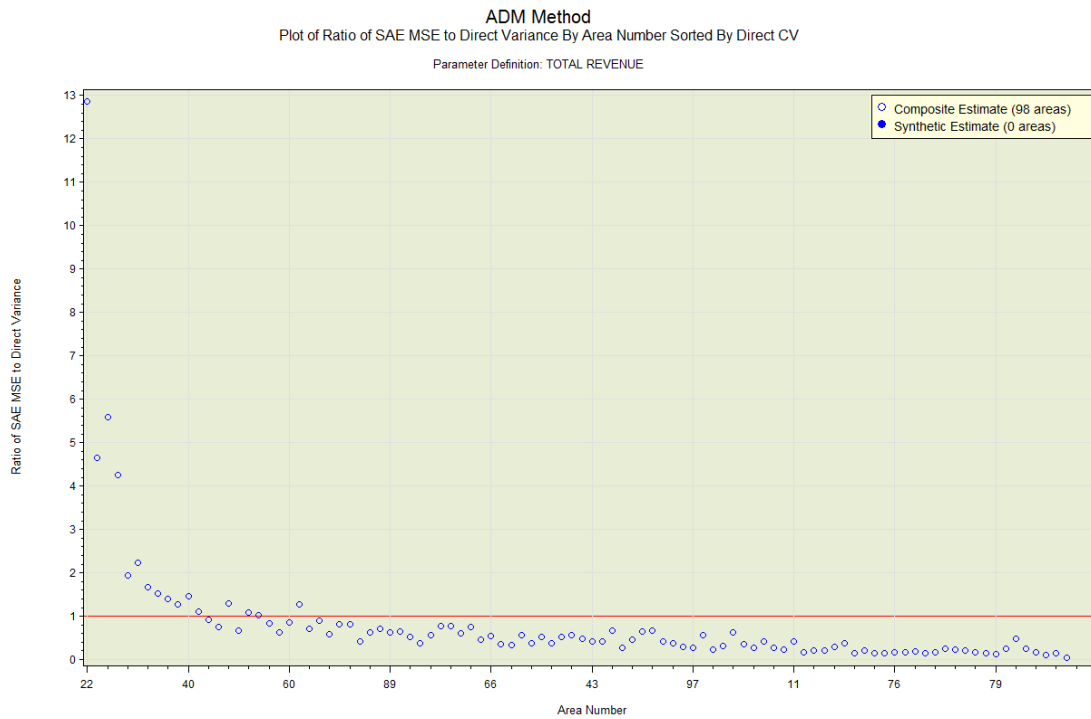


The red reference line on the plot has an intercept of 0 and a slope of 1. If the model fits well then the SAE MSE estimates should be generally smaller than the direct variances. This means most of points on the plot should be below the red line.

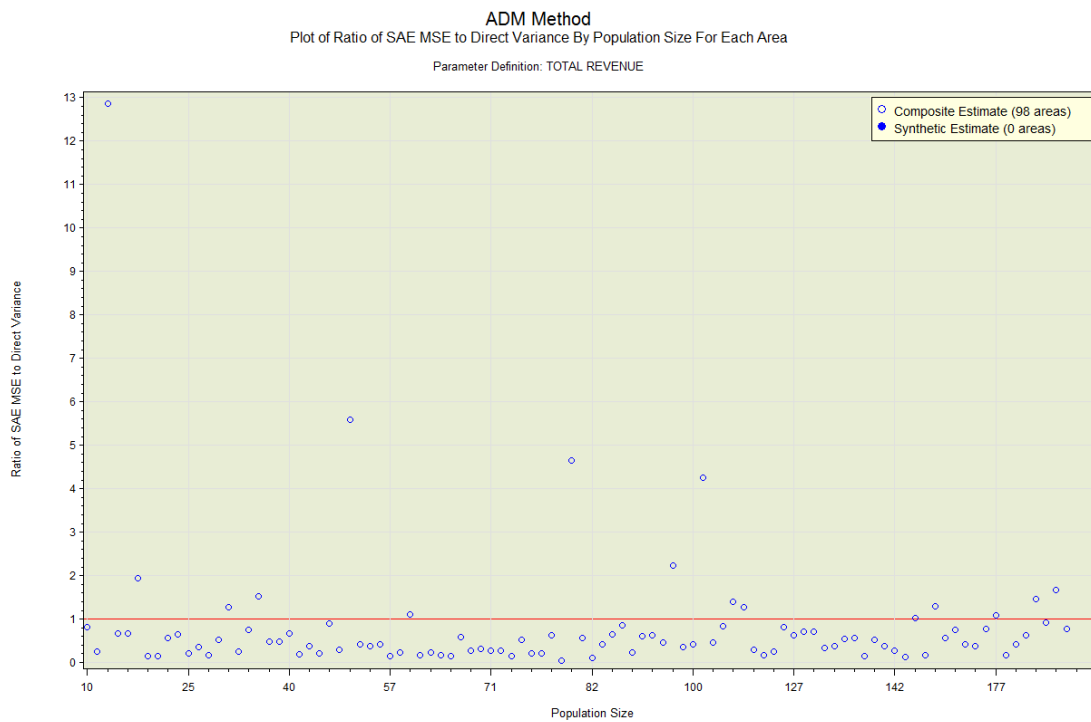
Ratio of SAE MSE to Direct Variance By Area Number For Each Area



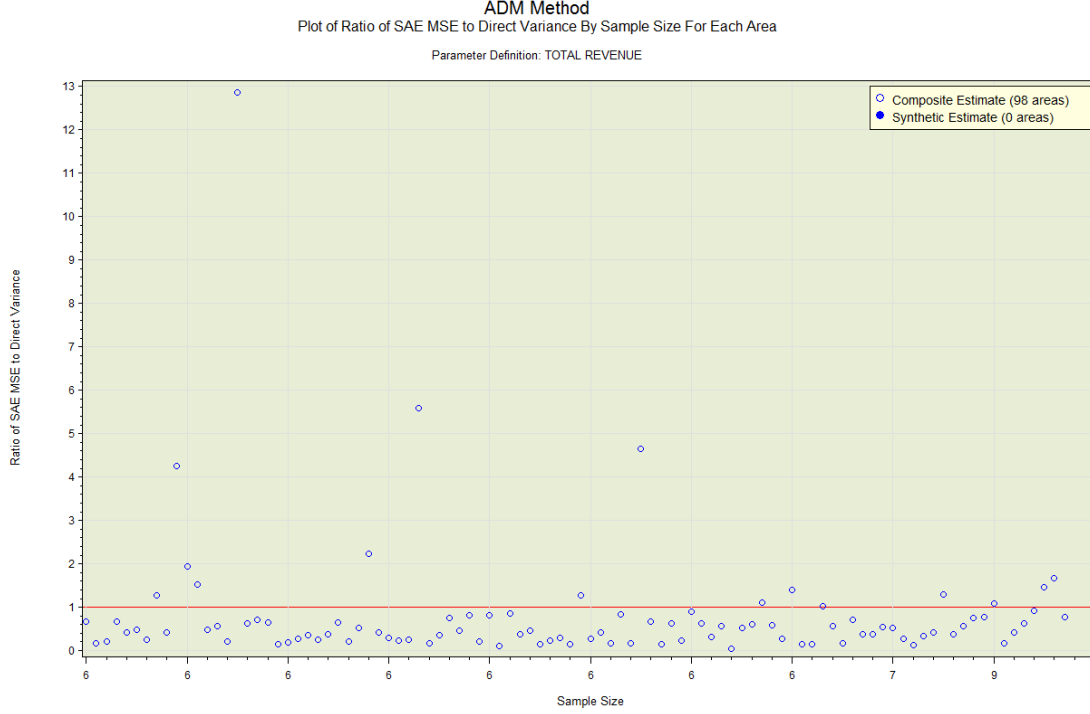
Ratio of SAE MSE to Direct Variance By Area Number Sorted By Direct CV



Ratio of SAE MSE to Direct Variance By Population Size for Each Area



Ratio of SAE MSE to Direct Variance By Sample Size for Each Area



Plots on the direct CVs and the SAE RRMSEs

These plots provide a comparison between the direct coefficients of variation (CV) calculated as

$\sqrt{\hat{\psi}_i} / \left| \hat{\theta}_i^{DIR} \right|$ and the estimated SAE root relative mean square errors (RRMSE) obtained as

$\sqrt{mse(\hat{\theta}_i^{EBLUP})} / \left| \hat{\theta}_i^{EBLUP} \right|$. The SAE RRMSEs are shown as blue circles for areas in area group A_{VAL}

and blue dots for areas in area group A_{NUL} . These are all connected by a blue line. The CVs are shown

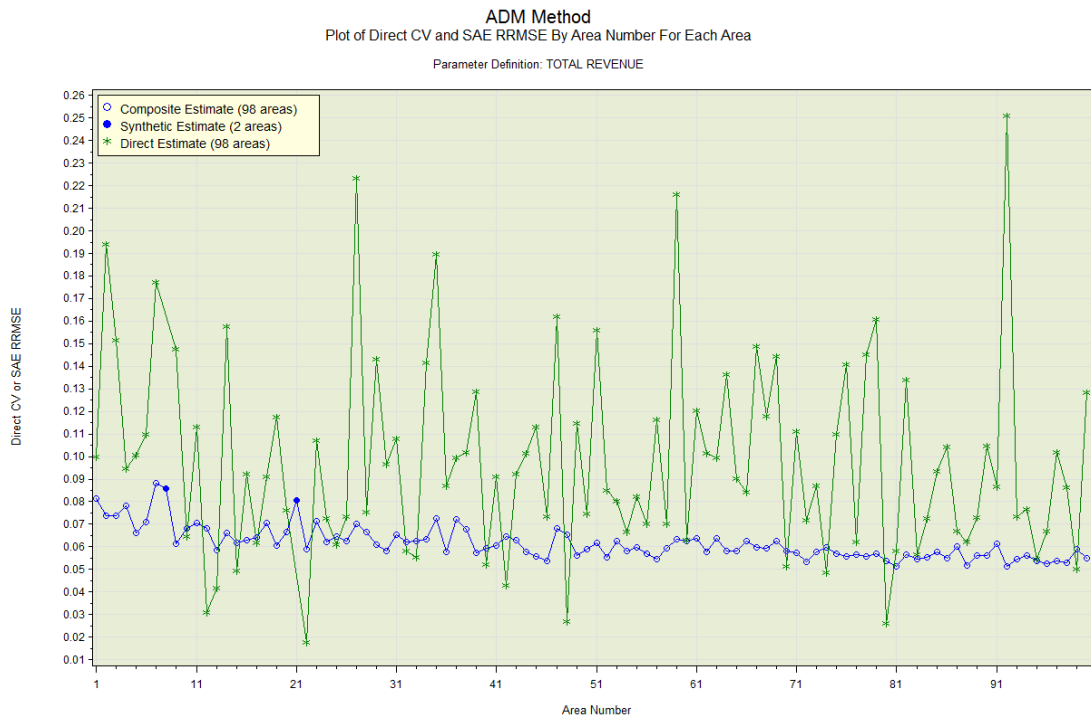
as asterisks (*) for the areas in area group A_{VAL} . These are connected by a green line. There are no CVs

displayed for areas in area group A_{NUL} . These areas have a missing value for the direct estimate

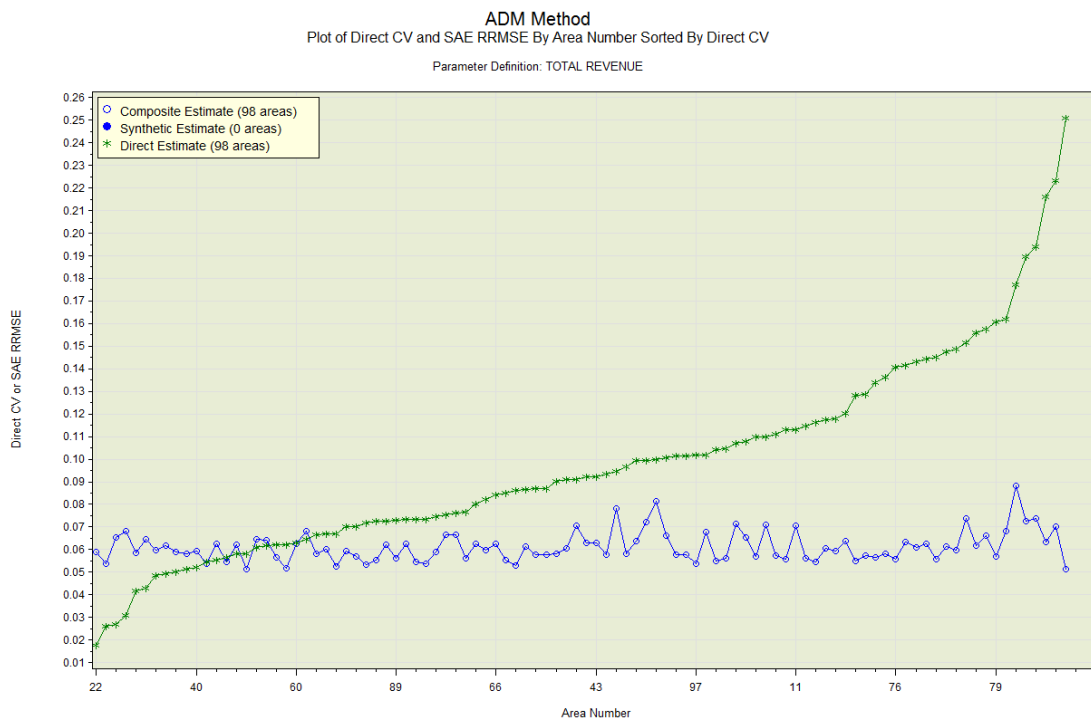
$(\hat{\theta}_i^{DIR})$ or the input variance $(\hat{\psi}_i)$ or both. However, neither one of these can be missing to produce a CV.

For each area in area group A_{VAL} , we expect the small area RRMSE to be generally smaller than the direct CV if the SAE model provides a good fit. However, if the direct estimate is based on calibration estimation with a sufficiently large number of sample units in the area then the direct CV should be comparable to the SAE RRMSE.

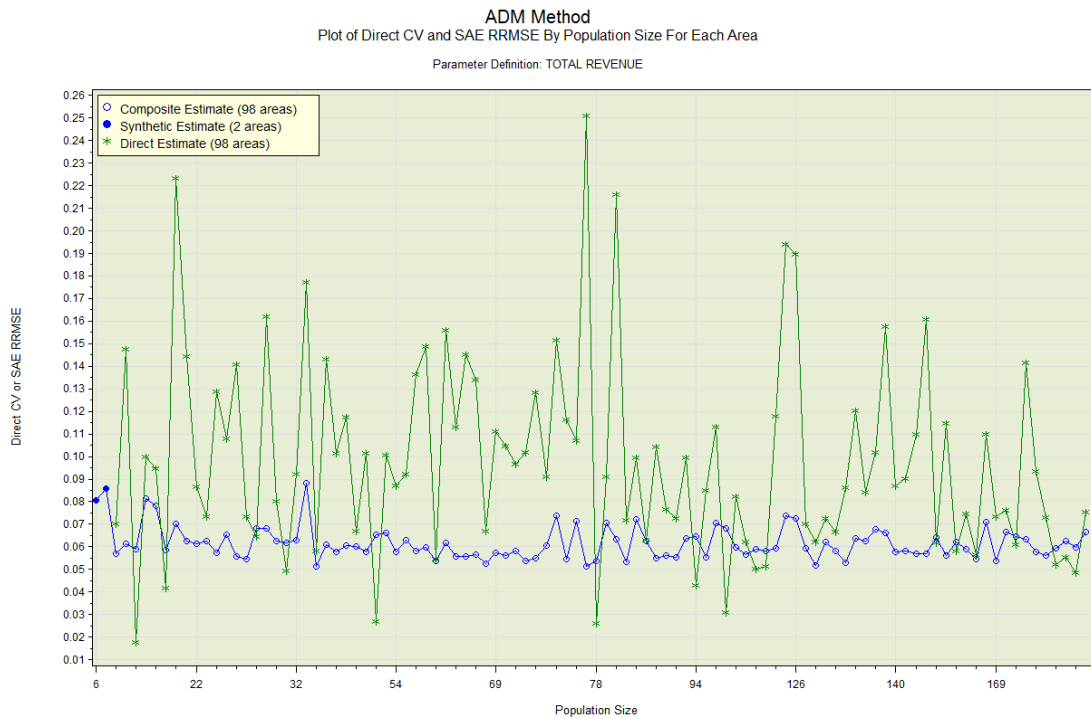
Direct CV and SAE RRMSE By Area Number For Each Area



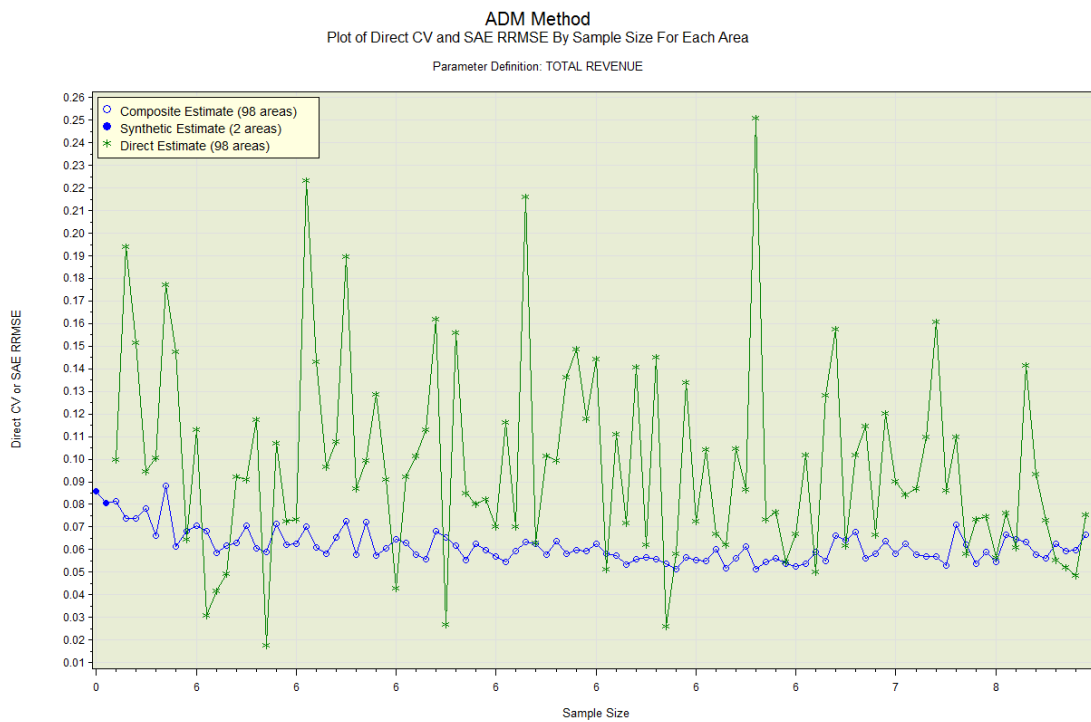
Direct CV and SAE RRMSE By Area Number Sorted By Direct CV



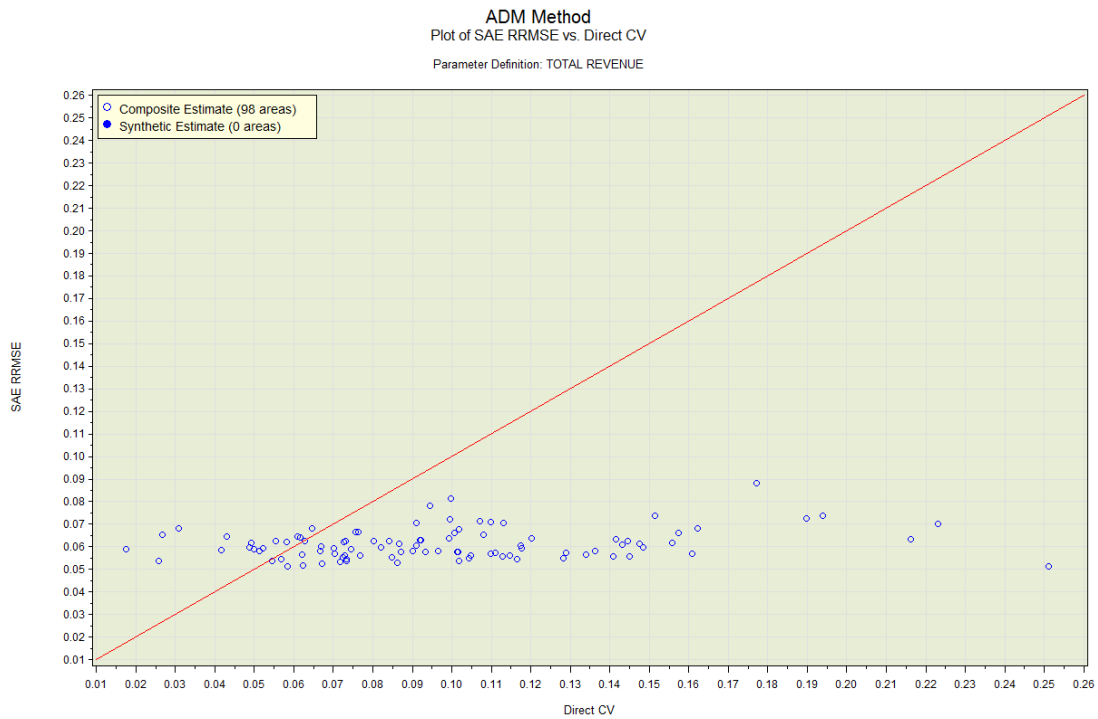
Direct CV and SAE RRMSE By Population Size for Each Area



Direct CV and SAE RRMSE By Sample Size for Each Area

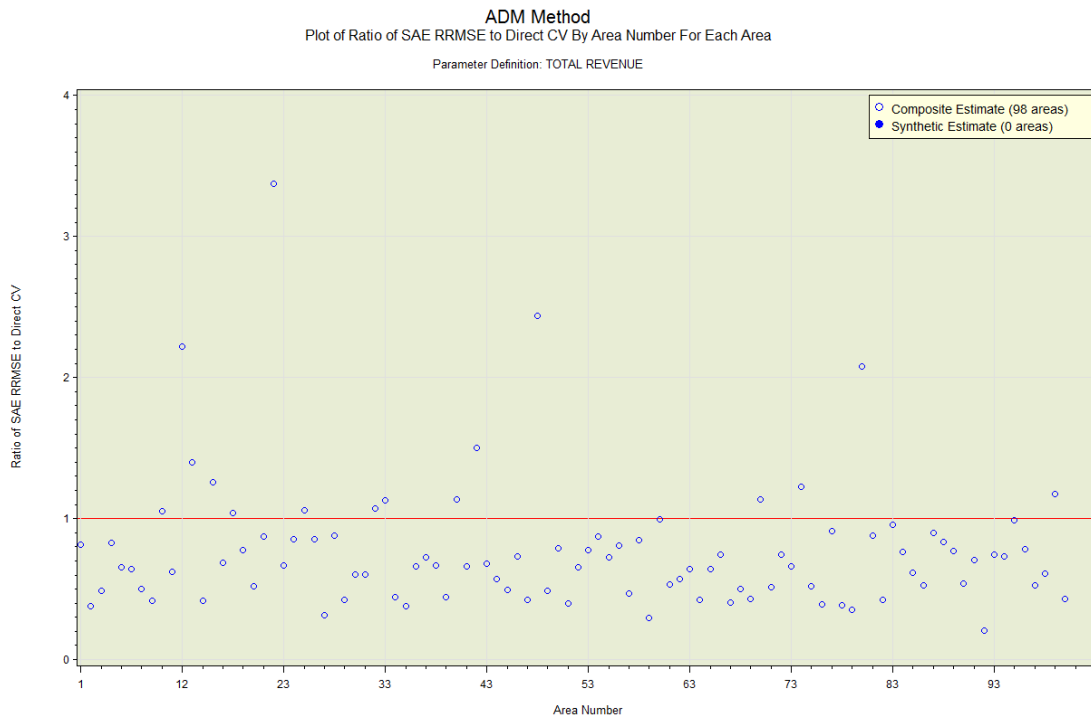


SAE RRMSE vs. Direct CV

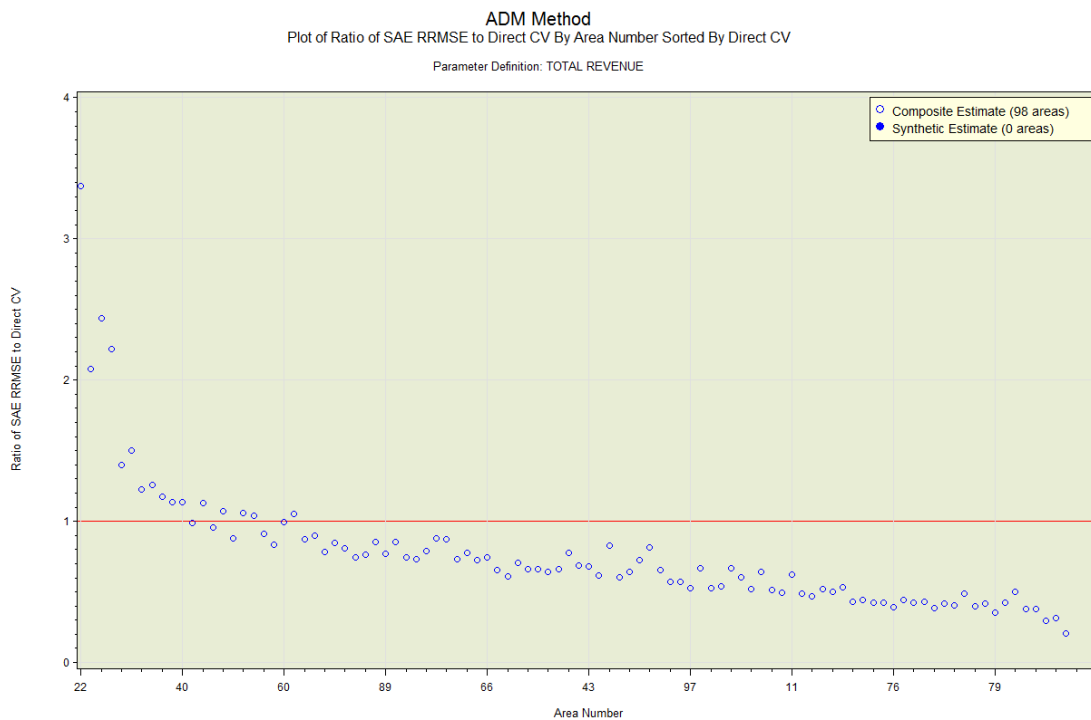


The red reference line on the plot has an intercept of 0 and a slope of 1. If the model fits well then the SAE RRMSE estimates should be generally smaller than the direct CVs. This means most of points on the plot should be below the red line.

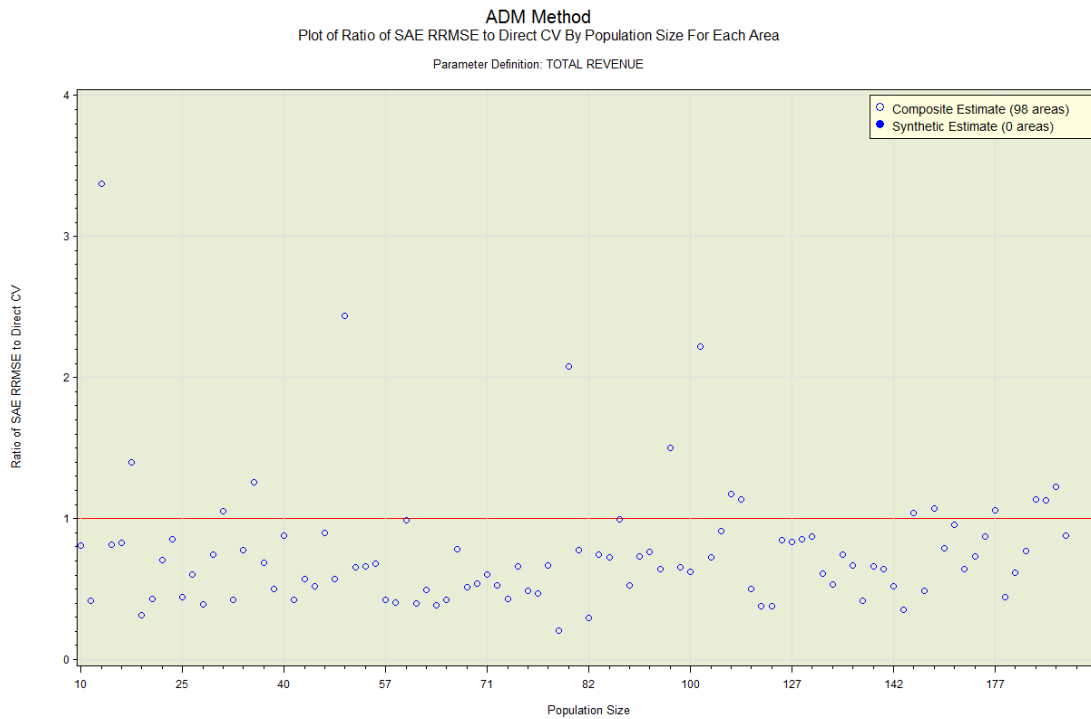
Ratio of SAE RRMSE to Direct CV By Area Number For Each Area



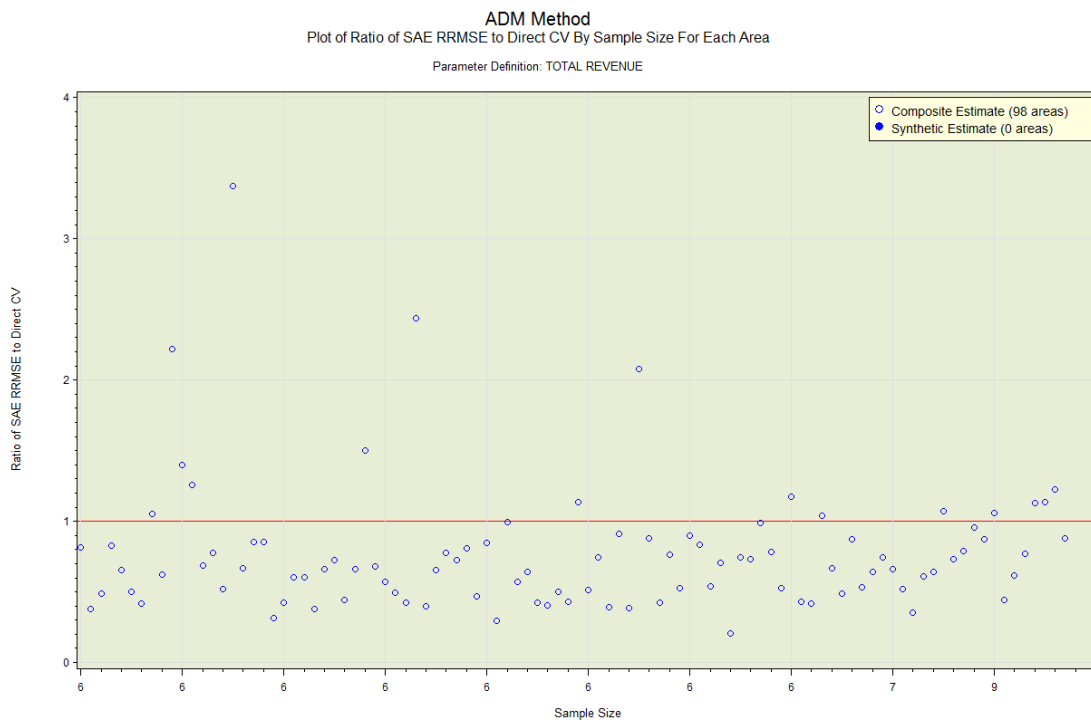
Ratio of SAE RRMSE to Direct CV By Area Number Sorted By Direct CV



Ratio of SAE RRMSE to Direct CV By Population Size for Each Area



Ratio of SAE RRMSE to Direct CV By Sample Size for Each Area



Plots on the fit of the model

These plots are used to provide an evaluation of the fit of the SAE model. We compute residuals and predicted values from the model. Then we use these to produce various plots and statistics to assess the fit of the model.

The SAE model is determined by the logic in steps 2 to 6 of each of the methods. It is important to remember the following distinction in the creation of the SAE model. If the parameter is a total and we apply a MEAN model then the SAE model begins as a transformed model as outlined in step 2. Otherwise, we apply a model without this transformation. In some of the plots, we compare the direct estimates and SAE estimates to the model predicted values. When we use a transformed model, we need to transform the input direct estimates ($\hat{\theta}_i^{DIR}$) and the resulting SAE estimates ($\hat{\theta}_i^{EBLUP}$) for the totals in order to have a meaningful comparison with the values from the transformed model. For these plots, we use transformed direct estimates ($\hat{\theta}_i^{DIR}/N_i$) and transformed SAE estimates ($\hat{\theta}_i^{EBLUP}/N_i$) which correspond to estimates for a mean.

The model residuals and predicted values are obtained from the SAE model as follows. We write the combined SAE model as $\hat{\theta}_i^{DIR} = \mathbf{z}_i^T \boldsymbol{\beta} + \xi_i$, where $\xi_i = (b_i v_i + e_i) \stackrel{iid}{\sim} (0, \psi_i + b_i^2 \sigma_v^2)$. The ξ_i are uncorrelated but unknown error terms. It is also useful to look at the standardized model which is obtained by dividing by $\sqrt{(\psi_i + b_i^2 \sigma_v^2)}$. In this model $\tilde{\theta}_i = \tilde{\mathbf{z}}_i^T \boldsymbol{\beta} + \tilde{\xi}_i$, we have $\tilde{\theta}_i = \hat{\theta}_i^{DIR} / \sqrt{(\psi_i + b_i^2 \sigma_v^2)}$, $\tilde{\mathbf{z}}_i = \mathbf{z}_i / \sqrt{(\psi_i + b_i^2 \sigma_v^2)}$ and $\tilde{\xi}_i = \xi_i / \sqrt{(\psi_i + b_i^2 \sigma_v^2)}$. The $\tilde{\xi}_i$ are standardized and uncorrelated error terms. Since $\boldsymbol{\beta}$ and σ_v^2 are unknown, we replace them with their corresponding estimates $\hat{\boldsymbol{\beta}}$ and $\hat{\sigma}_v^2$. This produces the model standardized residuals denoted by $\hat{\tilde{\xi}}_i = (\hat{\theta}_i^{DIR} - \mathbf{z}_i^T \hat{\boldsymbol{\beta}}) / \sqrt{(\psi_i + b_i^2 \hat{\sigma}_v^2)}$. The model predicted values are given by $\mathbf{z}_i^T \hat{\boldsymbol{\beta}}$ and the model standardized predicted values are $\mathbf{z}_i^T \hat{\boldsymbol{\beta}} / \sqrt{(\psi_i + b_i^2 \hat{\sigma}_v^2)}$. In practice, the ψ_i are also unknown so we use appropriate estimates as shown in the construction of the model for each of the four methods.

We also calculate an R^2 statistic to provide an indication of the strength of the correlation between the auxiliary variables and the parameter of interest. If the parameter of interest is a total and a MEAN model is applied then the R-Squared value is shown on the Plot of Transformed Direct Estimate vs. Model Predicted Value. Otherwise, it is shown on the Plot of Direct Estimate vs. Model Predicted Value. Details on the calculation of the R-Squared value are shown in the section for those plots.

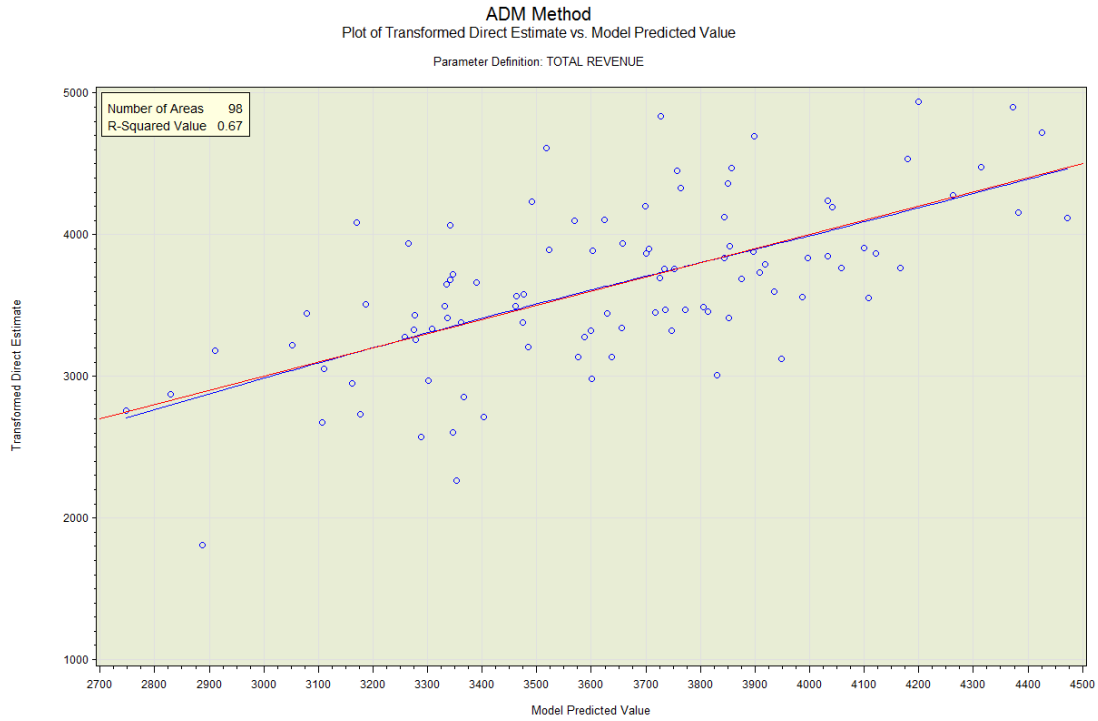
The following table summarizes the various statistics used in the axes of this group of plots. Other statistics shown in the insets of each plot are described later.

notation	description
$\hat{\theta}_i^{DIR} / N_i$	transformed direct estimate (for a MEAN model)
$\hat{\theta}_i^{EBLUP} / N_i$	transformed SAE estimate (for a MEAN model)
$(\hat{\theta}_i^{DIR} / N_i) - (\hat{\theta}_i^{EBLUP} / N_i)$	difference between transformed direct estimate and transformed SAE estimate
$\hat{\theta}_i^{DIR} - \hat{\theta}_i^{EBLUP}$	difference between direct estimate and SAE estimate
$\mathbf{z}_i^T \hat{\boldsymbol{\beta}}$	model predicted value
$\hat{\xi}_i = (\hat{\theta}_i^{DIR} - \mathbf{z}_i^T \hat{\boldsymbol{\beta}}) / \sqrt{(\psi_i + b_i^2 \hat{\sigma}_v^2)}$	model standardized residual
$\hat{\xi}_i^2 = (\hat{\theta}_i^{DIR} - \mathbf{z}_i^T \hat{\boldsymbol{\beta}})^2 / (\psi_i + b_i^2 \hat{\sigma}_v^2)$	square of model standardized residual
$\mathbf{z}_i^T \hat{\boldsymbol{\beta}} / \sqrt{(\psi_i + b_i^2 \hat{\sigma}_v^2)}$	model standardized predicted value
Note: We use appropriate estimates for the ψ_i in these formulas. This depends on the construction of the SAE model as shown in the description of each of the four methods.	

Using the standardized model we also compute two additional test statistics to assess the fit of the model. The first statistic is based on a test of the standardized residuals to see if they could be viewed as random values from a Normal(0,1) distribution. The second statistic comes from a test on the sign of the standardized residuals. On average, we expect the same number of positive and negative residuals, with an extreme imbalance indicating a lack of fit.

For each of these two tests we calculate the probability of observing a more extreme value of the test statistic. This is known as the p-value. Small p-values provide evidence against the null hypothesis of each test. These two p-values are displayed on the QQ-Plot of the Standardized Residuals vs. Standard Normal Quantiles. The two tests are described in more detail in the section for that plot.

Transformed Direct Estimate vs. Model Predicted Value



The red line on the plot is a reference line with an intercept of 0 and a slope of 1. The blue curve is a smooth spline between the points on the plot. If the model fits well, then the smooth spline should be close to the reference line.

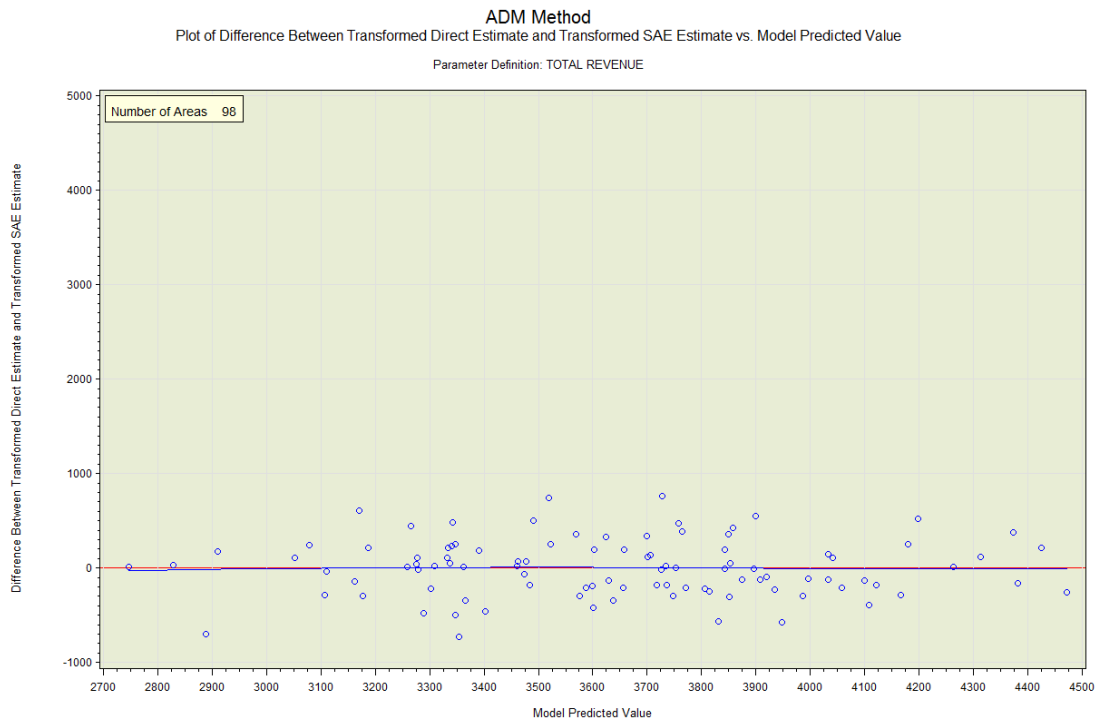
R-squared statistic

This statistic has the following form.

$$R^2 = 1 - \frac{\hat{\sigma}_v^2}{\hat{\sigma}_v^2 + S^2(\hat{\beta})} \quad \text{with} \quad S^2(\hat{\beta}) = \frac{\sum_{i \in A_{VAL}} (z_i^T \hat{\beta} - \sum_{i \in A_{VAL}} z_i^T \hat{\beta} / m)^2}{(m-1)}$$

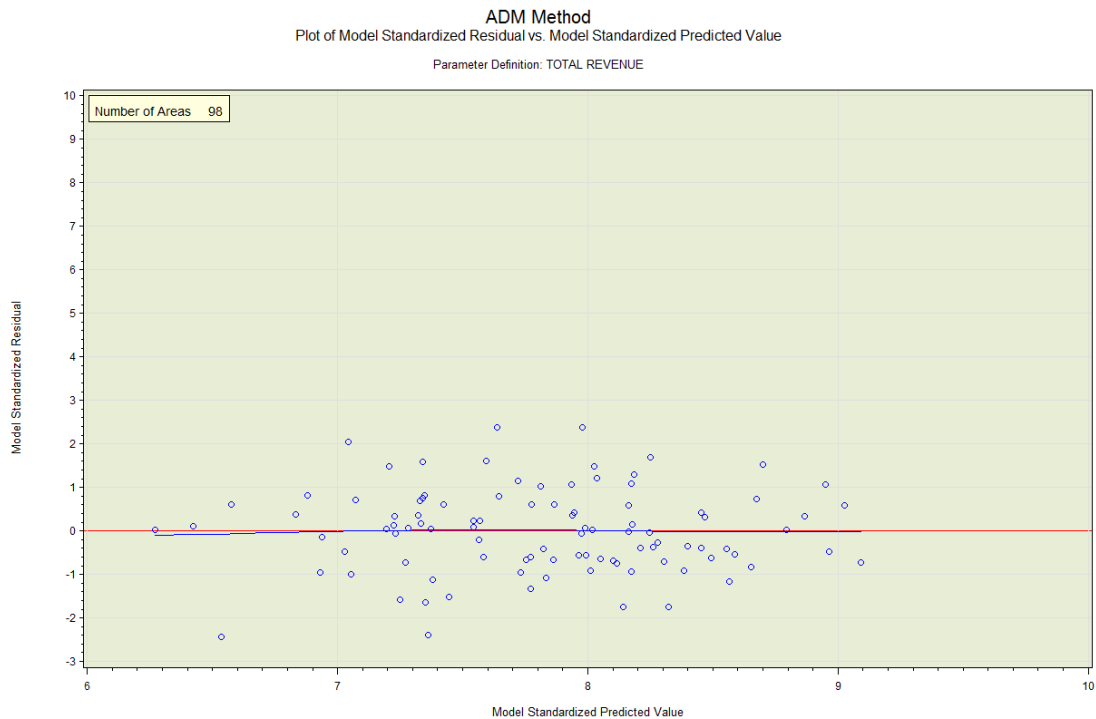
If the model fits well then $\hat{\sigma}_v^2$ should be relatively small compared to $S^2(\hat{\beta})$ and R^2 should be close to 1.

Difference Between Transformed Direct Estimate and Transformed SAE Estimate vs. Model Predicted Value



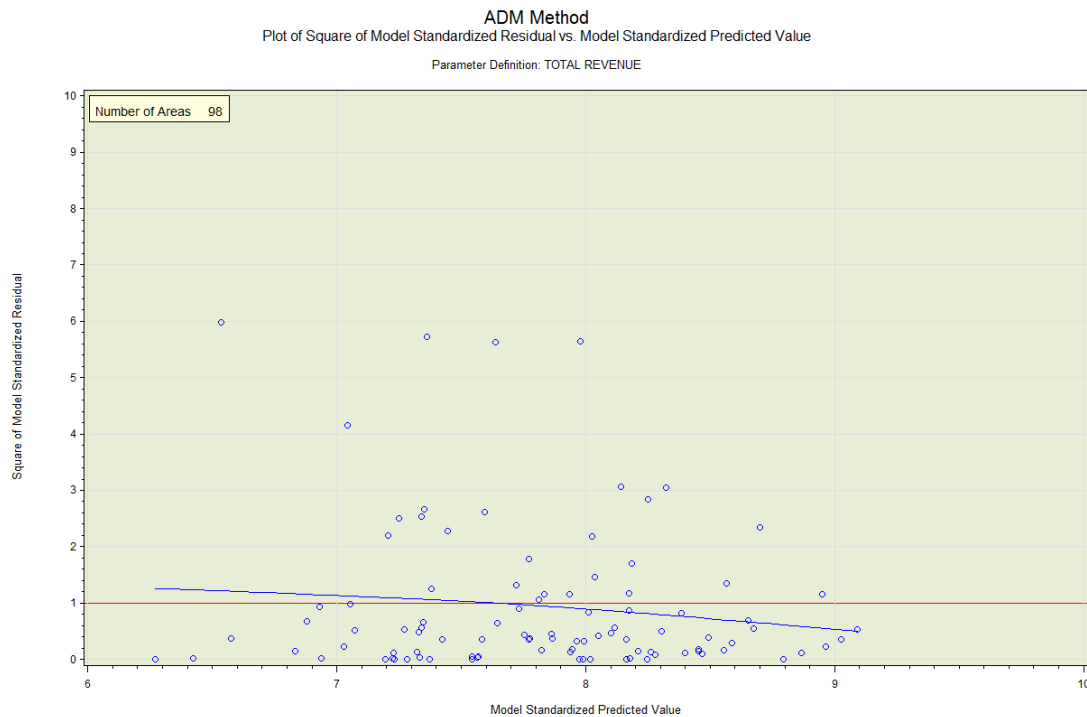
The red line is a reference line with value 0. The differences between the transformed direct estimate and the transformed SAE estimate should be close to this line. These differences should appear to be random. Any clearly distinct pattern indicates a lack of fit of the model. The blue curve is a smooth spline between the points on the plot. If the model fits well then the smooth spline should be close to the reference line.

Model Standardized Residual vs. Model Standardized Predicted Value



The red line is a reference line with value 0. The model standardized residual values should be close to this line. These differences should appear to be random. Any clearly distinct pattern indicates a lack of fit of the model. The blue curve is a smooth spline between the points on the plot. If the model fits well then the smooth spline should be close to the reference line.

Square of Model Standardized Residual vs. Model Standardized Predicted Value

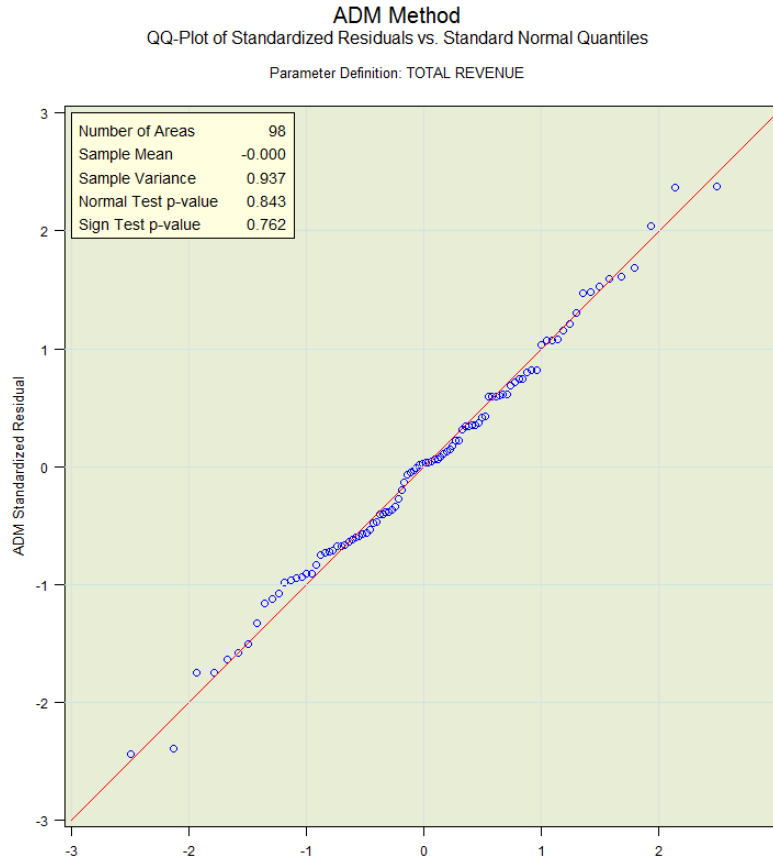


The red line is a reference line with value 1. The square of the model standardized residual values should be close to this line. These differences should appear to be random. Any clearly distinct pattern indicates a lack of fit of the model. The blue curve is a smooth spline between the points on the plot. If the model fits well, then the smooth spline should be close to the reference line.

QQ-plot of standardized residuals vs. standard Normal quantiles

This plot displays the standardized residuals from the SAE model against the corresponding quantiles from a standard Normal distribution. The red line on the plot has an intercept of 0 and a slope of 1. If the model fits well, then the points should be close to this line.

Two formal tests are shown in the inset. The first is a test of the hypothesis that the residuals come from a standard Normal distribution. This is done by calculating the Shapiro-Wilk test statistic. The p-value is the probability of observing a more extreme test statistic. A very small p-value is evidence against the hypothesis. The second test is a sign test on the standardized residuals. It is based on the observed number of positive and negative residuals. If the SAE model is good then we should expect to see about the same number of positive and negative residuals. Again, a small p-value provides evidence against this hypothesis.



Shapiro–Wilk W test of the standardized residuals

The Shapiro–Wilk test calculates a W statistic that tests whether the standardized residuals $\hat{\xi}_i$ for $i \in A_{VAL}$, follow a $N(0, 1)$ Normal distribution. Small values of W are evidence of departure from normality. The W statistic is calculated as follows:

$$W = \frac{\left(\sum_{i \in A_{VAL}} a_i \hat{\xi}_{(i)} \right)^2}{\sum_{i \in A_{VAL}} (\hat{\xi}_{(i)} - \bar{\hat{\xi}})^2}$$

where the $\hat{\xi}_{(i)}$ are the ordered values of the standardized residuals ($\hat{\xi}_{(1)}$ is the smallest), $\bar{\hat{\xi}}$ is their mean and the a_i are constants generated from the means, variances and covariances of the order statistics of a sample of size m from a normal distribution (see Pearson and Hartley (1972), Table 15).

Sign test on the standardized residuals

The sign test statistic M is based on the hypothesis that the median of the standardized residuals is 0, $H_0 : median = 0$. We let n^+ denote the number of residuals with values greater than 0 and n^- represent the number of residuals with values less than 0. Then we compute the observed test statistic $M_{obs} = (1/2)(n^+ - n^-)$. Under $H_0 : median = 0$, the p-value for the observed statistic is calculated as follows.

$$Prob\{|M| \geq |M_{obs}|\} = \left(\frac{1}{2}\right)^{n_t-1} \sum_{i=0}^{\min(n^+, n^-)} \binom{n_t}{i}$$

The quantity $n_t = n^+ + n^-$ is the number of positive and negative standardized residuals—those not equal to 0. A small p-value provides evidence against the hypothesis and the fit of the SAE model.

Influence measures

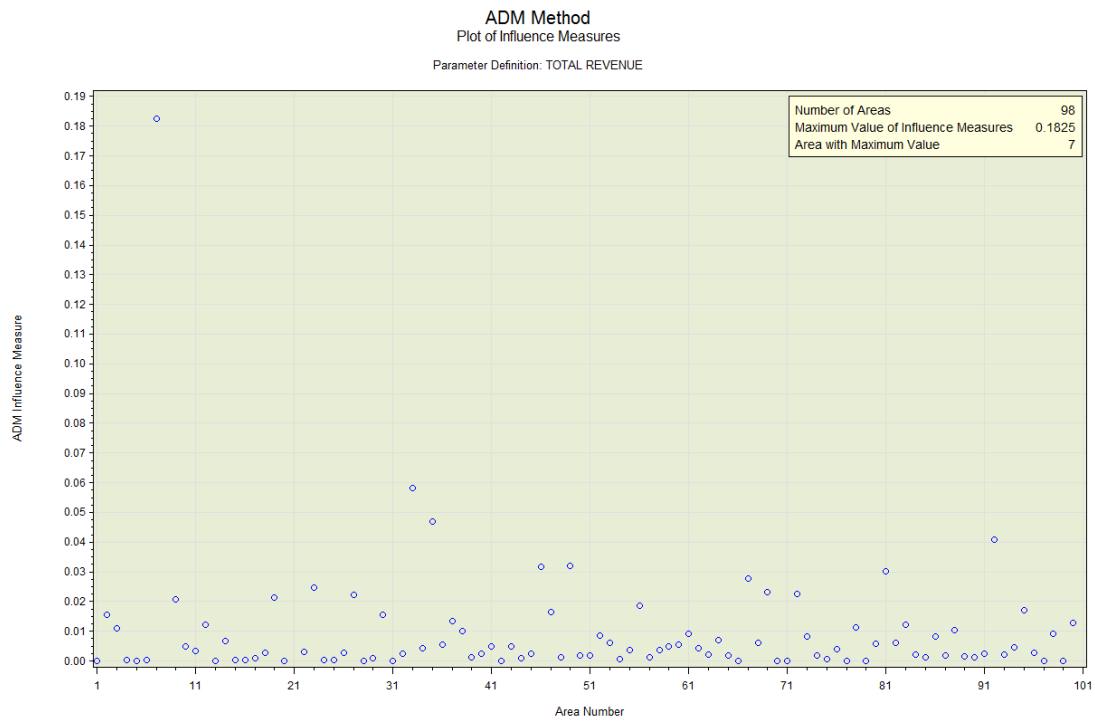
This plot provides an evaluation of the influence of each area on the fit of the SAE model. It is useful for determining domains that could be outliers.

Influence measure of each area

This test uses the Mahalanobis distance to measure the influence of each area on the fit of the small-area model. As described by Banerjee and Frees (1977), we compute following statistic for each area $i \in A_{VAL}$.

$$B_i(\hat{\beta}) = \frac{(\hat{\beta} - \hat{\beta}_{(i)})^T \left(\sum_{i \in A_{VAL}} \frac{\mathbf{z}_i \mathbf{z}_i^T}{(\psi_i + b_i^2 \hat{\sigma}_v^2)} \right) (\hat{\beta} - \hat{\beta}_{(i)})}{p}$$

Here, $\hat{\beta}_{(i)}$ is the estimate of β after dropping area i and p is the number of auxiliary variables in $\mathbf{z}_i$ (including the augmented value $\omega_i \tilde{\psi}_i$ or $\omega_i \hat{\psi}_i$ if the method of augment auxiliary vector is used). They note that the values of $B_i(\hat{\beta})$ can be compared to a χ_p^2 distribution, with extreme values indicating influential areas. A dot plot of the values $B_i(\hat{\beta})$ provides another approach to the identification of areas which have a significant influence of the model. The following plot shows the $B_i(\hat{\beta})$ values for 98 areas with the area number on the x -axis. In this example, area 7 (with an influence measure of 0.1825) appears as an outlier and is a possible influential area. One way to handle influential areas is to increase the b_i value for these areas so they have less effect on the regression coefficient of the model.



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